GAME MATHEMATICS

WITCHCRAFT

Mathematical model of the published game

96.70%	50,000×	256	8
RTP in every bet mode	max win per round	ways, board 4-2-4-2-4	bet modes, 5 purchasable

PREPARED FOR
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MODEL VERSION
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DATA BASIS
6,106,027 published rounds and their lookup tables

Contents

1 Executive summary 3
2 Game rules 4
3 Mathematical architecture 6
4 The eight modes compared 8
5 Base game (BASE) 12
6 Enhanced Spins 1 (ENHANCE1) 19
7 Enhanced Spins 2 (ENHANCE2) 25
8 Hexed Spins (HEXED) 31
9 Cursed Spins (CURSED) 37
10 Wicked Fortune buy (WICKED) 43
11 Grave Spell buy (GRAVE) 48
12 Mystery bonus (MYSTERY) 54
13 Inside the free spins 59
14 Sessions: waiting times and results over many rounds 64
15 Maximum win and win cap 70
16 Verification and consistency 72

A Conditional distributions by outcome class 78
B Way wins by symbol in the bonuses 83

Reading conventions

×	Multiple of the base bet (the price of one BASE spin, all 256 ways).
Round	One paid spin and any bonus it triggers, or one purchase. Probabilities are per round.
1 in N	Probability 1/N per round — an average rate, not a schedule.
RTP points	Percentage of the price returned; the points of a mode add up to 96.70.
Bands	"5–9.9" = every win from 5.0× to 9.9× (wins are multiples of 0.1×).
Labels	EXACT from the published files · EXACT FORMULA formula on exact values · MONTE CARLO simulation, 95% intervals · CONFIGURATION simulation input · SPECIFICATION design rule or target

CHAPTER 1

Executive summary

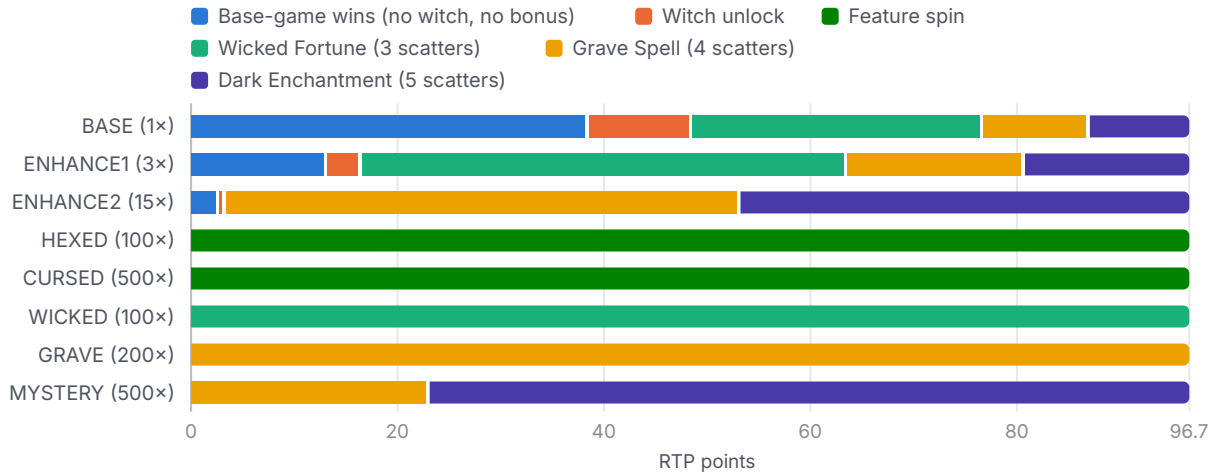
8 bet modes, each returning 96.70% of its price, with a 50,000× cap per round.

Table 1.1 Key indicators by bet mode EXACT

Mode	Price	RTP	Hit rate	Win ≥ price	Median win	SD / price	Max win: 1 in	Bonus
BASE	1×	96.70%	22.00%	7.630%	0×	45.72	6,243,175	bonus 1 in 175
ENHANCE1	3×	96.70%	15.49%	4.905%	0×	30.04	2,236,069	bonus 1 in 35.0
ENHANCE2	15×	96.70%	24.29%	3.343%	0×	13.20	262,948	bonus 1 in 35.0
HEXED	100×	96.70%	20.00%	12.26%	0×	9.55	15,000	one feature spin
CURSED	500×	96.70%	70.00%	17.46%	125.6×	4.07	3,100	one feature spin
WICKED	100×	96.70%	100%	22.08%	43.70×	6.09	40,700	every purchase
GRAVE	200×	96.70%	100%	25.05%	86.80×	3.52	44,700	every purchase
MYSTERY	500×	96.70%	35.00%	15.52%	0×	4.38	3,083	bonus 35.0%

Hit rate: P(win > 0). SD / price: standard deviation per unit staked (volatility). Max win: 50,000× exactly.

Figure 1.1 Where the return comes from EXACT



Each bar adds up to 96.70 points. In BASE, base game 50.0% and bonuses 50.0% of the return.

1.1 Key findings

- **BASE** — 22% of spins win; a bonus 1 in 175 spins; base game and bonuses share the return 50/50.
- **Special squares and witch** — a special symbol breaks through on 8.8% of BASE spins; the witch unlocks the squares 1 in 50.0 spins (median 4.80×) and drops the scatters of 30% of the bonuses.
- **Dark Enchantment** — 1 in 20,000 BASE spins, never below 1,000× (median 1,460×, mean 1,965×).
- **ENHANCE1 / ENHANCE2** — 3×: every tier 5× as frequent, bonuses 83% of the return; 15×: only 4+ scatters, a bonus 1 in 35.0 spins, Grave Spell averaging 296×.
- **Purchases** — bonus buys: median 43–44% of the price, Grave Spell pays its price back 1 in 3.99; HEXED pays nothing 80% of the time, CURSED 30%; Mystery: 65% losing spins; volatility within the brief rating for every rated mode (Chapter 4).
- **Verified** — 6,106,027 published rounds re-checked: 0 defects; every specification target met (Chapter 16).

CHAPTER 2

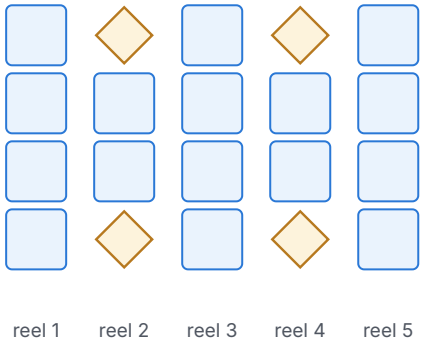
Game rules

The rules that every published round follows.

2.1 Board and ways

Five reels of 4, 2, 4, 2 and 4 cells (256 ways), plus four **special squares** above and below reels 2 and 4. A way is one cell per reel on consecutive reels from reel 1; it pays for 3 to 5 reels when every cell holds the symbol or a wild and at least one cell holds the symbol (a way made only of wilds pays nothing). There is no wild on the reels: wilds come from the special squares. There are no tumbles: every spin is evaluated once.

Figure 2.1 The board SPECIFICATION



Squares: ordinary cells. Diamonds: the four special squares (reel 2 top and bottom, reel 4 top and bottom), read in that order.

2.2 Paytable

Table 2.1 Win of one way (× base bet) SPECIFICATION

Code	Symbol	Class	3 reels	4 reels	5 reels
H1	Premium 1	Premium	1.0×	3.0×	10.0×
H2	Premium 2	Premium	0.8×	2.0×	6.0×
H3	Premium 3	Premium	0.6×	1.5×	4.0×
H4	Premium 4	Premium	0.5×	1.2×	3.0×
L1	A	Low	0.3×	0.6×	1.5×
L2	K	Low	0.3×	0.5×	1.2×
L3	Q	Low	0.2×	0.4×	1.0×
L4	J	Low	0.2×	0.3×	0.8×
L5	10	Low	0.1×	0.3×	0.6×

The order of the four premiums will be set after the artwork (brief): H1 is the best paid. Scatters (S) do not pay.

$$\text{win of a symbol} = \text{pay}(\text{symbol}, \text{reels}) \times \sum_{\text{ways}} \max\left(1,\ \sum \text{multipliers of the wilds on the way}\right)$$

Wild multipliers on the same way **add up** (wild ×5 + wild ×10 = ×15); a ×1 wild adds nothing and a way without a multiplied wild pays once.

2.3 Special squares

Table 2.2 The four special symbols SPECIFICATION

Symbol	One to three on the squares	All four squares
Broomstick	the board **splits**: every symbol and wild counts twice, so each way counts ×2 per reel (×8, ×16, ×32 on 3, 4, 5 reels); several broomsticks give the same split	the whole board lands again with premiums only, split, and always pays
Spellbook	1 to 6 ×1 wilds on random cells (never on a scatter or a wild)	only 2 **wild nudges** on 2 of reels 1, 3, 5: the wild slides to fill the reel, multiplier = number of steps (×1 to ×3)
Orb	a wild with a multiplier on a random cell: ×2, ×3, ×5, ×10, ×20, ×50, ×100, ×250, ×500, ×1,000 (a ×1 wild takes the multiplier)	4 wilds of ×20 or more
Mirror	nothing	**max win**: 50,000× at once

A square without a special symbol shows a skull. Resolution order: mirrors, spellbooks (or wild nudges), orbs, broomsticks, then the ways. Wild nudges never use a reel showing a scatter.

- **Base game** — the squares are locked; a special symbol breaks through from time to time.
- **Unlocked** — in the free spins, in the HEXED and CURSED spins, and when the witch unlocks them, special symbols are much more frequent.
- **Witch (base game)** — unlocks the four squares with at least 2 special symbols on 1 spin in 50, or drops three of the scatters of a bonus trigger (about 30% of the bonuses triggered in the base game).
- **Witch (free spins)** — at least 2 special symbols on random free spins.

2.4 Free-spins bonuses

Table 2.3 Bonus tiers SPECIFICATION

Tier	Scatters	Free spins	Special squares	Purchase
Wicked Fortune	3	10	unlocked for the whole bonus	WICKED, 100×
Grave Spell	4	10	at least 1 special symbol on every spin	GRAVE, 200×
Dark Enchantment	5	8	at least 3 special symbols on every spin	not purchasable

Dark Enchantment is a “1,000× equivalent”: it never pays less than 1,000×. Scatters: one per reel at most.

- **Retrigger** — in the free spins, every scatter on the board adds one free spin (at most 40 spins in a bonus).
- **End** — when the spins run out, or at once when the round reaches 50,000×.

2.5 Bet modes and win cap

Table 2.4 The 8 bet modes SPECIFICATION

Mode	Price	What the player gets
BASE	1×	One spin: ways, special squares, witch, bonuses.
ENHANCE1	3×	Same spin; each bonus tier 5× as frequent (any bonus 1 in 35.0).
ENHANCE2	15×	Same spin; only 4 or 5 scatters trigger (any bonus 1 in 35.0).
HEXED	100×	One spin, squares unlocked, at least 2 special symbols. No scatters; 80% of spins pay nothing.
CURSED	500×	One spin with 4 special symbols. No scatters; 30% of spins pay nothing.
WICKED	100×	Wicked Fortune bonus (3 scatters).
GRAVE	200×	Grave Spell bonus (4 scatters).
MYSTERY	500×	65% losing spin · 25% Grave Spell · 10% Dark Enchantment.

Win cap: 50,000× the base bet per round in every mode; the round stops as soon as it is reached.

CHAPTER 3

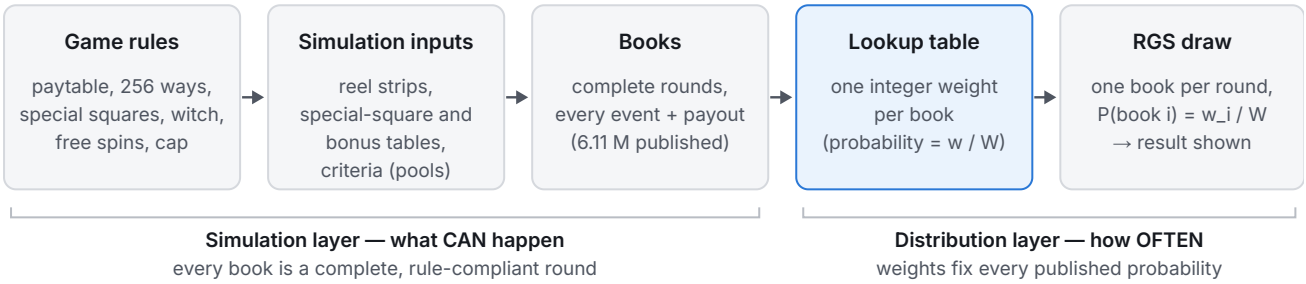
Mathematical architecture

How the game server draws a round, and which figures are exact.

3.1 How a round is drawn

Each mode is published as a **book** file (one complete pre-computed round per line) and a **lookup table** (one integer weight per book). For every round the game server draws one book, with probability proportional to its weight.

Figure 3.1 Books define what can happen; weights define how often — SPECIFICATION



$$P(\text{book } i) = \frac{w_i}{W}, \qquad W = \sum_j w_j, \qquad P(E) = \frac{1}{W} \sum_{i\,:\,E \text{ occurs in book } i} w_i$$

Every per-round figure in this document is such a finite weighted sum, computed exactly.

3.2 Published books

The game engine simulated each mode in pools (no win, base-game wins, witch, each bonus tier, high-win bands, max win), so that rare outcomes are carried by many distinct rounds.

Table 3.1 Books per mode — EXACT

Mode	Simulated	Published	Pools	Distinct payouts	Effective books	Heaviest book
BASE	1,300,000	1,266,915	28	155,484	383,688	1 in 62,615
ENHANCE1	1,300,000	1,266,729	28	155,562	374,054	1 in 74,537
ENHANCE2	1,000,000	984,947	21	130,048	356,216	1 in 52,812
HEXED	500,000	493,605	8	25,474	312,321	1 in 138,106
CURSED	500,000	495,382	8	27,266	219,305	1 in 81,486
WICKED	500,000	499,551	7	76,711	340,855	1 in 170,147
GRAVE	500,000	499,553	6	77,756	375,875	1 in 184,233
MYSTERY	600,000	599,345	13	134,862	295,821	1 in 172,657
Total	**6,200,000**	**6,106,027**	**119**			

Books with weight 0 can never be drawn and were not published. Effective books: $(\Sigma w)^2 / \Sigma w^2$. Heaviest book: probability of the most likely single round.

3.3 Lookup tables

Weights were computed deterministically from the specification's targets (bonus frequencies, medians, lowest possible wins, max-win rates), then tilted within each payout band to hit the RTP exactly. The most likely single book of any mode has probability 1 in 52,812. Chapter 16 checks every target.

3.4 Exact results, estimates and inputs

Table 3.2 Nature of the figures

Label	Covers	Precision
EXACT	Every per-round statistic: RTP, probabilities, moments, percentiles, bonus mechanics.	Exact for the published files; rounded for display.
EXACT FORMULA	Closed formulas on exact values, e.g. $1 - (1 - p)^N$.	Exact (rounds are independent).
MONTE CARLO	Session returns and bankroll survival.	Estimates; sessions simulated and 95% intervals stated.
CONFIGURATION	Reel strips, special-square tables, bonus profiles, pools.	Simulation inputs; they do not set frequencies.
SPECIFICATION	Rules and design targets.	Compared with measured values in Chapter 16.

Not available: a "theoretical" RTP from reel strips (strips only generate the books) and the uncapped value of max-win rounds (they stop at the cap, Chapter 15).

3.5 Reel strips

6 strip sets of five reels; no wild and no special symbol on any strip (special symbols land on the four special squares, drawn separately); a reel never shows two scatters; the feature-spin strips carry no scatter.

Table 3.3 Reel strip sets CONFIGURATION

Set	Role	Used by	Stops per reel	Premium	Low	Scatters per reel
BR0	Paid spins and bonus-triggering spins	BASE, ENHANCE1, ENHANCE2, bonus buys, MYSTERY	122	49.2%	49.2%	2
BRX	Base-game tail material: large wins and max-win spins	BASE, ENHANCE1, ENHANCE2	101	49.5%	49.5%	1
FR0	Free spins	all bonus tiers, all modes	241	49.8%	49.8%	1
FRX	Free-spin tail material: high bonus wins and max wins	all bonus tiers, all modes	201	49.8%	49.8%	1
FT0	Feature spins	HEXED, CURSED	120	50.0%	50.0%	0
FTX	Feature-spin tail material: wins ≥ 5,000× and max wins	HEXED, CURSED	100	50.0%	50.0%	0

Premium: share of stops holding H1–H4; Low: L1–L5; the rest are scatters. Strips only generate the books: played frequencies are set by the lookup tables.

CHAPTER 4

The eight modes compared

The eight modes side by side, relative to the price paid.

4.1 Moments and volatility indicators

For a round win X (in base bets) bought at price c, the table gives the mean $\mu = E[X]$, the RTP μ/c, the standard deviation σ, the coefficient of variation σ/μ, the volatility index $\mathrm{VI}_{90} = 1.645\,\sigma/c$ and the standardised skewness and excess kurtosis.

Table 4.1 Moments of the round win — EXACT

Mode	Price	Mean (×)	RTP	SD (×)	SD / price	CV	VI 90%	VI 95%	Skewness	Excess kurtosis
BASE	1×	0.9670	96.7000%	45.72	45.722	47.28	75.21	89.61	702.4	664,865
ENHANCE1	3×	2.9010	96.7000%	90.13	30.045	31.07	49.42	58.89	313.0	138,908
ENHANCE2	15×	14.5050	96.7000%	197.99	13.199	13.65	21.71	25.87	101.5	20,483
HEXED	100×	96.7000	96.7000%	954.78	9.548	9.87	15.70	18.71	35.8	1,522
CURSED	500×	483.5000	96.7000%	2,035.10	4.070	4.21	6.69	7.98	14.4	271
WICKED	100×	96.7000	96.7000%	608.92	6.089	6.30	10.02	11.93	52.5	3,409
GRAVE	200×	193.4000	96.7000%	703.79	3.519	3.64	5.79	6.90	32.9	1,688
MYSTERY	500×	483.5000	96.7000%	2,188.01	4.376	4.53	7.20	8.58	11.7	193

SD: standard deviation of the round win in base bets. CV: SD / mean. VI: volatility index per unit staked (z·σ/c with z = 1.645 or 1.960). Very large skewness and kurtosis are normal for slot distributions: most rounds pay little and a few pay a lot.

Table 4.2 Volatility against the brief — SPECIFICATION

Mode	Price	Brief rating	Target SD / price	Published SD / price	Status
ENHANCE1	3×	3.5/5	29.19 – 32.70	30.04	met
ENHANCE2	15×	4.5/5	11.12 – 13.21	13.20	met
HEXED	100×	5/5	≥ 8.88	9.55	met
CURSED	500×	4/5	4.03 – 4.26	4.07	met
WICKED	100×	5/5	≥ 5.80	6.09	met
GRAVE	200×	4/5	3.45 – 4.61	3.52	met
MYSTERY	500×	5/5	≥ 4.26	4.38	met

The brief rates the volatility of each mode out of 5. Each rating is a design target on the SD / price (standard deviation of the round win divided by the price of the mode), set for the price of the mode. BASE has no rating in the brief; the Dark Enchantment is not a bet mode: it is part of the modes that trigger it.

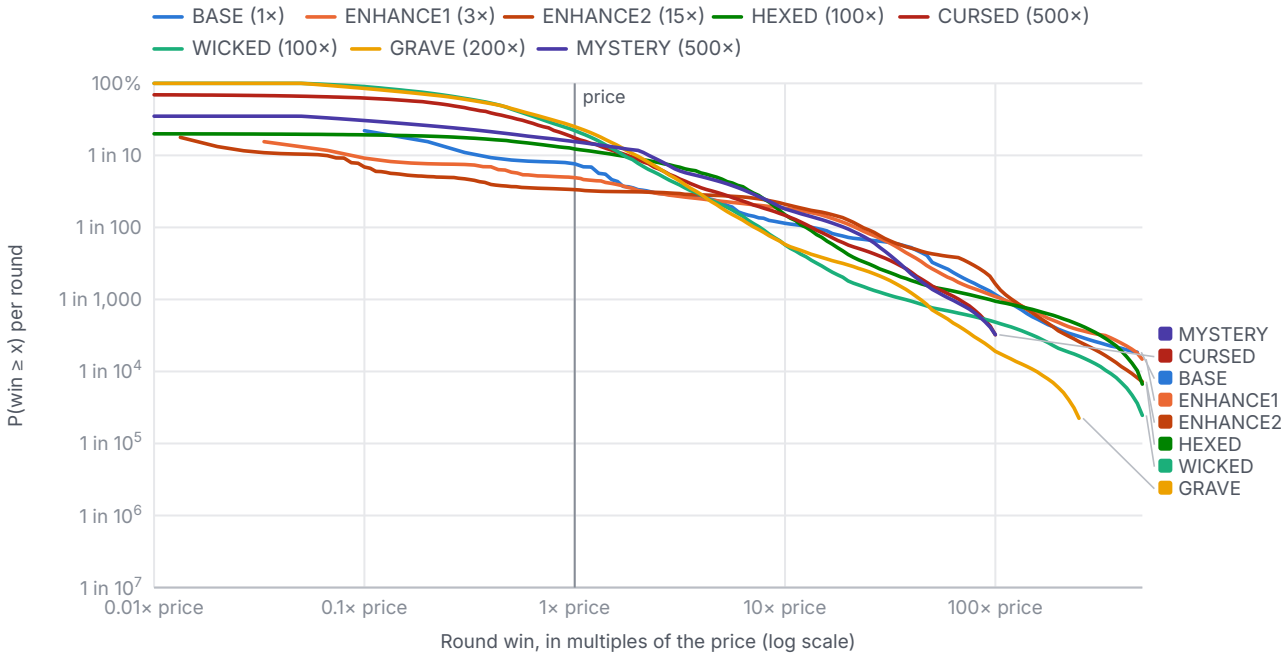
4.2 Distribution relative to the price

Table 4.3 Percentiles of the round win, in multiples of the price EXACT

Mode	P25	Median	P75	P90	P99	P99.9	P99.99	Maximum
BASE	0.000	0.000	0.000	0.300	13.500	109.200	742.400	50,000
ENHANCE1	0.000	0.000	0.000	0.067	21.600	108.333	597.567	16,667
ENHANCE2	0.000	0.000	0.000	0.060	23.387	119.860	431.707	3,333
HEXED	0.000	0.000	0.000	1.600	12.608	93.240	476.784	500
CURSED	0.000	0.251	0.710	1.779	13.376	67.210	100.000	100
WICKED	0.191	0.437	0.884	1.710	7.680	37.177	341.402	500
GRAVE	0.159	0.434	1.002	1.965	7.162	44.054	156.876	250
MYSTERY	0.000	0.000	0.231	2.251	19.106	61.709	100.000	100

Pxx: smallest win w such that P(X ≤ w) ≥ xx%. Read: in BASE, 90% of spins pay at most the P90 value. A value of 0 means that at least that share of rounds pays nothing.

Figure 4.1 Chance per round of winning at least x times the price EXACT



Each curve gives, for each multiple x of the price, the probability that one round pays at least that much. The paid spins (BASE, ENHANCE1, ENHANCE2) start lower (most spins pay nothing) but reach far higher multiples of their small price: the cap of 50,000× is 50,000 times the BASE price but only 100 times the CURSED and Mystery prices. Among the purchases, HEXED (80% zero) and the Mystery (65% zero) are all or nothing: most rounds pay nothing, the others pay well.

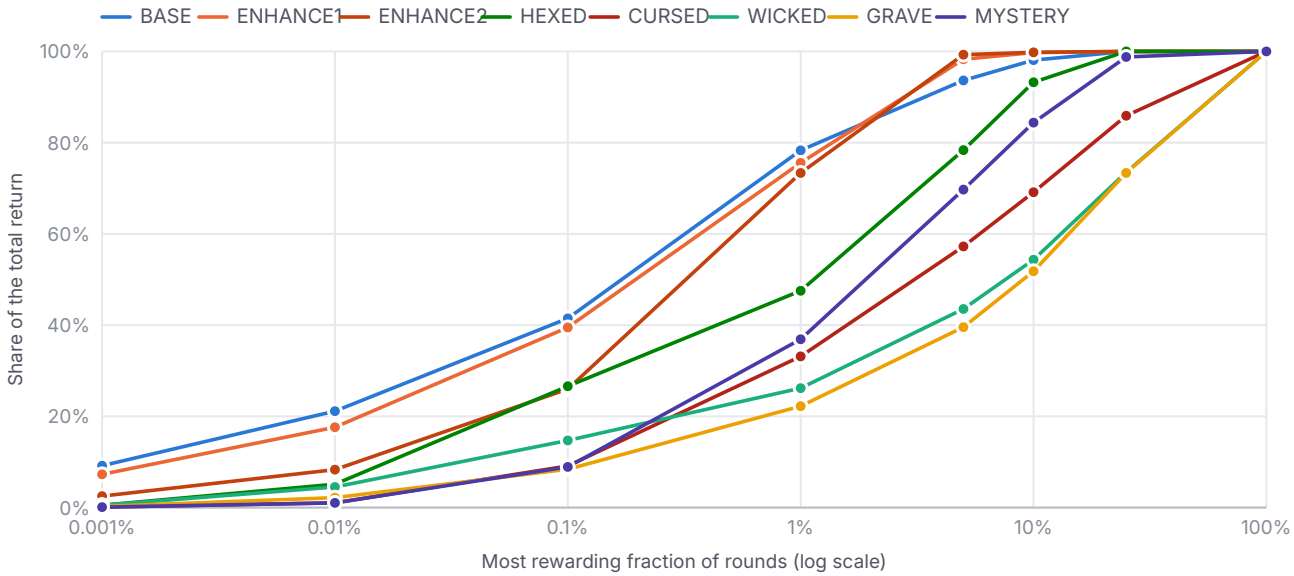
Table 4.4 Chance per round of a win of at least k times the price EXACT

Mode	≥ 0.5× price	≥ 1× price	≥ 2× price	≥ 5× price	≥ 10× price	≥ 20× price	≥ 50× price
BASE	8.62% 1 in 11.6	7.63% 1 in 13.1	3.34% 1 in 29.9	2.22% 1 in 45.1	1.14% 1 in 87.6	0.725% 1 in 138	0.333% 1 in 301
ENHANCE1	5.84% 1 in 17.1	4.90% 1 in 20.4	3.25% 1 in 30.7	2.30% 1 in 43.4	1.84% 1 in 54.3	1.10% 1 in 90.5	0.267% 1 in 375
ENHANCE2	3.69% 1 in 27.1	3.34% 1 in 29.9	3.12% 1 in 32.1	2.76% 1 in 36.2	2.10% 1 in 47.7	1.26% 1 in 79.2	0.430% 1 in 233
HEXED	15.7% 1 in 6.36	12.3% 1 in 8.15	9.00% 1 in 11.1	4.69% 1 in 21.3	1.50% 1 in 66.8	0.419% 1 in 239	0.149% 1 in 669
CURSED	33.5% 1 in 2.99	17.5% 1 in 5.73	8.74% 1 in 11.4	2.92% 1 in 34.3	1.47% 1 in 67.9	0.568% 1 in 176	0.154% 1 in 650
WICKED	43.9% 1 in 2.28	22.1% 1 in 4.53	7.73% 1 in 12.9	2.06% 1 in 48.4	0.572% 1 in 175	0.180% 1 in 556	0.0765% 1 in 1,308
GRAVE	44.7% 1 in 2.24	25.0% 1 in 3.99	9.68% 1 in 10.3	1.85% 1 in 54.0	0.580% 1 in 172	0.300% 1 in 333	0.0719% 1 in 1,392
MYSTERY	19.8% 1 in 5.06	15.5% 1 in 6.44	11.7% 1 in 8.52	4.25% 1 in 23.5	1.81% 1 in 55.4	0.947% 1 in 106	0.139% 1 in 722

"—": threshold above the 50,000× cap for this mode (impossible).

4.3 Concentration of the return

Figure 4.2 Share of the total return carried by the most rewarding rounds EXACT



Rounds are ranked from the highest win down; the curve gives the share of the mode's total return paid by the top fraction of rounds. In BASE, the top 1% of spins pays 78.3% of everything returned (the other 99% average 0.21× per spin); in the HEXED spins the top 1% pays 47.5%. The steeper the curve, the more the return depends on rare rounds.

4.4 Scatter anticipation

Table 4.5 Rounds whose reels slow down for the missing scatter EXACT

Mode	Rounds with anticipation	Frequency	A bonus follows
BASE	1.69%	1 in 59.0	23.6%
ENHANCE1	3.33%	1 in 30.0	60.2%
ENHANCE2	3.34%	1 in 30.0	61.6%
HEXED	never	—	—
CURSED	never	—	—
WICKED	never	—	—
GRAVE	never	—	—
MYSTERY	never	—	—

Anticipation as written in the published books (reveal event of the paid or bought spin: the reels after the second scatter), weighted by the lookup tables; never in the free spins. How it is shown is up to the game client. A bonus follows: share of the anticipated rounds that end in a bonus.

4.5 Maximum win by mode

Table 4.6 Probability of the 50,000× max win EXACT

Mode	Cap in prices	P(max win)	Frequency	RTP points at the cap	Books at the cap
BASE	50,000	1.60×10^{-5}%	1 in 6,243,175	0.801	19,012
ENHANCE1	16,667	4.47×10^{-5}%	1 in 2,236,069	0.745	19,031
ENHANCE2	3,333	0.0003803%	1 in 262,948	1.268	13,704
HEXED	500	0.006667%	1 in 15,000	3.333	10,290
CURSED	100	0.03226%	1 in 3,100	3.226	12,667
WICKED	500	0.002457%	1 in 40,700	1.229	12,705
GRAVE	250	0.002237%	1 in 44,700	0.559	13,062
MYSTERY	100	0.03243%	1 in 3,083	3.243	14,082

RTP points at the cap: 50,000 × P(max win) / price × 100. Every mode has more than 10,000 distinct max-win books, so no max-win outcome is a repeated replay.

CHAPTER 5

Base game (BASE)

The 1× spin: base game and bonuses share the return about equally.

5.1 At a glance

Return to player **96.7000%** mean win 0.97×	Hit rate **22.00%** rounds paying more than 0	Win ≥ price **7.630%** 1 in 13.1	Median win **0×** 0.0% of the price
Standard deviation **45.72 × price** 45.7× in base bets	Win of 100× price or more **0.117%** 1 in 857	Max win 50,000× **1 in 6,243,175** 1.60 × 10^{-5}%	Published books **1,266,915** 155,484 distinct payouts

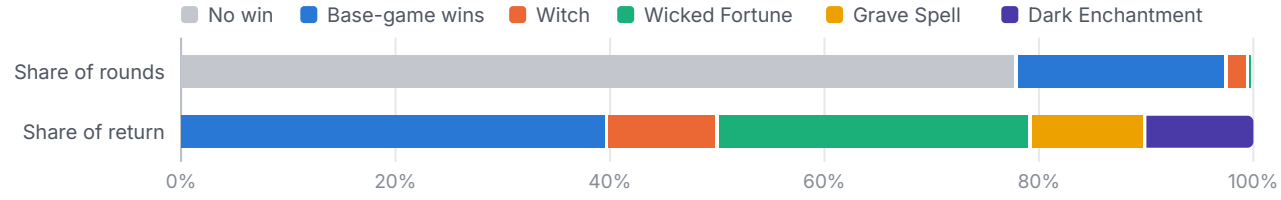
5.2 Outcome classes

Table 5.1 Outcome classes: frequency, value and conditional distribution — EXACT

Class	Probability per round	Frequency	RTP points	Share of RTP	Mean win	Median	P10 – P90	P(no win)	Max win within class	Books
No win	77.86%	1 in 1.28	0.00	0.0%	0×	0×	0× – 0×	100%	—	285,503
Base-game wins (no witch, no bonus)	19.57%	1 in 5.11	38.36	39.7%	1.96×	0.20×	0.10× – 1.60×	0%	1 in 271,935,707	298,488
Witch unlock	2.000%	1 in 50.0	9.99	10.3%	5.00×	4.80×	1.00× – 8.20×	7.14%	1 in 413,702	126,888
Wicked Fortune (3 scatters)	0.5000%	1 in 200	28.20	29.2%	56.39×	43.10×	8.00× – 114.2×	0%	1 in 100,000	176,441
Grave Spell (4 scatters)	0.06667%	1 in 1,500	10.33	10.7%	154.9×	85.50×	15.40× – 301.4×	0%	1 in 60,000	187,202
Dark Enchantment (5 scatters)	0.005000%	1 in 20,000	9.83	10.2%	1,965.3×	1,460.1×	1,088.6× – 2,906.4×	0%	1 in 1,000	192,393
All rounds	**100%**		**96.70**	**100%**	**0.97×**	**0×**	**0× – 0.30×**	**78.0%**	**1 in 6,243,175**	**1,266,915**

Every round belongs to exactly one class. Mean, median, P10–P90, P(no win) and the max-win frequency are conditional on the class (i.e. per round of that class). RTP points add up to the mode's RTP.

Figure 5.1 BASE: share of spins and share of the return by outcome class — EXACT

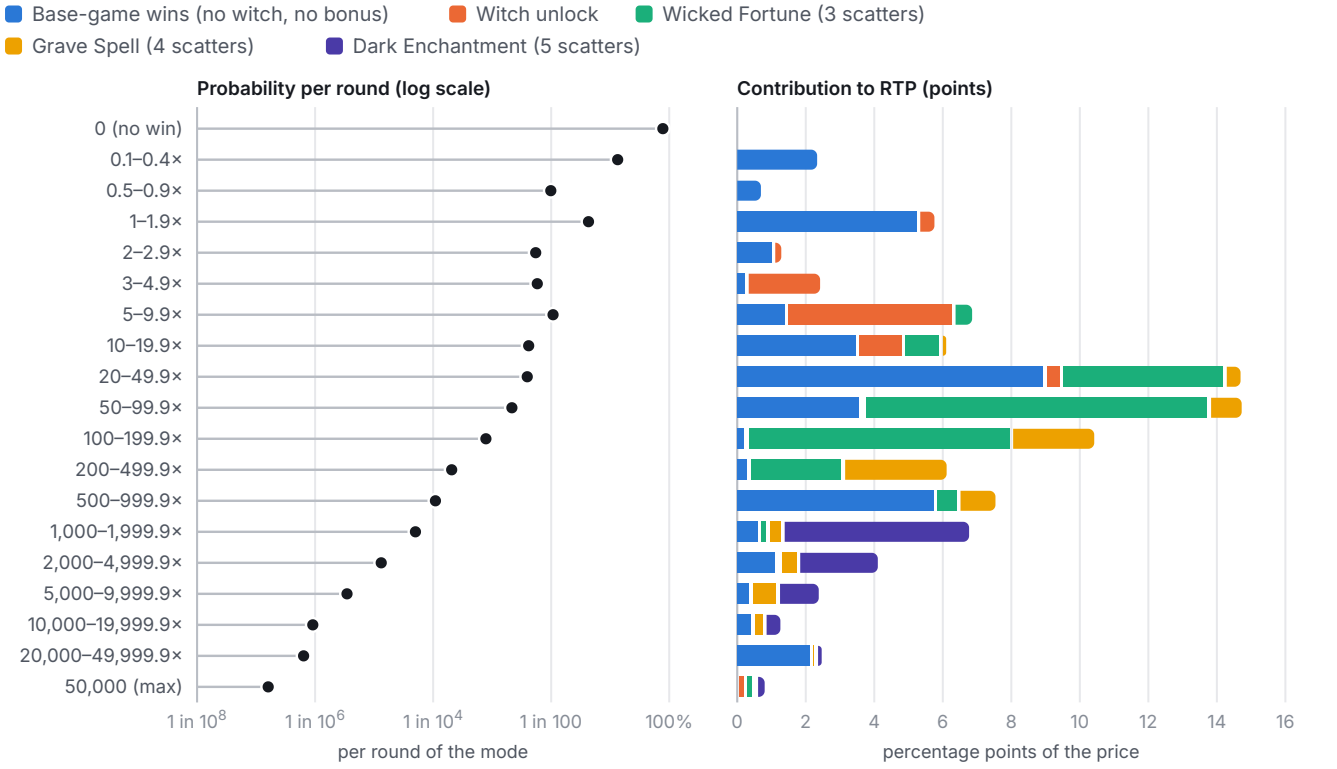


77.86% of spins pay nothing; base-game wins without the witch occur on 19.57% of spins and carry 39.7% of the return. Bonuses are 0.572% of spins but 50.0% of the return.

5.3 Payout distribution

Figure 5.2 BASE: probability and RTP contribution of each payout band

EXACT



Left: probability that a round's total win falls in the band (dots on a log scale; empty bands are omitted). Right: the band's contribution to the RTP, coloured by outcome class. Frequent small wins carry little value; rare large wins carry a large part of it.

Table 5.2 BASE: payout bands EXACT

Round win (× base bet)	Probability	Frequency	Per 10,000 rounds	RTP points	Mean in band	P(win ≥ band start)	Books
0 (no win)	78.00%	1 in 1.28	7,800	0.000	—	100%	319,557
0.1–0.4×	13.38%	1 in 7.47	1,338	2.332	0.17×	22.00%	81,323
0.5–0.9×	0.9854%	1 in 101	98.54	0.691	0.70×	8.616%	95,810
1–1.9×	4.286%	1 in 23.3	429	5.756	1.34×	7.630%	62,485
2–2.9×	0.5466%	1 in 183	54.66	1.286	2.35×	3.344%	14,771
3–4.9×	0.5814%	1 in 172	58.14	2.419	4.16×	2.798%	28,324
5–9.9×	1.075%	1 in 93.0	108	6.856	6.38×	2.216%	34,927
10–19.9×	0.4156%	1 in 241	41.56	6.099	14.67×	1.141%	42,771
20–49.9×	0.3928%	1 in 255	39.28	14.692	37.40×	0.7254%	69,022
50–99.9×	0.2159%	1 in 463	21.59	14.724	68.21×	0.3325%	50,399
100–199.9×	0.07830%	1 in 1,277	7.83	10.420	133.1×	0.1167%	40,248
200–499.9×	0.02060%	1 in 4,853	2.06	6.112	296.6×	0.03839%	36,606
500–999.9×	0.01093%	1 in 9,146	1.09	7.538	689.4×	0.01778%	18,692
1,000–1,999.9×	0.005011%	1 in 19,956	0.5011	6.768	1,350.6×	0.006848%	95,978
2,000–4,999.9×	0.001319%	1 in 75,831	0.1319	4.103	3,111.6×	0.001837%	99,443
5,000–9,999.9×	0.0003470%	1 in 288,159	0.0347	2.379	6,856.7×	0.0005181%	64,235
10,000–19,999.9×	9.13×10^{-5}%	1 in 1,095,004	0.0091	1.260	13,792.2×	0.0001711%	49,334
20,000–49,999.9×	6.38×10^{-5}%	1 in 1,568,069	0.0064	2.465	38,652.9×	7.98×10^{-5}%	43,978
50,000 (max)	1.60×10^{-5}%	1 in 6,243,175	0.0016	0.801	50,000×	1.60×10^{-5}%	19,012
Total	**100%**		**10,000**	**96.700**	**0.97×**		**1,266,915**

Bands are labelled by the smallest and largest win they can contain (wins are multiples of 0.1×). "P(win ≥ band start)" is the survival probability at the lower edge of the band. Per 10,000 rounds: expected number of rounds in the band.

5.4 Percentiles, thresholds and waiting times

Table 5.3 BASE: percentiles of the round win EXACT

Percentile	P25	P50	P75	P90	P95	P99	P99.5	P99.9	P99.99	P99.999
Win (× base bet)	0×	0×	0×	0.30×	1.40×	13.50×	41.00×	109.2×	742.4×	3,353.2×
Win (× price)	0.000	0.000	0.000	0.300	1.400	13.500	41.000	109.200	742.400	3,353.200
Base-game wins (no witch, no bonus) (×, conditional)	0.10×	0.20×	1.00×	1.60×	2.40×	40.00×	48.00×	80.00×	866.0×	4,375×
Witch unlock (×, conditional)	2.00×	4.80×	5.80×	8.20×	11.00×	19.40×	26.40×	51.60×	134.4×	384.0×
Wicked Fortune (3 scatters) (×, conditional)	15.80×	43.10×	71.80×	114.2×	146.3×	276.5×	385.2×	771.4×	1,896.9×	50,000×
Grave Spell (4 scatters) (×, conditional)	29.60×	85.50×	161.4×	301.4×	420.6×	966.7×	1,713×	7,680.5×	18,705.2×	50,000×
Dark Enchantment (5 scatters) (×, conditional)	1,223.9×	1,460.1×	1,826×	2,906.4×	4,647×	9,694.7×	13,270×	49,997.5×	50,000×	50,000×

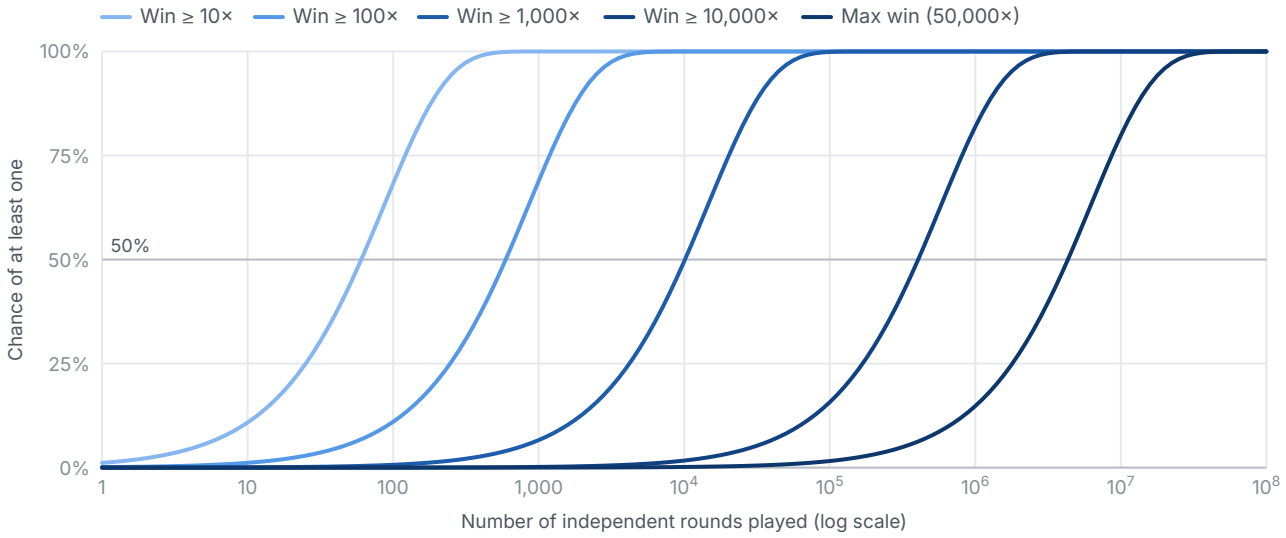
Pxx is the smallest win w with $P(X \le w) \ge xx\%$. Conditional rows give the percentiles within each outcome class.

Table 5.4 BASE: thresholds, frequencies and waiting times EXACT FORMULA

Event per round	Probability	Frequency	Median wait (rounds)	At least once in 100 rounds	At least once in 1,000 rounds	At least once in 10,000 rounds
Any win (> 0)	22.00%	1 in 4.55	3	100%	100%	100%
Win ≥ 1×	7.630%	1 in 13.1	9	100.0%	100%	100%
Win ≥ 2×	3.344%	1 in 29.9	21	96.7%	100%	100%
Win ≥ 5×	2.216%	1 in 45.1	31	89.4%	100%	100%
Win ≥ 10×	1.141%	1 in 87.6	61	68.3%	100.0%	100%
Win ≥ 20×	0.7254%	1 in 138	96	51.7%	99.9%	100%
Win ≥ 50×	0.3325%	1 in 301	209	28.3%	96.4%	100%
Win ≥ 100×	0.1167%	1 in 857	594	11.0%	68.9%	100.0%
Win ≥ 200×	0.03839%	1 in 2,605	1,806	3.77%	31.9%	97.8%
Win ≥ 500×	0.01778%	1 in 5,624	3,898	1.76%	16.3%	83.1%
Win ≥ 1,000×	0.006848%	1 in 14,603	10,122	0.682%	6.62%	49.6%
Win ≥ 2,000×	0.001837%	1 in 54,441	37,736	0.184%	1.82%	16.8%
Win ≥ 5,000×	0.0005181%	1 in 192,996	133,775	0.0518%	0.517%	5.05%
Win ≥ 10,000×	0.0001711%	1 in 584,405	405,079	0.0171%	0.171%	1.70%
Max win (50,000×)	1.60×10^{-5}%	1 in 6,243,175	4,327,439	0.00160%	0.0160%	0.160%

Probabilities are exact. Median wait: smallest n with P(event within n rounds) ≥ 50%, i.e. $\lceil \ln 0.5 / \ln(1-p) \rceil$. "At least once in n rounds" = $1-(1-p)^n$ (independent rounds).

Figure 5.3 BASE: chance of at least one event within n rounds EXACT FORMULA

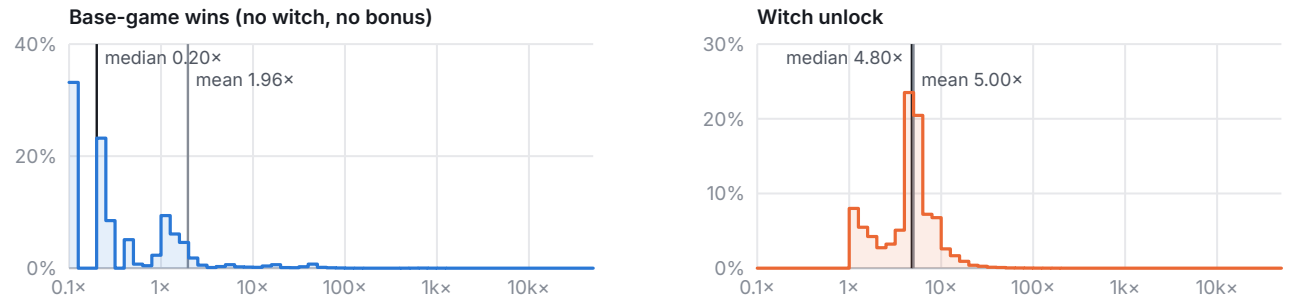


Curves $1-(1-p)^n$ for selected events of the table above (darker = rarer). Each curve crosses 50% at the median waiting time and 63% at n = 1/p rounds.

5.5 Base game without a bonus

99.43% of BASE spins do not trigger a bonus. Among them, wins without the witch occur 1 in 5.11 spins, with a median of 0.20× and a mean of 1.96×; special symbols breaking through the locked squares can lift them up to 50,000×. A win of 100× or more without a bonus occurs 1 in 8,000 spins.

Figure 5.4 BASE: distribution of the win within the two base-game classes EXACT



Conditional probability per tenth of a decade of win (log scale). Vertical lines mark the median (dark) and the mean (grey).

5.6 Special squares in the base game

All paid BASE spins (witch spins and bonus-triggering spins included, free spins excluded).

Table 5.5 BASE: special symbols per spin EXACT

Spins	With a special symbol	Exactly 1	2	3	4	Broomsticks per 100 spins	Spellbooks per 100 spins	Orbs per 100 spins	Mirrors per 100 spins	4 identical
BASE paid spins	8.80%	6.59%	1.77%	0.411%	0.0334%	2.34	3.51	2.36	3.29	1 in 421,981

Per spin played in that context, weighted by the published lookup tables. 4 identical: the four squares show the same special symbol (its “all four” effect).

Table 5.6 BASE: what the special symbols are worth EXACT

Spins	Mean win, no special symbol	Mean win, with one or more	Hit rate, no special symbol	Hit rate, with one or more	Share of the spin wins
BASE paid spins	0.10×	4.49×	19.1%	47.0%	81.3%

Mean win of a spin in base bets (before the round cap). Share of the spin wins: part of the total paid by the spins of that context that comes from spins showing at least one special symbol.

Table 5.7 BASE: effects of the special symbols EXACT

Spins	Board split (broomstick)	4 broomsticks: premium board	Wilds per spellbook	Wild nudges per 1,000 spins	Steps per nudge	Orb wilds per 100 spins	Mean orb multiplier
BASE paid spins	2.22%	1 in 3,540,949	2.32	0.00	1.88	2.36	7.2×

Board split: share of spins with at least one broomstick. Wild nudges come only from four spellbooks (2 per occurrence); the reel multiplier equals the number of steps. Weighted by the published lookup tables.

Table 5.8 BASE: multiplier of the orb wilds EXACT

Spins	×2	×3	×5	×10	×20	×50	×100	×250	×500	×1,000
BASE paid spins	42.4%	28.5%	15.8%	7.35%	3.24%	1.44%	0.824%	0.277%	0.115%	0.0551%

Share of the orb wilds by multiplier (four orbs included: ×20 or more each).

5.7 The witch in the base game

The witch unlocks the four squares on 1 in 50.0 BASE spins (2.000%), with at least two special symbols. The spin then pays in 92.86% of cases, with a median win of 4.80× and a mean of 5.00×; the smallest paying result is 1.00×. Her spins contribute 9.99 RTP points (10.3% of the mode’s RTP). She also drops three of the scatters of 30.1% of the bonuses triggered in BASE.

Table 5.9 BASE: win of a witch spin EXACT

Per witch spin	≥ 1×	≥ 2×	≥ 5×	≥ 10×	≥ 20×	≥ 50×	≥ 100×	≥ 1,000×	Max win
Probability	92.9%	75.1%	44.8%	6.14%	0.971%	0.109%	0.0210%	0.000311%	0.000242%
Per round of the mode	1 in 53.8	1 in 66.5	1 in 112	1 in 814	1 in 5,150	1 in 45,796	1 in 238,211	1 in 16,097,155	1 in 20,685,035

First row: conditional on the witch unlocking the squares (spins without a bonus). Second row: frequency per spin of the mode.

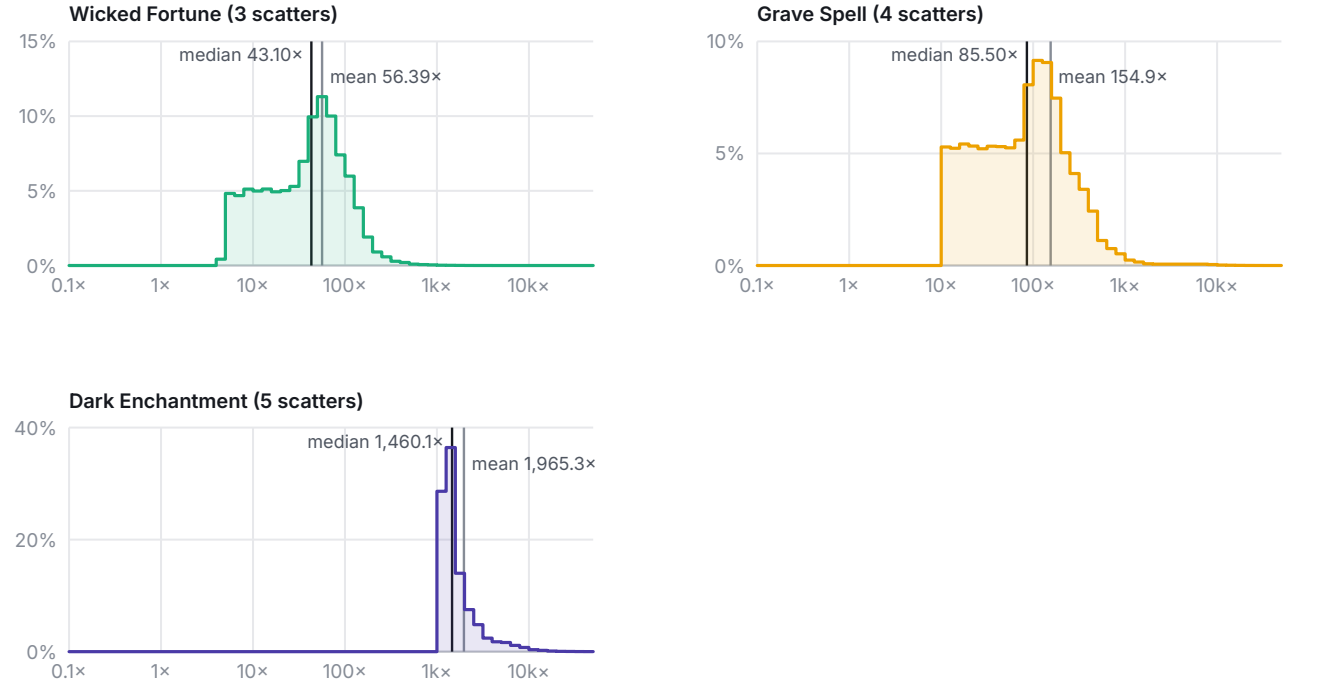
5.8 Bonus tiers as triggered in BASE

Table 5.10 BASE: bonus wins conditional on the tier EXACT

Tier	Frequency	Min	P10	P25	Median	Mean	P75	P90	P99	P(≥ 100×)	P(≥ 1,000×)	Max win
Wicked Fortune (3 scatters)	1 in 200	5.00×	8.00×	15.80×	43.10×	56.39×	71.80×	114.2×	276.5×	14.0%	0.0480%	1 in 100,000
Grave Spell (4 scatters)	1 in 1,500	10.00×	15.40×	29.60×	85.50×	154.9×	161.4×	301.4×	966.7×	44.0%	0.930%	1 in 60,000
Dark Enchantment (5 scatters)	1 in 20,000	1,000.1×	1,088.6×	1,223.9×	1,460.1×	1,965.3×	1,826×	2,906.4×	9,694.7×	100%	100%	1 in 1,000

All statistics except the frequency are per bonus of the tier (conditional on its start); wins in base bets, including the triggering spin.

Figure 5.5 BASE: distribution of the bonus win by tier (conditional on the tier) EXACT



Conditional probability per tenth of a decade of win. The max-win atom (50,000×) is not drawn: it has probability 1 in 100,000, 1 in 60,000, 1 in 1,000 per bonus of 3, 4 and 5 scatters respectively.

5.9 Way wins by symbol

Table 5.11 BASE: RTP points of base-game way wins by symbol (all paid spins, witch included) EXACT

Symbol	3 reels	4 reels	5 reels	Total (pts)	Symbol wins per 1,000 rounds	Mean multiplier
H1 Premium 1	2.981	1.376	5.029	9.386	12.81	1.45
H2 Premium 2	3.431	1.082	4.450	8.963	19.71	1.40
H3 Premium 3	3.264	1.845	4.820	9.928	25.91	1.37
H4 Premium 4	3.029	1.932	4.205	9.166	27.17	1.50
L1 A	1.056	0.298	1.136	2.490	12.49	1.72
L2 K	1.134	0.328	1.136	2.597	17.78	1.48
L3 Q	1.172	0.412	1.095	2.678	36.33	1.27
L4 J	1.296	0.316	1.105	2.716	37.09	1.24
L5 10	1.195	0.316	1.090	2.600	84.47	1.17
All symbols				**50.525**	**273.8**	

RTP points paid by the way wins of each symbol and length, wild multipliers and board splits included, before the round cap. Mean multiplier: multiplied ways divided by ways, i.e. the average wild multiplier of a winning way (1 without a multiplied wild; the board split doubles both and is not included).

Table 5.12 BASE: RTP points of free-spin way wins by symbol (bonuses triggered in BASE) EXACT

Symbol	3 reels	4 reels	5 reels	Total (pts)	Symbol wins per 1,000 rounds	Mean multiplier
H1 Premium 1	1.402	1.161	6.900	9.463	4.01	1.72
H2 Premium 2	1.536	1.113	6.976	9.625	5.14	1.73
H3 Premium 3	1.437	1.133	7.468	10.039	6.91	1.70
H4 Premium 4	1.518	1.320	8.597	11.435	8.89	1.66
L1 A	0.330	0.210	1.296	1.836	2.76	1.95
L2 K	0.364	0.211	1.331	1.906	3.36	1.87
L3 Q	0.306	0.210	1.512	2.027	4.21	1.89
L4 J	0.349	0.197	1.514	2.060	4.91	1.88
L5 10	0.228	0.247	1.699	2.173	5.96	1.89
All symbols				**50.564**	**46.2**	

RTP points paid by the way wins of each symbol and length, wild multipliers and board splits included, before the round cap. Mean multiplier: multiplied ways divided by ways, i.e. the average wild multiplier of a winning way (1 without a multiplied wild; the board split doubles both and is not included).

CHAPTER 6

Enhanced Spins 1 (ENHANCE1)

The 3× spin: same game, every bonus tier five times as frequent.

6.1 At a glance

Return to player **96.7000%** mean win 2.90×	Hit rate **15.49%** rounds paying more than 0	Win ≥ price **4.905%** 1 in 20.4	Median win **0×** 0.0% of the price
Standard deviation **30.04 × price** 90.1× in base bets	Win of 100× price or more **0.110%** 1 in 912	Max win 50,000× **1 in 2,236,069** 4.47×10^{-5}%	Published books **1,266,729** 155,562 distinct payouts

6.2 Outcome classes and comparison with BASE

Table 6.1 Outcome classes: frequency, value and conditional distribution — EXACT

Class	Probability per round	Frequency	RTP points	Share of RTP	Mean win	Median	P10 – P90	P(no win)	Max win within class	Books
No win	84.37%	1 in 1.19	0.00	0.0%	0×	0×	0× – 0×	100%	—	285,877
Base-game wins (no witch, no bonus)	10.77%	1 in 9.28	13.02	13.5%	3.63×	0.20×	0.10× – 1.60×	0%	1 in 165,137,835	298,245
Witch unlock	2.000%	1 in 50.0	3.33	3.4%	4.99×	4.70×	1.00× – 8.10×	7.14%	1 in 413,126	126,729
Wicked Fortune (3 scatters)	2.500%	1 in 40.0	47.03	48.6%	56.43×	43.10×	8.00× – 114.2×	0%	1 in 96,428	176,405
Grave Spell (4 scatters)	0.3333%	1 in 300	17.21	17.8%	154.9×	85.50×	15.40× – 301.6×	0%	1 in 60,000	186,762
Dark Enchantment (5 scatters)	0.02500%	1 in 4,000	16.11	16.7%	1,933.3×	1,460×	1,088.4× – 2,900.2×	0%	1 in 3,000	192,711
All rounds	**100%**		**96.70**	**100%**	**2.90×**	**0×**	**0× – 0.20×**	**84.5%**	**1 in 2,236,069**	**1,266,729**

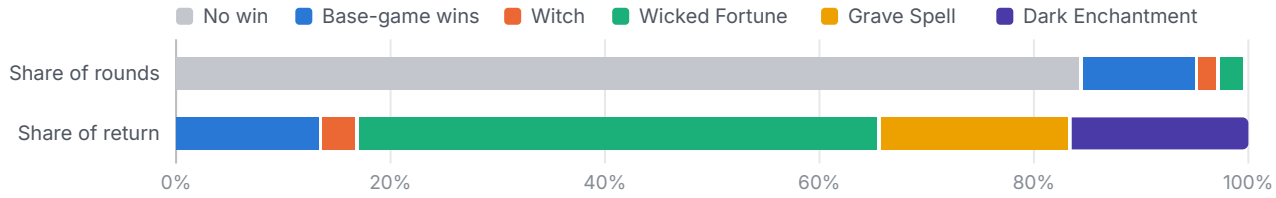
Every round belongs to exactly one class. Mean, median, P10–P90, P(no win) and the max-win frequency are conditional on the class (i.e. per round of that class). RTP points add up to the mode's RTP.

Table 6.2 BASE and ENHANCE1 side by side EXACT

Class	P per spin BASE	P per spin ENHANCE1	Ratio ENHANCE1 / BASE	Mean win BASE	Mean win ENHANCE1	Expected win per spin BASE (×)	Expected win per spin ENHANCE1 (×)
No win	77.86%	84.37%	1.084	0×	0×	0.0000	0.0000
Base-game wins (no witch, no bonus)	19.57%	10.77%	0.550	1.96×	3.63×	0.3836	0.3907
Witch unlock	2.000%	2.000%	1.000	5.00×	4.99×	0.0999	0.0999
Wicked Fortune (3 scatters)	0.5000%	2.500%	5.000	56.39×	56.43×	0.2820	1.4108
Grave Spell (4 scatters)	0.06667%	0.3333%	5.000	154.9×	154.9×	0.1033	0.5164
Dark Enchantment (5 scatters)	0.005000%	0.02500%	5.000	1,965.3×	1,933.3×	0.0983	0.4833
All rounds	**100%**	**100%**		**0.97×**	**2.90×**	**0.9670**	**2.9010**

Expected win per spin: class probability × class mean, in base bets. ENHANCE1 costs 3×: its expected win per spin (2.901×) is 3 times that of BASE (0.967×). Every bonus tier is exactly 5 times as frequent as in BASE; the witch keeps her BASE frequency.

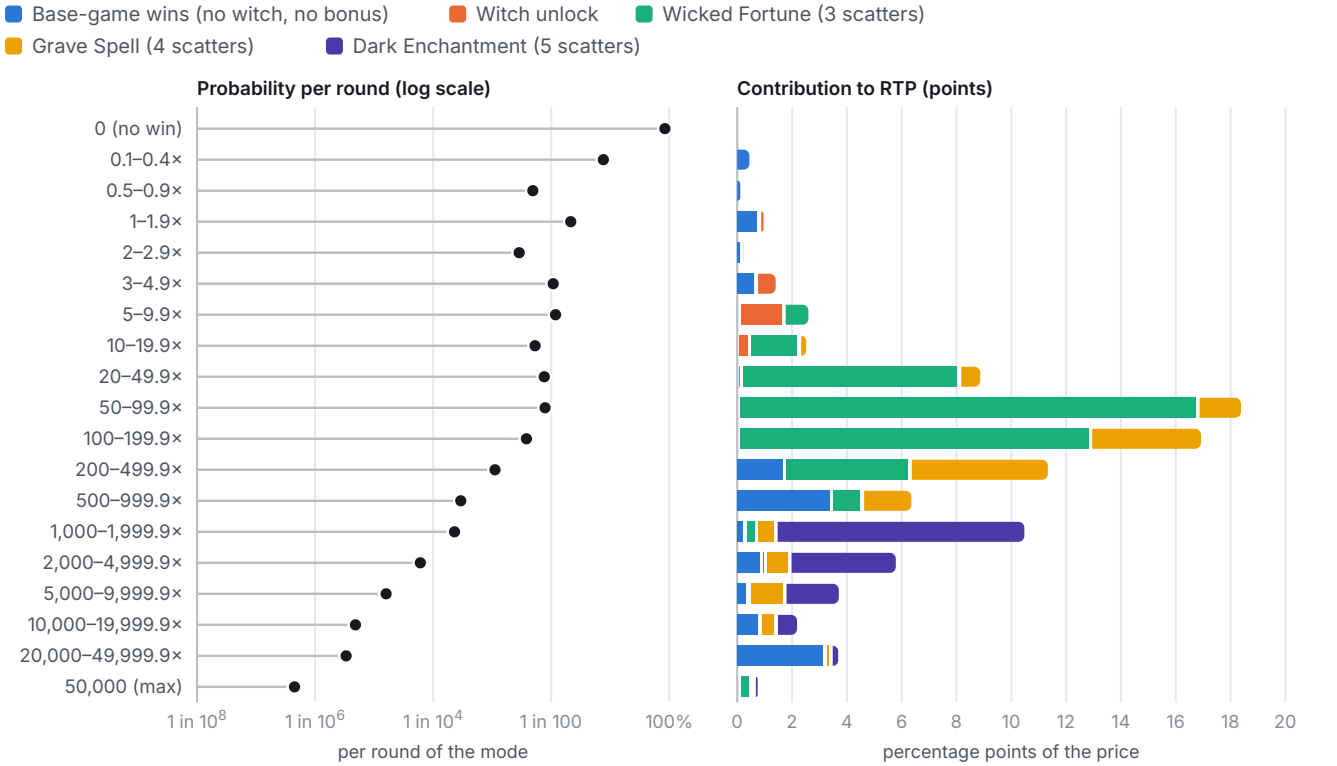
Figure 6.1 ENHANCE1: share of spins and share of the return by outcome class EXACT



Bonuses are 2.86% of ENHANCE1 spins and 83.1% of its return.

6.3 Payout distribution

Figure 6.2 ENHANCE1: probability and RTP contribution of each payout band EXACT



Left: probability that a round's total win falls in the band (dots on a log scale; empty bands are omitted). Right: the band's contribution to the RTP, coloured by outcome class. Frequent small wins carry little value; rare large wins carry a large part of it.

Table 6.3 ENHANCE1: payout bands EXACT

Round win (× base bet)	Probability	Frequency	Per 10,000 rounds	RTP points	Mean in band	P(win ≥ band start)	Books
0 (no win)	84.51%	1 in 1.18	8,451	0.000	—	100%	319,932
0.1–0.4×	7.663%	1 in 13.0	766	0.443	0.17×	15.49%	82,023
0.5–0.9×	0.4883%	1 in 205	48.83	0.111	0.68×	7.825%	95,783
1–1.9×	2.145%	1 in 46.6	215	0.959	1.34×	7.337%	61,767
2–2.9×	0.2868%	1 in 349	28.68	0.238	2.49×	5.192%	14,740
3–4.9×	1.081%	1 in 92.5	108	1.402	3.89×	4.905%	28,322
5–9.9×	1.186%	1 in 84.3	119	2.597	6.57×	3.823%	34,901
10–19.9×	0.5334%	1 in 187	53.34	2.516	14.15×	2.638%	42,606
20–49.9×	0.7636%	1 in 131	76.36	8.867	34.84×	2.104%	69,522
50–99.9×	0.7877%	1 in 127	78.77	18.380	70.00×	1.341%	50,520
100–199.9×	0.3804%	1 in 263	38.04	16.920	133.4×	0.5530%	40,202
200–499.9×	0.1117%	1 in 895	11.17	11.333	304.4×	0.1726%	36,182
500–999.9×	0.02939%	1 in 3,403	2.94	6.354	648.6×	0.06093%	18,616
1,000–1,999.9×	0.02302%	1 in 4,343	2.30	10.476	1,365.0×	0.03154%	96,068
2,000–4,999.9×	0.006059%	1 in 16,505	0.6059	5.785	2,864.6×	0.008516%	98,981
5,000–9,999.9×	0.001594%	1 in 62,718	0.1594	3.707	6,974.4×	0.002457%	64,138
10,000–19,999.9×	0.0004824%	1 in 207,298	0.0482	2.182	13,570.5×	0.0008625%	49,285
20,000–49,999.9×	0.0003354%	1 in 298,143	0.0335	3.684	32,951.2×	0.0003801%	44,110
50,000 (max)	4.47×10^{-5}%	1 in 2,236,069	0.0045	0.745	50,000×	4.47×10^{-5}%	19,031
Total	**100%**		**10,000**	**96.700**	**2.90×**		**1,266,729**

Bands are labelled by the smallest and largest win they can contain (wins are multiples of 0.1×). "P(win ≥ band start)" is the survival probability at the lower edge of the band. Per 10,000 rounds: expected number of rounds in the band.

6.4 Percentiles, thresholds and waiting times

Table 6.4 ENHANCE1: percentiles of the round win EXACT

Percentile	P25	P50	P75	P90	P95	P99	P99.5	P99.9	P99.99	P99.999
Win (× base bet)	0×	0×	0×	0.20×	2.70×	64.80×	105.9×	325.0×	1,792.7×	9,285.2×
Win (× price)	0.000	0.000	0.000	0.067	0.900	21.600	35.300	108.333	597.567	3,095.067
Base-game wins (no witch, no bonus) (×, conditional)	0.10×	0.20×	0.90×	1.60×	3.10×	4.60×	5.40×	596.8×	2,591.2×	38,712×
Witch unlock (×, conditional)	2.00×	4.70×	5.80×	8.10×	11.00×	19.40×	26.40×	51.70×	134.4×	384.0×
Wicked Fortune (3 scatters) (×, conditional)	15.80×	43.10×	71.90×	114.2×	146.8×	276.6×	385.0×	771.7×	1,896×	50,000×
Grave Spell (4 scatters) (×, conditional)	29.60×	85.50×	161.4×	301.6×	420.7×	966.3×	1,712×	7,692×	18,737.6×	50,000×
Dark Enchantment (5 scatters) (×, conditional)	1,224.2×	1,460×	1,822.8×	2,900.2×	4,608.5×	9,440.2×	12,549.5×	24,721.6×	50,000×	50,000×

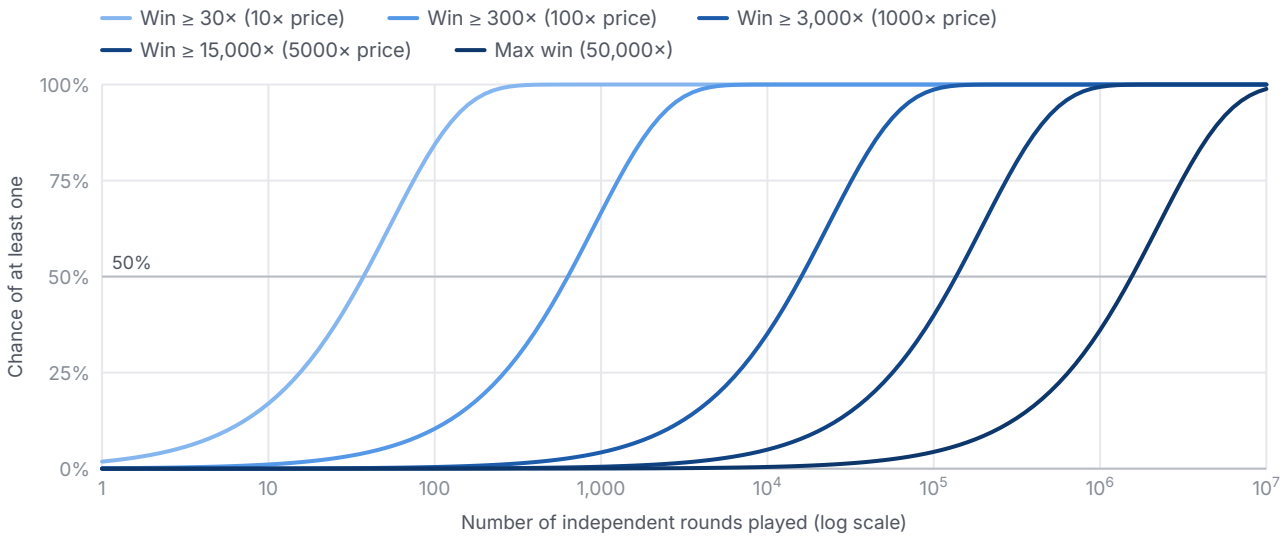
Pxx is the smallest win w with $P(X \le w) \ge xx\%$. Conditional rows give the percentiles within each outcome class.

Table 6.5 ENHANCE1: thresholds, frequencies and waiting times EXACT FORMULA

Event per round	Probability	Frequency	Median wait (rounds)	At least once in 100 rounds	At least once in 1,000 rounds	At least once in 10,000 rounds
Any win (> 0)	15.49%	1 in 6.46	5	100%	100%	100%
Win ≥ 3× (1× price)	4.905%	1 in 20.4	14	99.3%	100%	100%
Win ≥ 6× (2× price)	3.254%	1 in 30.7	21	96.3%	100%	100%
Win ≥ 15× (5× price)	2.304%	1 in 43.4	30	90.3%	100%	100%
Win ≥ 30× (10× price)	1.841%	1 in 54.3	38	84.4%	100%	100%
Win ≥ 60× (20× price)	1.104%	1 in 90.5	63	67.1%	100.0%	100%
Win ≥ 150× (50× price)	0.2669%	1 in 375	260	23.4%	93.1%	100%
Win ≥ 300× (100× price)	0.1097%	1 in 912	632	10.4%	66.6%	100.0%
Win ≥ 600× (200× price)	0.04789%	1 in 2,088	1,448	4.68%	38.1%	99.2%
Win ≥ 1,500× (500× price)	0.01467%	1 in 6,817	4,725	1.46%	13.6%	76.9%
Win ≥ 3,000× (1000× price)	0.004329%	1 in 23,099	16,011	0.432%	4.24%	35.1%
Win ≥ 15,000× (5000× price)	0.0005088%	1 in 196,553	136,240	0.0509%	0.507%	4.96%
Max win (50,000×)	4.47×10^{-5}%	1 in 2,236,069	1,549,925	0.00447%	0.0447%	0.446%

Probabilities are exact. Median wait: smallest n with P(event within n rounds) ≥ 50%, i.e. $\lceil \ln 0.5 / \ln(1-p) \rceil$. "At least once in n rounds" = $1-(1-p)^n$ (independent rounds).

Figure 6.3 ENHANCE1: chance of at least one event within n rounds EXACT FORMULA



Curves $1-(1-p)^n$ for selected events of the table above (darker = rarer). Each curve crosses 50% at the median waiting time and 63% at $n = 1/p$ rounds.

6.5 Bonus tiers as triggered in ENHANCE1

Table 6.6 ENHANCE1: bonus wins conditional on the tier EXACT

Tier	Frequency	Min	P10	P25	Median	Mean	P75	P90	P99	P(≥ 100×)	P(≥ 1,000×)	Max win
Wicked Fortune (3 scatters)	1 in 40.0	5.00×	8.00×	15.80×	43.10×	56.43×	71.90×	114.2×	276.6×	14.0%	0.0480%	1 in 96,428
Grave Spell (4 scatters)	1 in 300	10.00×	15.40×	29.60×	85.50×	154.9×	161.4×	301.6×	966.3×	44.0%	0.930%	1 in 60,000
Dark Enchantment (5 scatters)	1 in 4,000	1,000.1×	1,088.4×	1,224.2×	1,460×	1,933.3×	1,822.8×	2,900.2×	9,440.2×	100%	100%	1 in 3,000

All statistics except the frequency are per bonus of the tier (conditional on its start); wins in base bets, including the triggering spin.

Medians of the triggered tiers: 43.10×, 85.50×, 1,460× (BASE: 43.10×, 85.50×, 1,460.1×); max win per bonus 1 in 96,428, 1 in 60,000, 1 in 3,000 (BASE: 1 in 100,000, 1 in 60,000, 1 in 1,000), so that the mode's chance of the cap stays in line with its price.

6.6 Way wins by symbol

Table 6.7 ENHANCE1: RTP points of base-game way wins by symbol EXACT

Symbol	3 reels	4 reels	5 reels	Total (pts)	Symbol wins per 1,000 rounds	Mean multiplier
H1 Premium 1	0.879	0.405	2.400	3.685	8.19	2.02
H2 Premium 2	0.874	0.368	1.961	3.202	11.14	1.83
H3 Premium 3	0.930	0.417	1.747	3.094	15.72	1.63
H4 Premium 4	0.934	0.461	1.714	3.109	19.19	1.76
L1 A	0.289	0.199	0.535	1.024	8.59	1.95
L2 K	0.323	0.105	0.489	0.917	11.47	1.72
L3 Q	0.299	0.104	0.415	0.817	22.20	1.41
L4 J	0.316	0.092	0.356	0.764	23.14	1.37
L5 10	0.257	0.112	0.338	0.708	50.63	1.23
All symbols				**17.321**	**170.3**	

RTP points paid by the way wins of each symbol and length, wild multipliers and board splits included, before the round cap. Mean multiplier: multiplied ways divided by ways, i.e. the average wild multiplier of a winning way (1 without a multiplied wild; the board split doubles both and is not included).

CHAPTER 7

Enhanced Spins 2 (ENHANCE2)

The 15× spin: only Grave Spell or Dark Enchantment can trigger, a bonus 1 in 35.0 spins.

7.1 At a glance

Return to player **96.7000%** mean win 14.50×	Hit rate **24.29%** rounds paying more than 0	Win ≥ price **3.343%** 1 in 29.9	Median win **0×** 0.0% of the price
Standard deviation **13.20 × price** 198.0× in base bets	Win of 100× price or more **0.164%** 1 in 610	Max win 50,000× **1 in 262,948** 0.000380%	Published books **984,947** 130,048 distinct payouts

7.2 Outcome classes and comparison with BASE

Table 7.1 Outcome classes: frequency, value and conditional distribution EXACT

Class	Probability per round	Frequency	RTP points	Share of RTP	Mean win	Median	P10 – P90	P(no win)	Max win within class	Books
No win	75.57%	1 in 1.32	0.00	0.0%	0×	0×	0× – 0×	100%	—	253,804
Base-game wins (no witch, no bonus)	19.57%	1 in 5.11	2.55	2.6%	1.96×	0.20×	0.10× – 1.60×	0%	1 in 311,297,583	265,408
Witch unlock	2.000%	1 in 50.0	0.67	0.7%	5.00×	4.80×	1.00× – 8.20×	7.14%	1 in 412,925	112,502
Grave Spell (4 scatters)	2.524%	1 in 39.6	49.81	51.5%	296.0×	230.1×	98.40× – 512.4×	0%	1 in 60,000	182,246
Dark Enchantment (5 scatters)	0.3333%	1 in 300	43.67	45.2%	1,965.2×	1,460.4×	1,088× – 2,909.7×	0%	1 in 1,000	170,987
All rounds	**100%**		**96.70**	**100%**	**14.50×**	**0×**	**0× – 0.90×**	**75.7%**	**1 in 262,948**	**984,947**

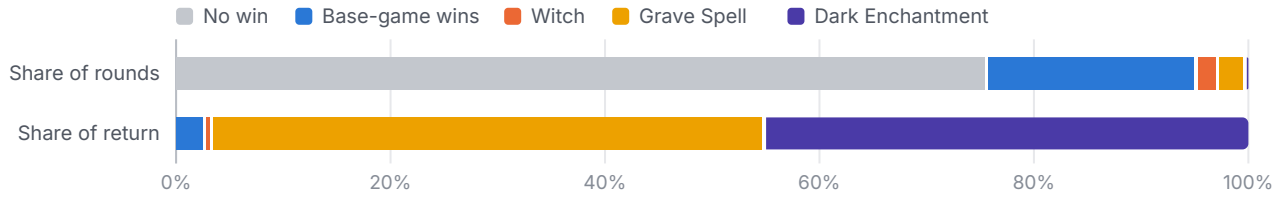
Every round belongs to exactly one class. Mean, median, P10–P90, P(no win) and the max-win frequency are conditional on the class (i.e. per round of that class). RTP points add up to the mode's RTP.

Table 7.2 BASE and ENHANCE2 side by side EXACT

Class	P per spin BASE	P per spin ENHANCE2	Ratio ENHANCE2 / BASE	Mean win BASE	Mean win ENHANCE2	Expected win per spin BASE (×)	Expected win per spin ENHANCE2 (×)
No win	77.86%	75.57%	0.971	0×	0×	0.0000	0.0000
Base-game wins (no witch, no bonus)	19.57%	19.57%	1.000	1.96×	1.96×	0.3836	0.3830
Witch unlock	2.000%	2.000%	1.000	5.00×	5.00×	0.0999	0.1000
Wicked Fortune (3 scatters)	0.5000%	never	—	56.39×	—	0.2820	—
Grave Spell (4 scatters)	0.06667%	2.524%	37.857	154.9×	296.0×	0.1033	7.4712
Dark Enchantment (5 scatters)	0.005000%	0.3333%	66.667	1,965.3×	1,965.2×	0.0983	6.5508
All rounds	**100%**	**100%**		**0.97×**	**14.50×**	**0.9670**	**14.5050**

Expected win per spin: class probability × class mean, in base bets. ENHANCE2 costs 15×: its expected win per spin (14.505×) is 15 times that of BASE (0.967×). Three scatters never trigger in ENHANCE2 (brief: “can NOT land 3 scatter”); the Grave Spell has its own, richer distribution (average 296×, BASE 155×), set so that a bonus lands 1 in 35.0 spins (brief: “around 1 in ~35”).

Figure 7.1 ENHANCE2: share of spins and share of the return by outcome class EXACT



Bonuses are 2.86% of ENHANCE2 spins (1 in 35.0) and 96.7% of its return.

7.3 Result relative to the price

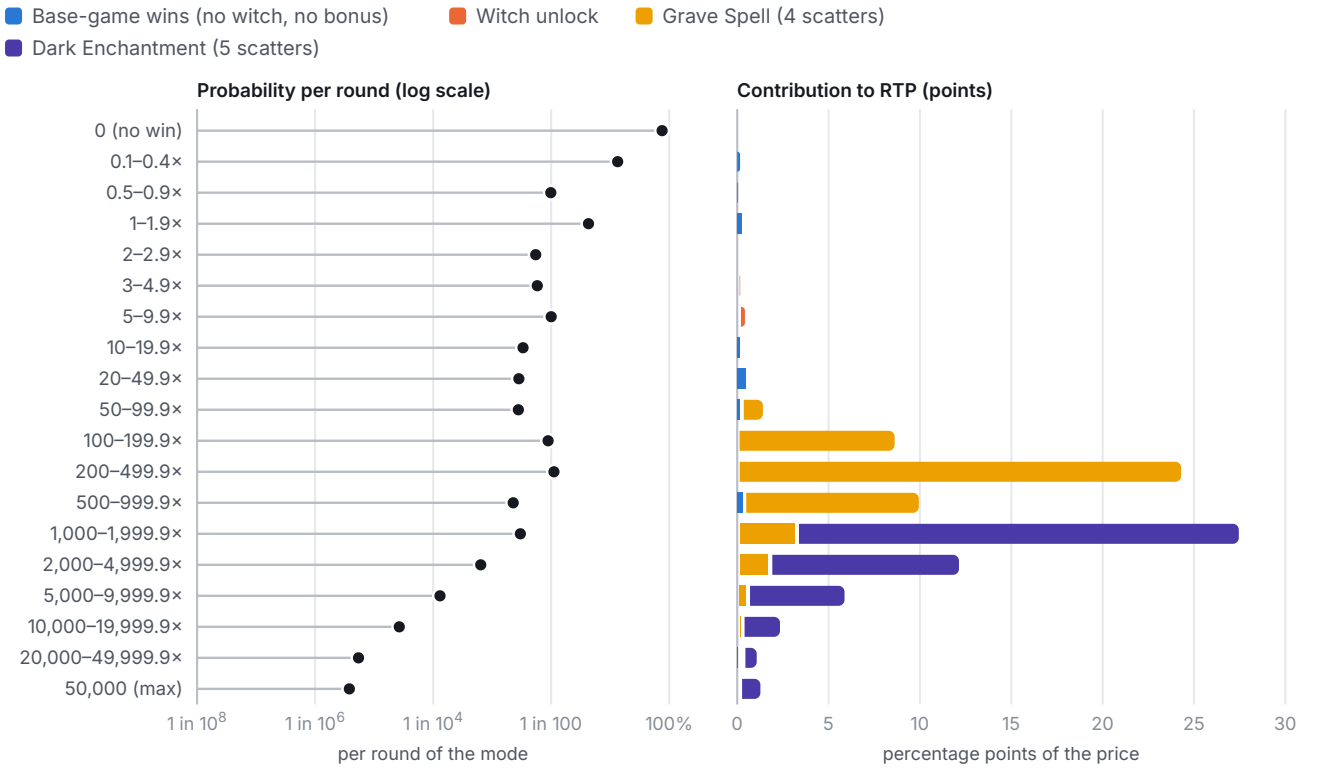
Table 7.3 ENHANCE2: result relative to the price EXACT

Threshold	P(win below)	P(win at or above)	Frequency (at or above)
0.1× price (2×)	93.10%	6.895%	1 in 14.5
0.25× price (4×)	95.06%	4.943%	1 in 20.2
0.5× price (8×)	96.31%	3.687%	1 in 27.1
0.75× price (11×)	96.55%	3.455%	1 in 28.9
1× price (15×)	96.66%	3.343%	1 in 29.9
1.5× price (23×)	96.85%	3.148%	1 in 31.8
2× price (30×)	96.88%	3.116%	1 in 32.1
3× price (45×)	97.03%	2.974%	1 in 33.6
5× price (75×)	97.24%	2.762%	1 in 36.2
10× price (150×)	97.90%	2.096%	1 in 47.7
20× price (300×)	98.74%	1.263%	1 in 79.2
50× price (750×)	99.57%	0.4298%	1 in 233
100× price (1,500×)	99.84%	0.1639%	1 in 610

Price: 15×. Mean win 14.50× = 96.70% of the price.

7.4 Payout distribution

Figure 7.2 ENHANCE2: probability and RTP contribution of each payout band EXACT



Left: probability that a round's total win falls in the band (dots on a log scale; empty bands are omitted). Right: the band's contribution to the RTP, coloured by outcome class. Frequent small wins carry little value; rare large wins carry a large part of it.

Table 7.4 ENHANCE2: payout bands EXACT

Round win (× base bet)	Probability	Frequency	Per 10,000 rounds	RTP points	Mean in band	P(win ≥ band start)	Books
0 (no win)	75.71%	1 in 1.32	7,571	0.000	—	100%	284,130
0.1–0.4×	13.38%	1 in 7.47	1,338	0.155	0.17×	24.29%	72,764
0.5–0.9×	0.9855%	1 in 101	98.55	0.046	0.70×	10.90%	85,539
1–1.9×	4.286%	1 in 23.3	429	0.384	1.34×	9.916%	54,751
2–2.9×	0.5466%	1 in 183	54.66	0.086	2.35×	5.630%	13,084
3–4.9×	0.5814%	1 in 172	58.14	0.161	4.16×	5.083%	25,267
5–9.9×	1.000%	1 in 100.0	100	0.421	6.31×	4.502%	17,657
10–19.9×	0.3331%	1 in 300	33.31	0.328	14.75×	3.502%	21,191
20–49.9×	0.2832%	1 in 353	28.32	0.730	38.67×	3.169%	39,801
50–99.9×	0.2779%	1 in 360	27.79	1.406	75.89×	2.885%	31,722
100–199.9×	0.8859%	1 in 113	88.59	8.614	145.9×	2.607%	25,756
200–499.9×	1.113%	1 in 89.8	111	24.297	327.4×	1.721%	23,334
500–999.9×	0.2272%	1 in 440	22.72	9.938	656.1×	0.6081%	12,270
1,000–1,999.9×	0.3003%	1 in 333	30.03	27.451	1,371.3×	0.3809%	70,004
2,000–4,999.9×	0.06397%	1 in 1,563	6.40	12.136	2,845.5×	0.08062%	74,354
5,000–9,999.9×	0.01306%	1 in 7,660	1.31	5.872	6,746.5×	0.01664%	52,870
10,000–19,999.9×	0.002664%	1 in 37,532	0.2664	2.340	13,173.1×	0.003588%	39,886
20,000–49,999.9×	0.0005438%	1 in 183,905	0.0544	1.067	29,421.8×	0.0009241%	26,863
50,000 (max)	0.0003803%	1 in 262,948	0.0380	1.268	50,000×	0.0003803%	13,704
Total	**100%**		**10,000**	**96.700**	**14.50×**		**984,947**

Bands are labelled by the smallest and largest win they can contain (wins are multiples of 0.1×). "P(win ≥ band start)" is the survival probability at the lower edge of the band. Per 10,000 rounds: expected number of rounds in the band.

7.5 Percentiles, thresholds and waiting times

Table 7.5 ENHANCE2: percentiles of the round win EXACT

Percentile	P25	P50	P75	P90	P95	P99	P99.5	P99.9	P99.99	P99.999
Win (× base bet)	0×	0×	0×	0.90×	3.30×	350.8×	617.6×	1,797.9×	6,475.6×	19,067.4×
Win (× price)	0.000	0.000	0.000	0.060	0.220	23.387	41.173	119.860	431.707	1,271.160
Base-game wins (no witch, no bonus) (×, conditional)	0.10×	0.20×	1.00×	1.60×	2.40×	40.00×	48.00×	80.00×	870.4×	4,406.4×
Witch unlock (×, conditional)	2.00×	4.80×	5.80×	8.20×	11.00×	19.20×	26.40×	51.60×	134.4×	384.0×
Grave Spell (4 scatters) (×, conditional)	137.1×	230.1×	357.3×	512.4×	671.3×	1,354.7×	1,872.9×	4,178.7×	14,030.2×	50,000×
Dark Enchantment (5 scatters) (×, conditional)	1,224×	1,460.4×	1,824.6×	2,909.7×	4,646.9×	9,691.3×	13,250.5×	50,000×	50,000×	50,000×

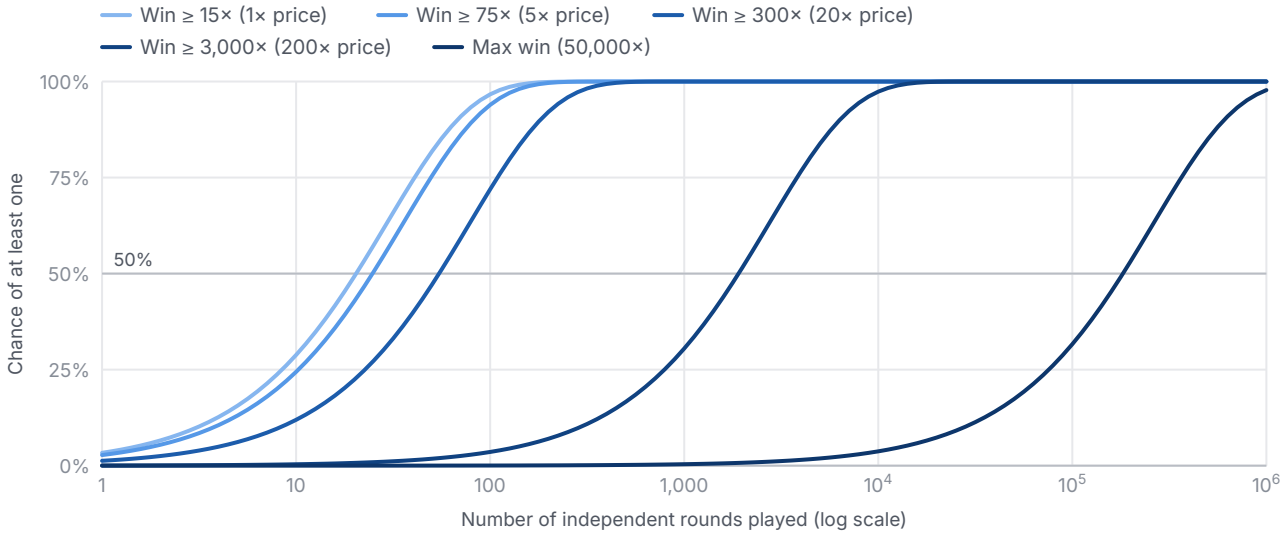
Pxx is the smallest win w with $P(X \le w) \ge xx\%$. Conditional rows give the percentiles within each outcome class.

Table 7.6 ENHANCE2: thresholds, frequencies and waiting times EXACT FORMULA

Event per round	Probability	Frequency	Median wait (rounds)	At least once in 10 rounds	At least once in 100 rounds	At least once in 1,000 rounds
Any win (> 0)	24.29%	1 in 4.12	3	93.8%	100%	100%
Win ≥ 15× (1× price)	3.343%	1 in 29.9	21	28.8%	96.7%	100%
Win ≥ 30× (2× price)	3.116%	1 in 32.1	22	27.1%	95.8%	100%
Win ≥ 75× (5× price)	2.762%	1 in 36.2	25	24.4%	93.9%	100%
Win ≥ 150× (10× price)	2.096%	1 in 47.7	33	19.1%	88.0%	100%
Win ≥ 300× (20× price)	1.263%	1 in 79.2	55	11.9%	72.0%	100.0%
Win ≥ 750× (50× price)	0.4298%	1 in 233	161	4.22%	35.0%	98.7%
Win ≥ 1,500× (100× price)	0.1639%	1 in 610	423	1.63%	15.1%	80.6%
Win ≥ 3,000× (200× price)	0.03637%	1 in 2,750	1,906	0.363%	3.57%	30.5%
Win ≥ 7,500× (500× price)	0.007203%	1 in 13,884	9,624	0.0720%	0.718%	6.95%
Win ≥ 15,000× (1000× price)	0.001516%	1 in 65,966	45,724	0.0152%	0.151%	1.50%
Max win (50,000×)	0.0003803%	1 in 262,948	182,262	0.00380%	0.0380%	0.380%

Probabilities are exact. Median wait: smallest n with P(event within n rounds) ≥ 50%, i.e. $\lceil \ln 0.5 / \ln(1-p) \rceil$. "At least once in n rounds" = $1-(1-p)^n$ (independent rounds).

Figure 7.3 ENHANCE2: chance of at least one event within n rounds EXACT FORMULA



Curves $1-(1-p)^n$ for selected events of the table above (darker = rarer). Each curve crosses 50% at the median waiting time and 63% at $n = 1/p$ rounds.

7.6 Bonus tiers as triggered in ENHANCE2

Table 7.7 ENHANCE2: bonus wins conditional on the tier EXACT

Tier	Frequency	Min	P10	P25	Median	Mean	P75	P90	P99	P(≥ 100×)	P(≥ 1,000×)	Max win
Grave Spell (4 scatters)	1 in 39.6	10.00×	98.40×	137.1×	230.1×	296.0×	357.3×	512.4×	1,354.7×	89.6%	1.85%	1 in 60,000
Dark Enchantment (5 scatters)	1 in 300	1,000.1×	1,088×	1,224×	1,460.4×	1,965.2×	1,824.6×	2,909.7×	9,691.3×	100%	100%	1 in 1,000

All statistics except the frequency are per bonus of the tier (conditional on its start); wins in base bets, including the triggering spin.

Table 7.8 ENHANCE2: its two tiers against the ones triggered in BASE EXACT

Per bonus (conditional)	Grave Spell ENHANCE2	Grave Spell BASE	Dark Enchantment ENHANCE2	Dark Enchantment BASE
Frequency per round of its mode	1 in 39.6	1 in 1,500	1 in 300	1 in 20,000
Minimum win	10.00×	10.00×	1,000.1×	1,000.1×
P25	137.1×	29.60×	1,224×	1,223.9×
Median win	230.1×	85.50×	1,460.4×	1,460.1×
Mean win	296.0×	154.9×	1,965.2×	1,965.3×
P75	357.3×	161.4×	1,824.6×	1,826×
P90	512.4×	301.4×	2,909.7×	2,906.4×
P99	1,354.7×	966.7×	9,691.3×	9,694.7×
P(win ≥ 100×)	89.6%	44.0%	100%	100%
P(win ≥ 1,000×)	1.85%	0.930%	100%	100%
P(win ≥ 10,000×)	0.0172%	0.0476%	0.920%	0.920%
P(max win)	1 in 60,000	1 in 60,000	1 in 1,000	1 in 1,000
Standard deviation	438.0×	545.5×	2,340.8×	2,341.2×

Wins in base bets. Each column is the distribution of the same bonus tier as published in a given mode.

CHAPTER 8

Hexed Spins (HEXED)

One spin for 100× with the squares unlocked and at least two special symbols; 80% of spins pay nothing.

8.1 At a glance

Return to player **96.7000%** mean win 96.70×	Hit rate **20.00%** rounds paying more than 0	Win ≥ price **12.26%** 1 in 8.15	Median win **0×** 0.0% of the price
Standard deviation **9.55 × price** 954.8× in base bets	Win of 100× price or more **0.0945%** 1 in 1,058	Max win 50,000× **1 in 15,000** 0.00667%	Published books **493,605** 25,474 distinct payouts

8.2 Result relative to the price

Table 8.1 HEXED: result relative to the price — EXACT

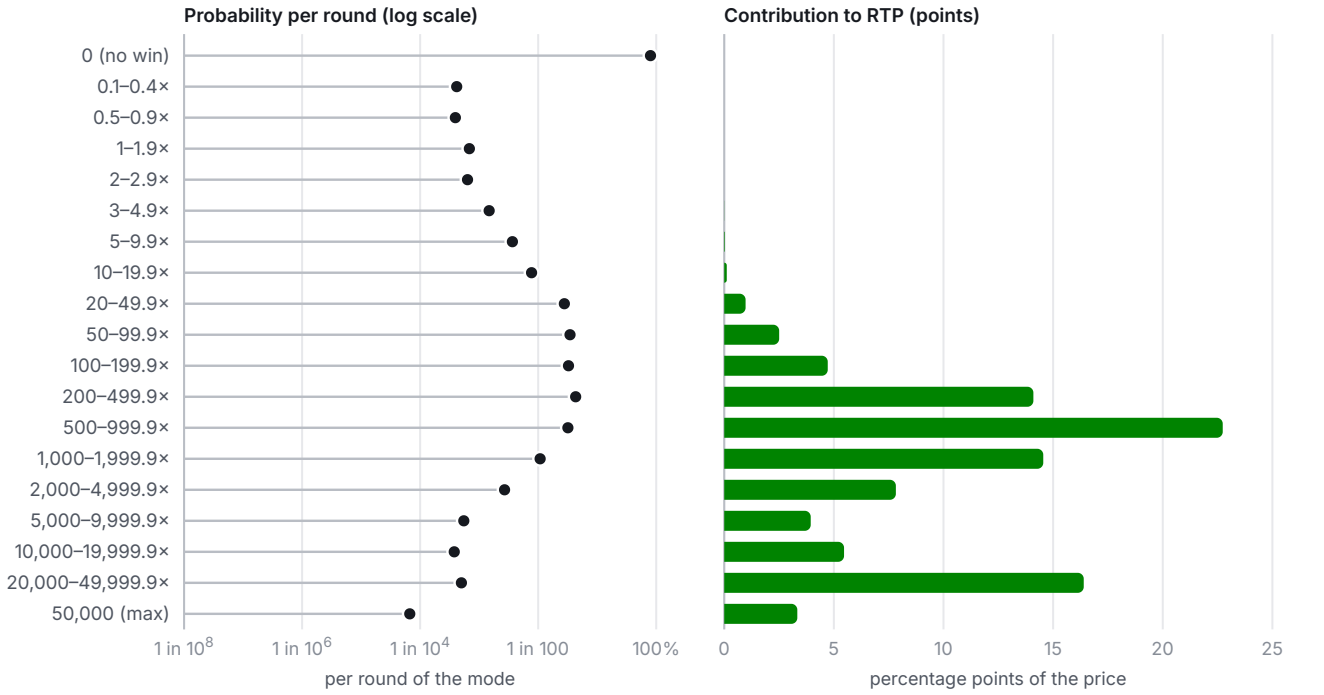
Threshold	P(win below)	P(win at or above)	Frequency (at or above)
0.1× price (10×)	80.73%	19.27%	1 in 5.19
0.25× price (25×)	81.93%	18.07%	1 in 5.53
0.5× price (50×)	84.27%	15.73%	1 in 6.36
0.75× price (75×)	86.30%	13.70%	1 in 7.30
1× price (100×)	87.74%	12.26%	1 in 8.15
1.5× price (150×)	89.62%	10.38%	1 in 9.63
2× price (200×)	91.00%	9.000%	1 in 11.1
3× price (300×)	92.92%	7.079%	1 in 14.1
5× price (500×)	95.31%	4.686%	1 in 21.3
10× price (1,000×)	98.50%	1.498%	1 in 66.8
20× price (2,000×)	99.58%	0.4186%	1 in 239
50× price (5,000×)	99.85%	0.1494%	1 in 669
100× price (10,000×)	99.91%	0.09449%	1 in 1,058

Price: 100×. Mean win 96.70× = 96.70% of the price.

8.3 Payout distribution

Figure 8.1 HEXED: probability and RTP contribution of each payout band

EXACT



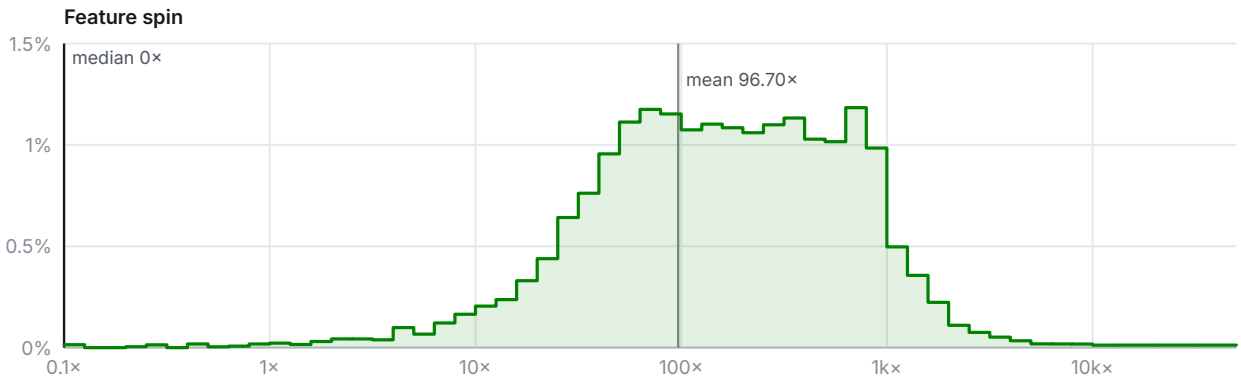
Left: probability that a round's total win falls in the band (dots on a log scale; empty bands are omitted). Right: the band's contribution to the RTP, coloured by outcome class. Frequent small wins carry little value; rare large wins carry a large part of it.

Table 8.2 HEXED: payout bands EXACT

Round win (× base bet)	Probability	Frequency	Per 10,000 rounds	RTP points	Mean in band	P(win ≥ band start)	Books
0 (no win)	80.00%	1 in 1.25	8,000	0.000	—	100%	234,100
0.1–0.4×	0.04162%	1 in 2,403	4.16	0.000	0.24×	20.00%	5,371
0.5–0.9×	0.03950%	1 in 2,531	3.95	0.000	0.71×	19.96%	8,231
1–1.9×	0.06823%	1 in 1,466	6.82	0.001	1.46×	19.92%	13,394
2–2.9×	0.06348%	1 in 1,575	6.35	0.002	2.41×	19.85%	8,893
3–4.9×	0.1476%	1 in 678	14.76	0.006	3.94×	19.79%	18,037
5–9.9×	0.3656%	1 in 273	36.56	0.027	7.47×	19.64%	21,842
10–19.9×	0.7721%	1 in 130	77.21	0.115	14.89×	19.27%	21,871
20–49.9×	2.776%	1 in 36.0	278	0.973	35.06×	18.50%	27,196
50–99.9×	3.463%	1 in 28.9	346	2.505	72.34×	15.73%	17,445
100–199.9×	3.263%	1 in 30.6	326	4.720	144.6×	12.26%	14,609
200–499.9×	4.314%	1 in 23.2	431	14.099	326.8×	9.000%	14,032
500–999.9×	3.188%	1 in 31.4	319	22.736	713.2×	4.686%	8,036
1,000–1,999.9×	1.079%	1 in 92.7	108	14.550	1,348.4×	1.498%	18,065
2,000–4,999.9×	0.2692%	1 in 372	26.92	7.827	2,908.0×	0.4186%	16,934
5,000–9,999.9×	0.05493%	1 in 1,820	5.49	3.944	7,180.3×	0.1494%	12,706
10,000–19,999.9×	0.03782%	1 in 2,644	3.78	5.464	14,446.0×	0.09449%	11,543
20,000–49,999.9×	0.05000%	1 in 2,000	5.00	16.396	32,792.7×	0.05667%	11,010
50,000 (max)	0.006667%	1 in 15,000	0.6667	3.333	50,000×	0.006667%	10,290
Total	**100%**		**10,000**	**96.700**	**96.70×**		**493,605**

Bands are labelled by the smallest and largest win they can contain (wins are multiples of 0.1×). "P(win ≥ band start)" is the survival probability at the lower edge of the band. Per 10,000 rounds: expected number of rounds in the band.

Figure 8.2 HEXED: distribution of the win EXACT



Probability per tenth of a decade of win (log scale), median (dark) and mean (grey); the 80% of spins paying nothing are not drawn. The max win (50,000×) is an additional atom of probability 0.00667% (1 in 15,000).

8.4 Percentiles, thresholds and waiting times

Table 8.3 HEXED: percentiles of the round win EXACT

Percentile	P25	P50	P75	P90	P95	P99	P99.5	P99.9	P99.99	P99.999
Win (× base bet)	0×	0×	0×	160.0×	470.4×	1,260.8×	1,807.2×	9,324×	47,678.4×	50,000×
Win (× price)	0.000	0.000	0.000	1.600	4.704	12.608	18.072	93.240	476.784	500.000

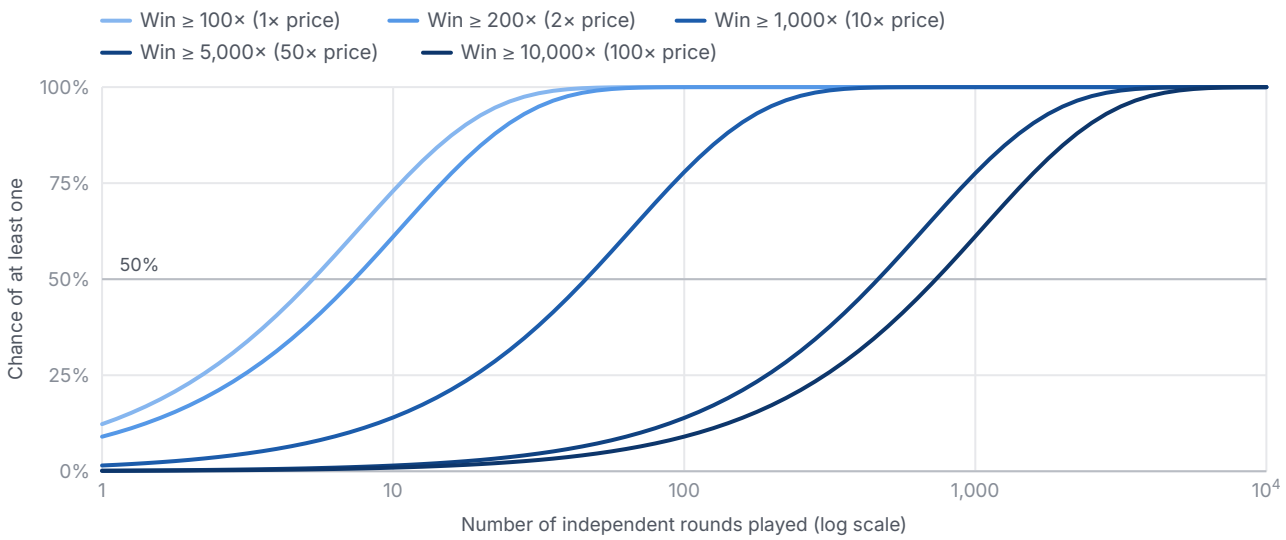
Pxx is the smallest win w with $P(X \le w) \ge xx\%$. Conditional rows give the percentiles within each outcome class.

Table 8.4 HEXED: thresholds, frequencies and waiting times EXACT FORMULA

Event per round	Probability	Frequency	Median wait (rounds)	At least once in 10 rounds	At least once in 100 rounds	At least once in 1,000 rounds
Any win (> 0)	20.00%	1 in 5.00	4	89.3%	100%	100%
Win ≥ 10× (0.1× price)	19.27%	1 in 5.19	4	88.2%	100%	100%
Win ≥ 25× (0.25× price)	18.07%	1 in 5.53	4	86.4%	100%	100%
Win ≥ 50× (0.5× price)	15.73%	1 in 6.36	5	81.9%	100%	100%
Win ≥ 100× (1× price)	12.26%	1 in 8.15	6	73.0%	100.0%	100%
Win ≥ 200× (2× price)	9.000%	1 in 11.1	8	61.1%	100.0%	100%
Win ≥ 500× (5× price)	4.686%	1 in 21.3	15	38.1%	99.2%	100%
Win ≥ 1,000× (10× price)	1.498%	1 in 66.8	46	14.0%	77.9%	100%
Win ≥ 2,000× (20× price)	0.4186%	1 in 239	166	4.11%	34.3%	98.5%
Win ≥ 5,000× (50× price)	0.1494%	1 in 669	464	1.48%	13.9%	77.6%
Win ≥ 10,000× (100× price)	0.09449%	1 in 1,058	734	0.941%	9.02%	61.1%
Max win (50,000×)	0.006667%	1 in 15,000	10,397	0.0666%	0.664%	6.45%

Probabilities are exact. Median wait: smallest n with P(event within n rounds) ≥ 50%, i.e. $\lceil \ln 0.5 / \ln(1-p) \rceil$. "At least once in n rounds" = $1-(1-p)^n$ (independent rounds).

Figure 8.3 HEXED: chance of at least one event within n rounds EXACT FORMULA



Curves $1 - (1 - p)^n$ for selected events of the table above (darker = rarer). Each curve crosses 50% at the median waiting time and 63% at $n = 1/p$ rounds.

8.5 Special symbols of the spin

No scatter can land. Four mirrors give 3.4% of the max wins of HEXED (1 in 447,391 spins); the others come from the ways.

Table 8.5 HEXED: special symbols per spin EXACT

Spins	With a special symbol	Exactly 1	2	3	4	Broomsticks per 100 spins	Spellbooks per 100 spins	Orbs per 100 spins	Mirrors per 100 spins	4 identical
HEXED spins	100%	0	58.4%	34.2%	7.41%	79.94	49.92	45.29	73.83	1 in 1,106

Per spin played in that context, weighted by the published lookup tables. 4 identical: the four squares show the same special symbol (its “all four” effect).

Table 8.6 HEXED: what the special symbols are worth EXACT

Spins	Mean win, no special symbol	Mean win, with one or more	Hit rate, no special symbol	Hit rate, with one or more	Share of the spin wins
HEXED spins	—	96.70×	—	20.0%	100.0%

Mean win of a spin in base bets (before the round cap). Share of the spin wins: part of the total paid by the spins of that context that comes from spins showing at least one special symbol.

Table 8.7 HEXED: effects of the special symbols EXACT

Spins	Board split (broomstick)	4 broomsticks: premium board	Wilds per spellbook	Wild nudges per 1,000 spins	Steps per nudge	Orb wilds per 100 spins	Mean orb multiplier
HEXED spins	61.9%	1 in 6,596	2.34	1.27	2.02	45.29	8.5×

Board split: share of spins with at least one broomstick. Wild nudges come only from four spellbooks (2 per occurrence); the reel multiplier equals the number of steps. Weighted by the published lookup tables.

Table 8.8 HEXED: multiplier of the orb wilds EXACT

Spins	×2	×3	×5	×10	×20	×50	×100	×250	×500	×1,000
HEXED spins	38.5%	27.5%	16.4%	9.51%	4.52%	2.15%	0.939%	0.323%	0.137%	0.0754%

Share of the orb wilds by multiplier (four orbs included: ×20 or more each).

8.6 Way wins by symbol

Table 8.9 HEXED: RTP points of way wins by symbol EXACT

Symbol	3 reels	4 reels	5 reels	Total (pts)	Symbol wins per 1,000 rounds	Mean multiplier
H1 Premium 1	1.117	2.052	29.244	32.412	68.22	4.67
H2 Premium 2	1.023	1.600	22.891	25.514	76.31	4.62
H3 Premium 3	0.842	1.514	21.026	23.381	86.36	4.52
H4 Premium 4	0.715	1.339	20.893	22.947	95.70	4.57
L1 A	0.232	0.370	5.072	5.674	49.99	5.20
L2 K	0.243	0.326	4.597	5.167	54.51	5.09
L3 Q	0.163	0.262	4.113	4.538	59.26	5.00
L4 J	0.184	0.199	3.722	4.105	62.52	4.84
L5 10	0.088	0.237	3.256	3.581	66.46	4.55
All symbols				**127.321**	**619.3**	

RTP points paid by the way wins of each symbol and length, wild multipliers and board splits included, before the round cap. Mean multiplier: multiplied ways divided by ways, i.e. the average wild multiplier of a winning way (1 without a multiplied wild; the board split doubles both and is not included).

CHAPTER 9

Cursed Spins (CURSED)

One spin for 500× with four special symbols; 30% of spins pay nothing.

9.1 At a glance

Return to player **96.7000%** mean win 483.5×	Hit rate **70.00%** rounds paying more than 0	Win ≥ price **17.46%** 1 in 5.73	Median win **125.6×** 25.1% of the price
Standard deviation **4.07 × price** 2,035.1× in base bets	Win of 100× price or more **0.0323%** 1 in 3,100	Max win 50,000× **1 in 3,100** 0.0323%	Published books **495,382** 27,266 distinct payouts

9.2 Result relative to the price

Table 9.1 CURSED: result relative to the price — EXACT

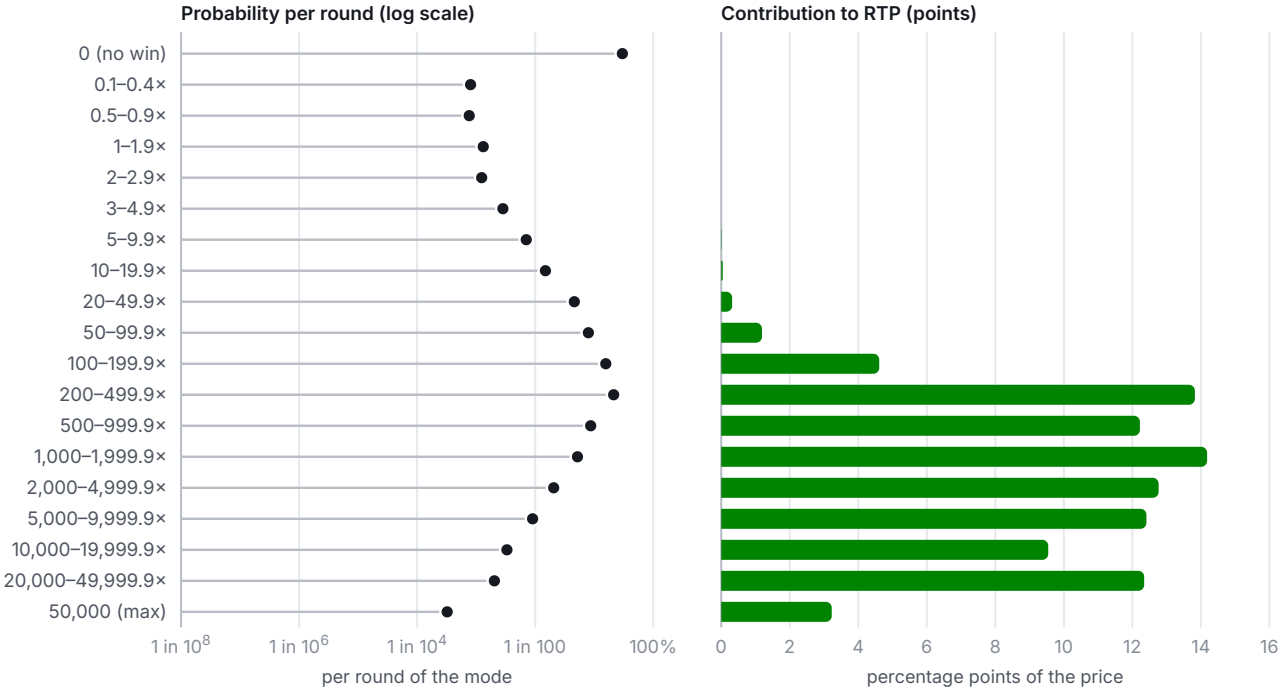
Threshold	P(win below)	P(win at or above)	Frequency (at or above)
0.1× price (50×)	37.50%	62.50%	1 in 1.60
0.25× price (125×)	49.90%	50.10%	1 in 2.00
0.5× price (250×)	66.54%	33.46%	1 in 2.99
0.75× price (375×)	75.90%	24.10%	1 in 4.15
1× price (500×)	82.54%	17.46%	1 in 5.73
1.5× price (750×)	88.19%	11.81%	1 in 8.47
2× price (1,000×)	91.26%	8.739%	1 in 11.4
3× price (1,500×)	94.93%	5.067%	1 in 19.7
5× price (2,500×)	97.08%	2.920%	1 in 34.3
10× price (5,000×)	98.53%	1.472%	1 in 67.9
20× price (10,000×)	99.43%	0.5679%	1 in 176
50× price (25,000×)	99.85%	0.1538%	1 in 650
100× price (50,000×)	99.97%	0.03226%	1 in 3,100

Price: 500×. Mean win 483.5× = 96.70% of the price.

9.3 Payout distribution

Figure 9.1 CURSED: probability and RTP contribution of each payout band

EXACT



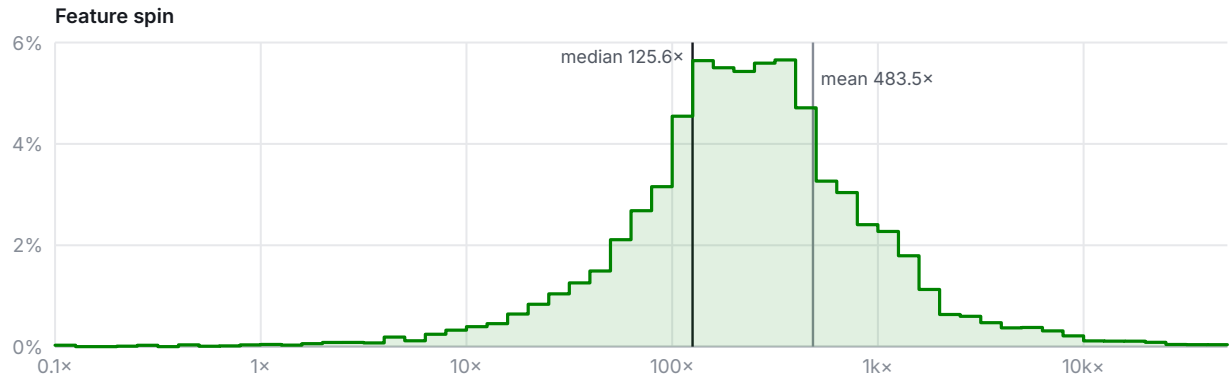
Left: probability that a round's total win falls in the band (dots on a log scale; empty bands are omitted). Right: the band's contribution to the RTP, coloured by outcome class. Frequent small wins carry little value; rare large wins carry a large part of it.

Table 9.2 CURSED: payout bands EXACT

Round win (× base bet)	Probability	Frequency	Per 10,000 rounds	RTP points	Mean in band	P(win ≥ band start)	Books
0 (no win)	30.00%	1 in 3.33	3,000	0.000	—	100%	205,639
0.1–0.4×	0.08037%	1 in 1,244	8.04	0.000	0.24×	70.00%	1,586
0.5–0.9×	0.07629%	1 in 1,311	7.63	0.000	0.71×	69.92%	3,457
1–1.9×	0.1318%	1 in 759	13.18	0.000	1.46×	69.84%	6,942
2–2.9×	0.1239%	1 in 807	12.39	0.001	2.42×	69.71%	5,282
3–4.9×	0.2837%	1 in 353	28.37	0.002	3.94×	69.59%	12,648
5–9.9×	0.7061%	1 in 142	70.61	0.011	7.50×	69.30%	16,952
10–19.9×	1.491%	1 in 67.1	149	0.044	14.92×	68.60%	19,816
20–49.9×	4.605%	1 in 21.7	460	0.319	34.63×	67.11%	27,859
50–99.9×	7.970%	1 in 12.5	797	1.189	74.57×	62.50%	21,181
100–199.9×	15.72%	1 in 6.36	1,572	4.612	146.7×	54.53%	21,099
200–499.9×	21.35%	1 in 4.68	2,135	13.826	323.7×	38.82%	24,918
500–999.9×	8.723%	1 in 11.5	872	12.221	700.4×	17.46%	16,360
1,000–1,999.9×	5.208%	1 in 19.2	521	14.182	1,361.6×	8.739%	26,703
2,000–4,999.9×	2.059%	1 in 48.6	206	12.767	3,101.1×	3.531%	27,241
5,000–9,999.9×	0.9044%	1 in 111	90.44	12.410	6,861.5×	1.472%	18,379
10,000–19,999.9×	0.3320%	1 in 301	33.20	9.545	14,376.8×	0.5679%	14,272
20,000–49,999.9×	0.2037%	1 in 491	20.37	12.345	30,304.6×	0.2359%	12,381
50,000 (max)	0.03226%	1 in 3,100	3.23	3.226	50,000×	0.03226%	12,667
Total	**100%**		**10,000**	**96.700**	**483.5×**		**495,382**

Bands are labelled by the smallest and largest win they can contain (wins are multiples of 0.1×). "P(win ≥ band start)" is the survival probability at the lower edge of the band. Per 10,000 rounds: expected number of rounds in the band.

Figure 9.2 CURSED: distribution of the win EXACT



Probability per tenth of a decade of win (log scale), median (dark) and mean (grey); the 30% of spins paying nothing are not drawn. The max win (50,000×) is an additional atom of probability 0.0323% (1 in 3,100).

9.4 Percentiles, thresholds and waiting times

Table 9.3 CURSED: percentiles of the round win EXACT

Percentile	P25	P50	P75	P90	P95	P99	P99.5	P99.9	P99.99	P99.999
Win (× base bet)	0×	125.6×	355.2×	889.6×	1,536×	6,688×	11,456×	33,604.8×	50,000×	50,000×
Win (× price)	0.000	0.251	0.710	1.779	3.072	13.376	22.912	67.210	100.000	100.000

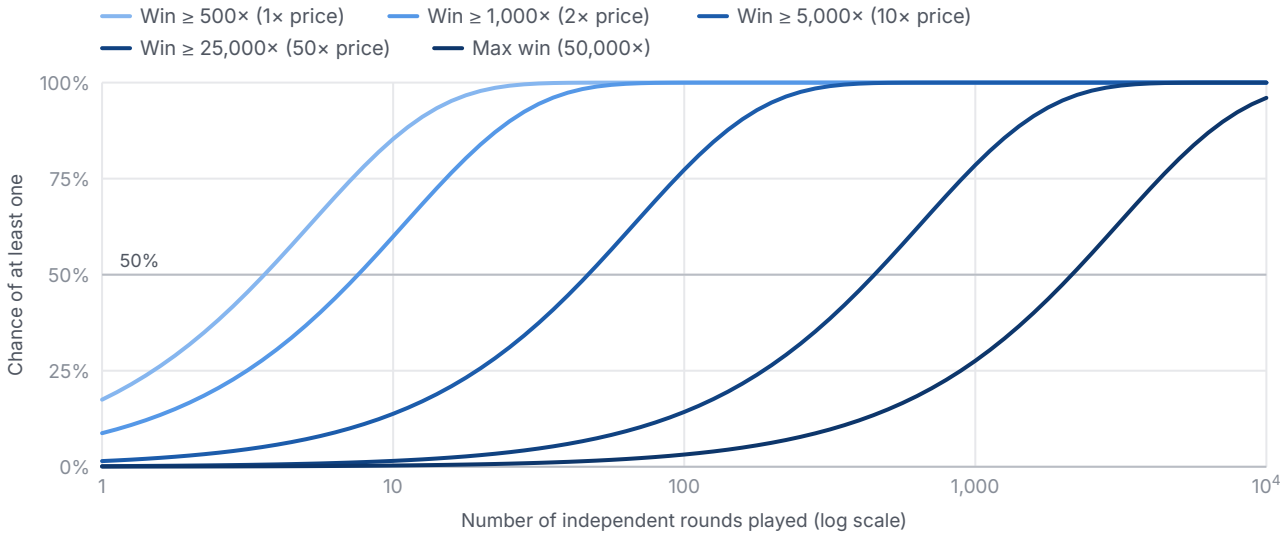
Pxx is the smallest win w with $P(X \le w) \ge xx\%$. Conditional rows give the percentiles within each outcome class.

Table 9.4 CURSED: thresholds, frequencies and waiting times EXACT FORMULA

Event per round	Probability	Frequency	Median wait (rounds)	At least once in 10 rounds	At least once in 100 rounds	At least once in 1,000 rounds
Any win (> 0)	70.00%	1 in 1.43	1	100.0%	100%	100%
Win ≥ 50× (0.1× price)	62.50%	1 in 1.60	1	100.0%	100%	100%
Win ≥ 125× (0.25× price)	50.10%	1 in 2.00	1	99.9%	100%	100%
Win ≥ 250× (0.5× price)	33.46%	1 in 2.99	2	98.3%	100%	100%
Win ≥ 500× (1× price)	17.46%	1 in 5.73	4	85.3%	100%	100%
Win ≥ 1,000× (2× price)	8.739%	1 in 11.4	8	59.9%	100.0%	100%
Win ≥ 2,500× (5× price)	2.920%	1 in 34.3	24	25.6%	94.8%	100%
Win ≥ 5,000× (10× price)	1.472%	1 in 67.9	47	13.8%	77.3%	100%
Win ≥ 10,000× (20× price)	0.5679%	1 in 176	122	5.54%	43.4%	99.7%
Win ≥ 25,000× (50× price)	0.1538%	1 in 650	451	1.53%	14.3%	78.5%
Max win (50,000×)	0.03226%	1 in 3,100	2,149	0.322%	3.17%	27.6%

Probabilities are exact. Median wait: smallest n with P(event within n rounds) ≥ 50%, i.e. $\lceil \ln 0.5 / \ln(1-p) \rceil$. "At least once in n rounds" = $1-(1-p)^n$ (independent rounds).

Figure 9.3 CURSED: chance of at least one event within n rounds EXACT FORMULA



Curves $1-(1-p)^n$ for selected events of the table above (darker = rarer). Each curve crosses 50% at the median waiting time and 63% at $n = 1/p$ rounds.

9.5 Special symbols of the spin

No scatter can land. Four mirrors give 19.8% of the max wins of CURSED (1 in 15,688 spins); the others come from the ways.

Table 9.5 CURSED: special symbols per spin EXACT

Spins	With a special symbol	Exactly 1	2	3	4	Broomsticks per 100 spins	Spellbooks per 100 spins	Orbs per 100 spins	Mirrors per 100 spins	4 identical
CURSED spins	100%	0	0	0	100%	99.49	115.42	76.04	109.05	1 in 109

Per spin played in that context, weighted by the published lookup tables. 4 identical: the four squares show the same special symbol (its “all four” effect).

Table 9.6 CURSED: what the special symbols are worth EXACT

Spins	Mean win, no special symbol	Mean win, with one or more	Hit rate, no special symbol	Hit rate, with one or more	Share of the spin wins
CURSED spins	—	483.5×	—	70.0%	100.0%

Mean win of a spin in base bets (before the round cap). Share of the spin wins: part of the total paid by the spins of that context that comes from spins showing at least one special symbol.

Table 9.7 CURSED: effects of the special symbols EXACT

Spins	Board split (broomstick)	4 broomsticks: premium board	Wilds per spellbook	Wild nudges per 1,000 spins	Steps per nudge	Orb wilds per 100 spins	Mean orb multiplier
CURSED spins	68.6%	1 in 419	2.59	10.80	2.10	76.04	8.7×

Board split: share of spins with at least one broomstick. Wild nudges come only from four spellbooks (2 per occurrence); the reel multiplier equals the number of steps. Weighted by the published lookup tables.

Table 9.8 CURSED: multiplier of the orb wilds EXACT

Spins	×2	×3	×5	×10	×20	×50	×100	×250	×500	×1,000
CURSED spins	45.7%	27.7%	13.4%	6.55%	3.67%	1.54%	0.766%	0.357%	0.210%	0.141%

Share of the orb wilds by multiplier (four orbs included: ×20 or more each).

9.6 Way wins by symbol

Table 9.9 CURSED: RTP points of way wins by symbol EXACT

Symbol	3 reels	4 reels	5 reels	Total (pts)	Symbol wins per 1,000 rounds	Mean multiplier
H1 Premium 1	1.182	1.766	26.205	29.153	238.12	4.80
H2 Premium 2	1.070	1.454	21.151	23.675	268.70	4.76
H3 Premium 3	0.856	1.375	19.380	21.610	302.51	4.59
H4 Premium 4	0.746	1.137	18.944	20.828	336.35	4.54
L1 A	0.269	0.374	5.514	6.156	193.62	5.41
L2 K	0.281	0.339	4.964	5.585	213.09	5.22
L3 Q	0.200	0.295	4.482	4.977	227.77	5.08
L4 J	0.195	0.230	4.175	4.601	239.47	4.93
L5 10	0.099	0.227	3.550	3.876	253.63	4.85
All symbols				**120.461**	**2,273.3**	

RTP points paid by the way wins of each symbol and length, wild multipliers and board splits included, before the round cap. Mean multiplier: multiplied ways divided by ways, i.e. the average wild multiplier of a winning way (1 without a multiplied wild; the board split doubles both and is not included).

CHAPTER 10

Wicked Fortune buy (WICKED)

Purchase of the 3-scatter Wicked Fortune for 100×; it never pays less than 5×.

10.1 At a glance

Return to player **96.7000%** mean win 96.70×	Hit rate **100%** rounds paying more than 0	Win ≥ price **22.08%** 1 in 4.53	Median win **43.70×** 43.7% of the price
Standard deviation **6.09 × price** 608.9× in base bets	Win of 100× price or more **0.0484%** 1 in 2,066	Max win 50,000× **1 in 40,700** 0.00246%	Published books **499,551** 76,711 distinct payouts

10.2 Result relative to the price

Table 10.1 WICKED: result relative to the price — EXACT

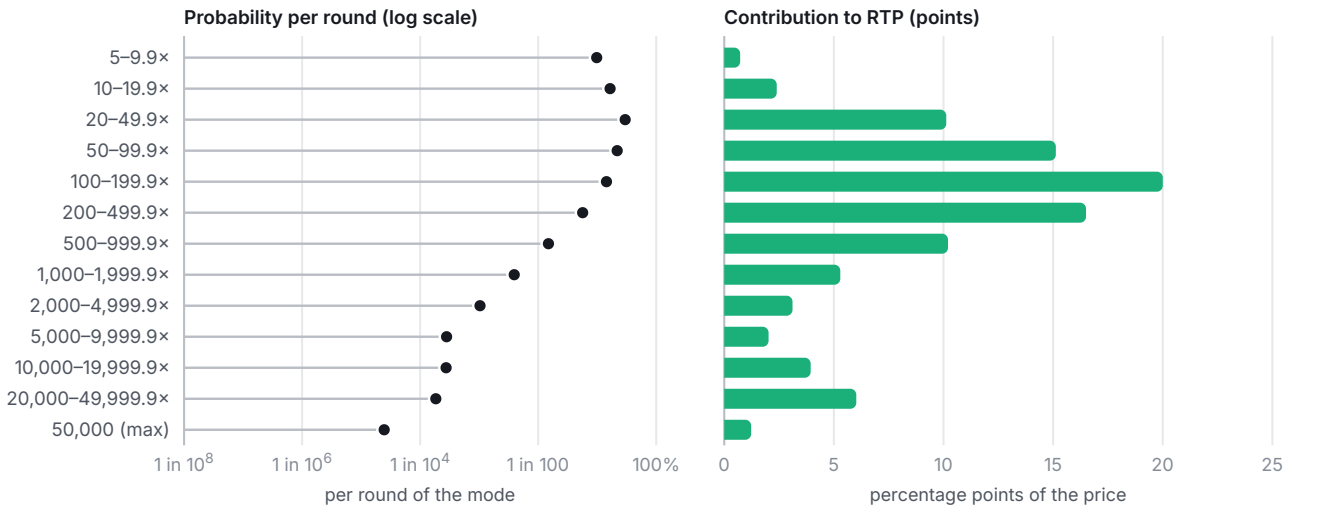
Threshold	P(win below)	P(win at or above)	Frequency (at or above)
0.1× price (10×)	9.807%	90.19%	1 in 1.11
0.25× price (25×)	32.73%	67.27%	1 in 1.49
0.5× price (50×)	56.12%	43.88%	1 in 2.28
0.75× price (75×)	70.73%	29.27%	1 in 3.42
1× price (100×)	77.92%	22.08%	1 in 4.53
1.5× price (150×)	87.44%	12.56%	1 in 7.96
2× price (200×)	92.27%	7.733%	1 in 12.9
3× price (300×)	95.85%	4.153%	1 in 24.1
5× price (500×)	97.94%	2.064%	1 in 48.4
10× price (1,000×)	99.43%	0.5724%	1 in 175
20× price (2,000×)	99.82%	0.1798%	1 in 556
50× price (5,000×)	99.92%	0.07647%	1 in 1,308
100× price (10,000×)	99.95%	0.04840%	1 in 2,066

Price: 100×. Mean win 96.70× = 96.70% of the price.

10.3 Payout distribution

Figure 10.1 WICKED: probability and RTP contribution of each payout band

EXACT



Left: probability that a round's total win falls in the band (dots on a log scale; empty bands are omitted). Right: the band's contribution to the RTP, coloured by outcome class. Frequent small wins carry little value; rare large wins carry a large part of it.

Table 10.2 WICKED: payout bands

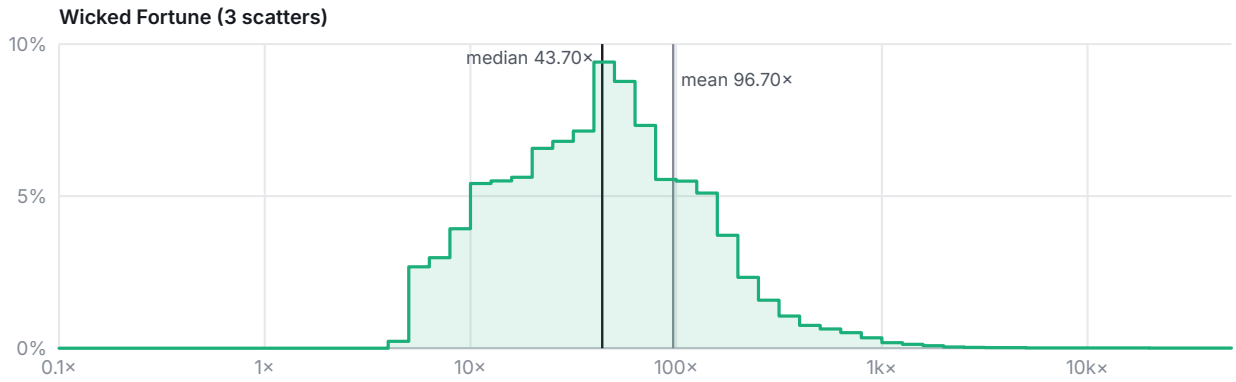
EXACT

Round win (× base bet)	Probability	Frequency	Per 10,000 rounds	RTP points	Mean in band	P(win ≥ band start)	Books
5–9.9×	9.807%	1 in 10.2	981	0.726	7.40×	100%	58,293
10–19.9×	16.54%	1 in 6.05	1,654	2.394	14.48×	90.19%	72,625
20–49.9×	29.77%	1 in 3.36	2,977	10.126	34.01×	73.66%	95,363
50–99.9×	21.80%	1 in 4.59	2,180	15.123	69.37×	43.88%	57,710
100–199.9×	14.35%	1 in 6.97	1,435	20.001	139.4×	22.08%	44,462
200–499.9×	5.669%	1 in 17.6	567	16.500	291.0×	7.733%	40,263
500–999.9×	1.492%	1 in 67.0	149	10.206	684.1×	2.064%	18,889
1,000–1,999.9×	0.3926%	1 in 255	39.26	5.293	1,348.1×	0.5724%	27,322
2,000–4,999.9×	0.1033%	1 in 968	10.33	3.114	3,014.3×	0.1798%	23,441
5,000–9,999.9×	0.02807%	1 in 3,562	2.81	2.022	7,203.9×	0.07647%	17,436
10,000–19,999.9×	0.02751%	1 in 3,635	2.75	3.943	14,330.5×	0.04840%	15,750
20,000–49,999.9×	0.01843%	1 in 5,427	1.84	6.023	32,685.3×	0.02088%	15,292
50,000 (max)	0.002457%	1 in 40,700	0.2457	1.229	50,000×	0.002457%	12,705
Total	**100%**		**10,000**	**96.700**	**96.70×**		**499,551**

Bands are labelled by the smallest and largest win they can contain (wins are multiples of 0.1×). "P(win ≥ band start)" is the survival probability at the lower edge of the band. Per 10,000 rounds: expected number of rounds in the band.

Figure 10.2 WICKED: distribution of the win

EXACT



Probability per tenth of a decade of win (log scale), median (dark) and mean (grey). The max win (50,000×) is an additional atom of probability 0.00246% (1 in 40,700).

10.4 Percentiles, thresholds and waiting times

Table 10.3 WICKED: percentiles of the round win

EXACT

Percentile	P25	P50	P75	P90	P95	P99	P99.5	P99.9	P99.99	P99.999
Win (× base bet)	19.10×	43.70×	88.40×	171.0×	264.8×	768.0×	1,076.4×	3,717.7×	34,140.2×	50,000×
Win (× price)	0.191	0.437	0.884	1.710	2.648	7.680	10.764	37.177	341.402	500.000

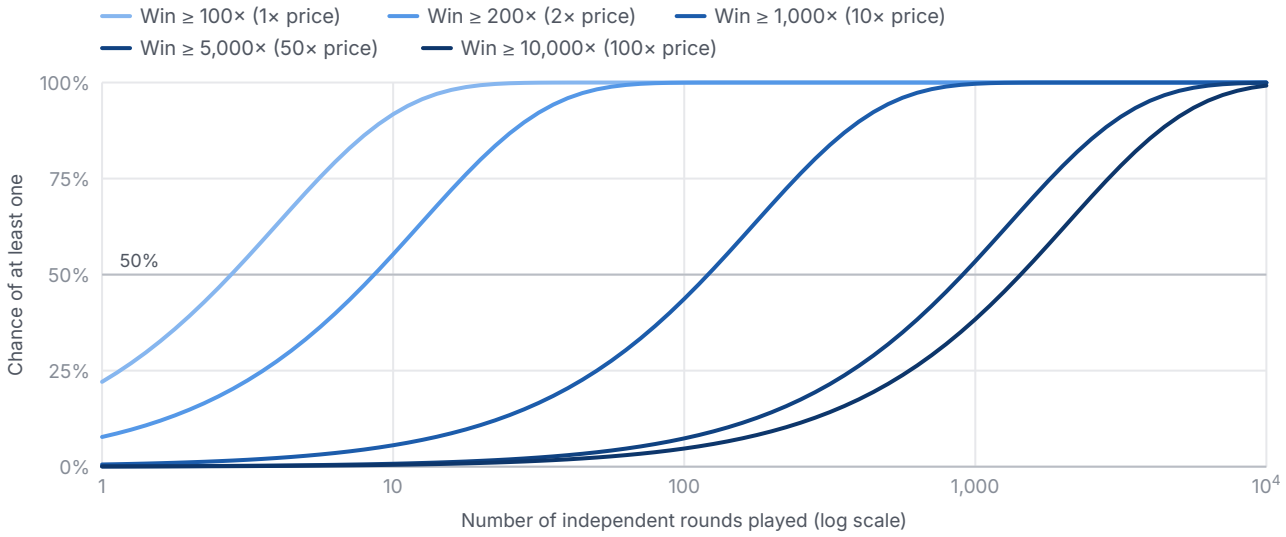
Pxx is the smallest win w with P(X ≤ w) ≥ xx%. Conditional rows give the percentiles within each outcome class.

Table 10.4 WICKED: thresholds, frequencies and waiting times EXACT FORMULA

Event per round	Probability	Frequency	Median wait (rounds)	At least once in 10 rounds	At least once in 100 rounds	At least once in 1,000 rounds
Any win (> 0)	100%	every round	0	100%	100%	100%
Win ≥ 10× (0.1× price)	90.19%	1 in 1.11	1	100%	100%	100%
Win ≥ 25× (0.25× price)	67.27%	1 in 1.49	1	100.0%	100%	100%
Win ≥ 50× (0.5× price)	43.88%	1 in 2.28	2	99.7%	100%	100%
Win ≥ 100× (1× price)	22.08%	1 in 4.53	3	91.8%	100%	100%
Win ≥ 200× (2× price)	7.733%	1 in 12.9	9	55.3%	100.0%	100%
Win ≥ 500× (5× price)	2.064%	1 in 48.4	34	18.8%	87.6%	100%
Win ≥ 1,000× (10× price)	0.5724%	1 in 175	121	5.58%	43.7%	99.7%
Win ≥ 2,000× (20× price)	0.1798%	1 in 556	386	1.78%	16.5%	83.5%
Win ≥ 5,000× (50× price)	0.07647%	1 in 1,308	907	0.762%	7.36%	53.5%
Win ≥ 10,000× (100× price)	0.04840%	1 in 2,066	1,432	0.483%	4.73%	38.4%
Max win (50,000×)	0.002457%	1 in 40,700	28,211	0.0246%	0.245%	2.43%

Probabilities are exact. Median wait: smallest n with P(event within n rounds) ≥ 50%, i.e. $\lceil \ln 0.5 / \ln(1-p) \rceil$. "At least once in n rounds" = $1-(1-p)^n$ (independent rounds).

Figure 10.3 WICKED: chance of at least one event within n rounds EXACT FORMULA



Curves $1-(1-p)^n$ for selected events of the table above (darker = rarer). Each curve crosses 50% at the median waiting time and 63% at $n = 1/p$ rounds.

10.5 Inside the bonus

Table 10.5 Wicked Fortune from the inside, in each mode where it occurs EXACT

Per bonus (conditional on its start)	WICKED (bought)	BASE (triggered)	ENHANCE1 (triggered)
Free spins played (mean)	10.73	10.73	10.73
P(at least one retrigger)	49.8%	49.9%	49.5%
Extra spins from retriggers (mean)	0.73	0.73	0.73
Special symbols per bonus (mean)	7.06	6.90	6.89
Special symbols per free spin (mean)	0.657	0.643	0.643
P(the witch casts at least once)	77.8%	76.1%	76.2%
P(4 identical special symbols at least once)	0.0697%	0.0622%	0.0559%
P(no orb wild in the bonus)	23.8%	24.5%	24.7%
P(an orb wild of ×10 or more)	20.2%	19.7%	19.2%
P(an orb wild of ×100 or more)	1.41%	1.18%	1.13%
Win of the triggering spin (mean)	0×	0.33×	0.34×
Win of the free spins (mean)	96.70×	56.06×	56.09×
P(bonus ends on the 50,000× cap)	0.00246% (1 in 40,700)	0.00100% (1 in 100,000)	0.00104% (1 in 96,428)

Weighted by the published lookup tables. A bought bonus has no triggering spin (0). Special symbols: those shown on the four squares during the free spins (four mirrors end the round at the cap).

10.6 Bought and triggered bonuses compared

Table 10.6 Wicked Fortune in each mode where it occurs EXACT

Per bonus (conditional)	WICKED (bought)	BASE (triggered)	ENHANCE1 (triggered)
Frequency per round of its mode	every purchase	1 in 200	1 in 40.0
Minimum win	5.00×	5.00×	5.00×
P25	19.10×	15.80×	15.80×
Median win	43.70×	43.10×	43.10×
Mean win	96.70×	56.39×	56.43×
P75	88.40×	71.80×	71.90×
P90	171.0×	114.2×	114.2×
P99	768.0×	276.5×	276.6×
P(win ≥ 100×)	22.1%	14.0%	14.0%
P(win ≥ 1,000×)	0.572%	0.0480%	0.0480%
P(win ≥ 10,000×)	0.0484%	0.00115%	0.00119%
P(max win)	1 in 40,700	1 in 100,000	1 in 96,428
Standard deviation	608.9×	174.2×	176.9×

Wins in base bets. Each column is the distribution of the same bonus tier as published in a given mode.

Bought: median 43.70×, minimum 5.00×, mean 96.70×; triggered in BASE: median 43.10×, minimum 5.00×, mean 56.39×. A triggered bonus is part of a 1× spin and is paid for by the base-game RTP split; a bought bonus returns 96.7% of its 100× price on its own.

CHAPTER 11

Grave Spell buy (GRAVE)

Purchase of the 4-scatter Grave Spell for 200×; it never pays less than 10×.

11.1 At a glance

Return to player **96.7000%** mean win 193.4×	Hit rate **100%** rounds paying more than 0	Win ≥ price **25.05%** 1 in 3.99	Median win **86.80×** 43.4% of the price
Standard deviation **3.52 × price** 703.8× in base bets	Win of 100× price or more **0.0190%** 1 in 5,259	Max win 50,000× **1 in 44,700** 0.00224%	Published books **499,553** 77,756 distinct payouts

11.2 Result relative to the price

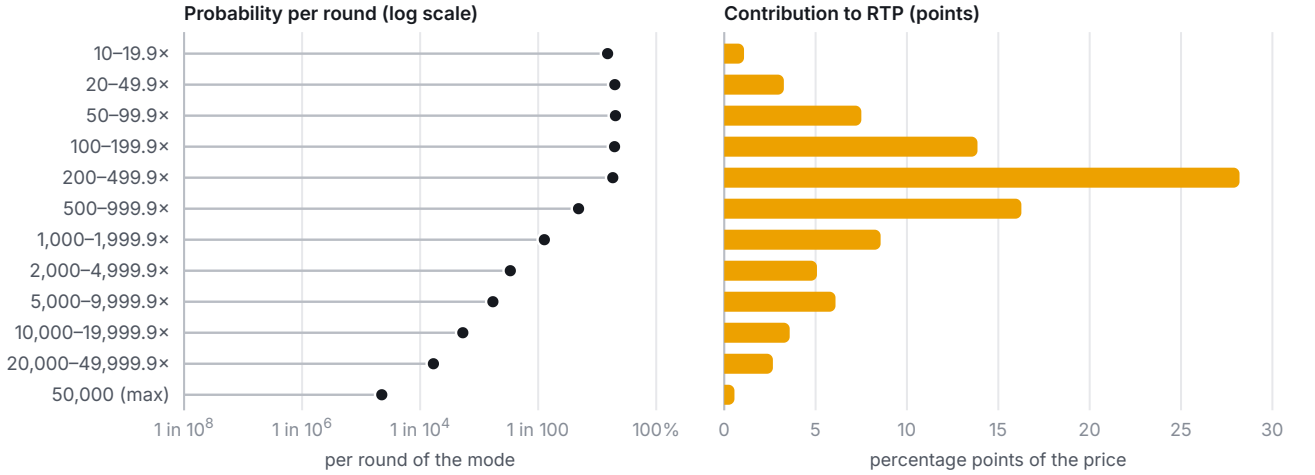
Table 11.1 GRAVE: result relative to the price — EXACT

Threshold	P(win below)	P(win at or above)	Frequency (at or above)
0.1× price (20×)	15.01%	84.99%	1 in 1.18
0.25× price (50×)	34.90%	65.10%	1 in 1.54
0.5× price (100×)	55.28%	44.72%	1 in 2.24
0.75× price (150×)	67.73%	32.27%	1 in 3.10
1× price (200×)	74.95%	25.05%	1 in 3.99
1.5× price (300×)	84.96%	15.04%	1 in 6.65
2× price (400×)	90.32%	9.678%	1 in 10.3
3× price (600×)	95.18%	4.815%	1 in 20.8
5× price (1,000×)	98.15%	1.852%	1 in 54.0
10× price (2,000×)	99.42%	0.5798%	1 in 172
20× price (4,000×)	99.70%	0.3002%	1 in 333
50× price (10,000×)	99.93%	0.07186%	1 in 1,392
100× price (20,000×)	99.98%	0.01902%	1 in 5,259

Price: 200×. Mean win 193.4× = 96.70% of the price.

11.3 Payout distribution

Figure 11.1 GRAVE: probability and RTP contribution of each payout band EXACT



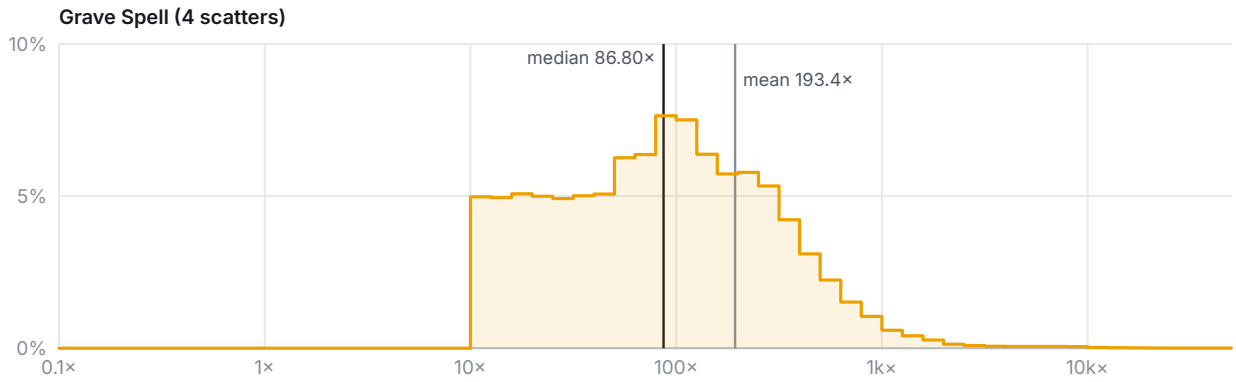
Left: probability that a round's total win falls in the band (dots on a log scale; empty bands are omitted). Right: the band's contribution to the RTP, coloured by outcome class. Frequent small wins carry little value; rare large wins carry a large part of it.

Table 11.2 GRAVE: payout bands EXACT

Round win (× base bet)	Probability	Frequency	Per 10,000 rounds	RTP points	Mean in band	P(win ≥ band start)	Books
10–19.9×	15.01%	1 in 6.66	1,501	1.082	14.43×	100%	38,690
20–49.9×	19.89%	1 in 5.03	1,989	3.262	32.80×	84.99%	99,456
50–99.9×	20.38%	1 in 4.91	2,038	7.504	73.63×	65.10%	89,745
100–199.9×	19.67%	1 in 5.08	1,967	13.860	140.9×	44.72%	73,262
200–499.9×	18.37%	1 in 5.45	1,837	28.201	307.1×	25.05%	64,776
500–999.9×	4.833%	1 in 20.7	483	16.262	673.0×	6.685%	27,627
1,000–1,999.9×	1.272%	1 in 78.6	127	8.559	1,345.9×	1.852%	16,733
2,000–4,999.9×	0.3371%	1 in 297	33.71	5.075	3,010.6×	0.5798%	26,107
5,000–9,999.9×	0.1708%	1 in 586	17.08	6.090	7,131.2×	0.2426%	18,110
10,000–19,999.9×	0.05284%	1 in 1,892	5.28	3.582	13,557.9×	0.07186%	16,267
20,000–49,999.9×	0.01678%	1 in 5,960	1.68	2.663	31,740.0×	0.01902%	15,718
50,000 (max)	0.002237%	1 in 44,700	0.2237	0.559	50,000×	0.002237%	13,062
Total	**100%**		**10,000**	**96.700**	**193.4×**		**499,553**

Bands are labelled by the smallest and largest win they can contain (wins are multiples of 0.1×). "P(win ≥ band start)" is the survival probability at the lower edge of the band. Per 10,000 rounds: expected number of rounds in the band.

Figure 11.2 GRAVE: distribution of the win EXACT



Probability per tenth of a decade of win (log scale), median (dark) and mean (grey). The max win (50,000×) is an additional atom of probability 0.00224% (1 in 44,700).

11.4 Percentiles, thresholds and waiting times

Table 11.3 GRAVE: percentiles of the round win EXACT

Percentile	P25	P50	P75	P90	P95	P99	P99.5	P99.9	P99.99	P99.999
Win (× base bet)	31.80×	86.80×	200.4×	392.9×	588.2×	1,432.4×	2,285.7×	8,810.8×	31,375.2×	50,000×
Win (× price)	0.159	0.434	1.002	1.965	2.941	7.162	11.429	44.054	156.876	250.000

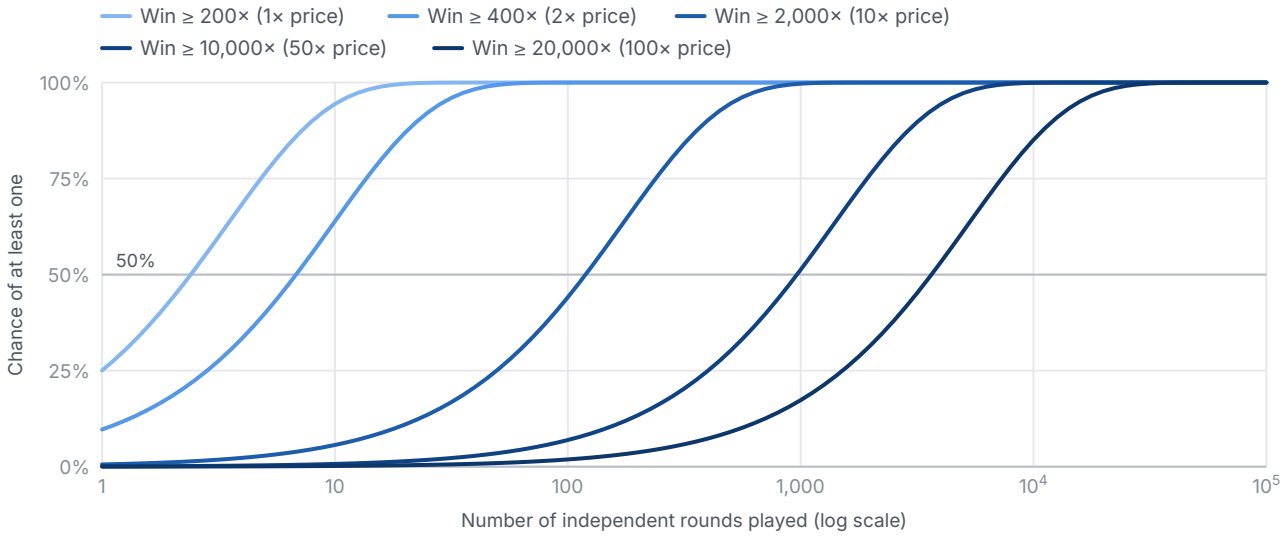
Pxx is the smallest win w with P(X ≤ w) ≥ xx%. Conditional rows give the percentiles within each outcome class.

Table 11.4 GRAVE: thresholds, frequencies and waiting times EXACT FORMULA

Event per round	Probability	Frequency	Median wait (rounds)	At least once in 10 rounds	At least once in 100 rounds	At least once in 1,000 rounds
Any win (> 0)	100%	every round	0	100%	100%	100%
Win ≥ 20× (0.1× price)	84.99%	1 in 1.18	1	100%	100%	100%
Win ≥ 50× (0.25× price)	65.10%	1 in 1.54	1	100.0%	100%	100%
Win ≥ 100× (0.5× price)	44.72%	1 in 2.24	2	99.7%	100%	100%
Win ≥ 200× (1× price)	25.05%	1 in 3.99	3	94.4%	100%	100%
Win ≥ 400× (2× price)	9.678%	1 in 10.3	7	63.9%	100.0%	100%
Win ≥ 1,000× (5× price)	1.852%	1 in 54.0	38	17.0%	84.6%	100%
Win ≥ 2,000× (10× price)	0.5798%	1 in 172	120	5.65%	44.1%	99.7%
Win ≥ 4,000× (20× price)	0.3002%	1 in 333	231	2.96%	26.0%	95.1%
Win ≥ 10,000× (50× price)	0.07186%	1 in 1,392	965	0.716%	6.94%	51.3%
Win ≥ 20,000× (100× price)	0.01902%	1 in 5,259	3,645	0.190%	1.88%	17.3%
Max win (50,000×)	0.002237%	1 in 44,700	30,984	0.0224%	0.223%	2.21%

Probabilities are exact. Median wait: smallest n with P(event within n rounds) ≥ 50%, i.e. $\lceil \ln 0.5 / \ln(1-p) \rceil$. "At least once in n rounds" = $1-(1-p)^n$ (independent rounds).

Figure 11.3 GRAVE: chance of at least one event within n rounds EXACT FORMULA



Curves $1-(1-p)^n$ for selected events of the table above (darker = rarer). Each curve crosses 50% at the median waiting time and 63% at $n = 1/p$ rounds.

11.5 Inside the bonus

Table 11.5 Grave Spell from the inside, in each mode where it occurs EXACT

Per bonus (conditional on its start)	GRAVE (bought)	BASE (triggered)	ENHANCE1 (triggered)	ENHANCE2 (triggered)	MYSTERY
Free spins played (mean)	10.71	10.71	10.71	10.77	10.75
P(at least one retrigger)	48.9%	48.8%	48.8%	51.2%	50.5%
Extra spins from retriggers (mean)	0.71	0.71	0.71	0.77	0.75
Special symbols per bonus (mean)	13.91	13.85	13.84	14.48	14.33
Special symbols per free spin (mean)	1.298	1.293	1.293	1.345	1.334
P(the witch casts at least once)	74.0%	73.4%	73.2%	80.6%	78.4%
P(4 identical special symbols at least once)	0.0621%	0.0652%	0.0749%	0.0618%	0.0881%
P(no orb wild in the bonus)	4.94%	5.12%	4.95%	4.38%	4.47%
P(an orb wild of ×10 or more)	35.9%	35.6%	35.3%	41.4%	39.5%
P(an orb wild of ×100 or more)	3.18%	2.98%	2.99%	4.29%	4.58%
Win of the triggering spin (mean)	0×	0.23×	0.23×	0.34×	0×
Win of the free spins (mean)	193.4×	154.7×	154.7×	295.7×	458.4×
P(bonus ends on the 50,000× cap)	0.00224% (1 in 44,700)	0.00167% (1 in 60,000)	0.00167% (1 in 60,000)	0.00167% (1 in 60,000)	0.00417% (1 in 24,000)

Weighted by the published lookup tables. A bought bonus has no triggering spin (0). Special symbols: those shown on the four squares during the free spins (four mirrors end the round at the cap).

11.6 Bought and triggered bonuses compared

Table 11.6 Grave Spell in each mode where it occurs EXACT

Per bonus (conditional)	GRAVE (bought)	BASE (triggered)	ENHANCE1 (triggered)	ENHANCE2 (triggered)	MYSTERY
Frequency per round of its mode	every purchase	1 in 1,500	1 in 300	1 in 39.6	1 in 4.00
Minimum win	10.00×	10.00×	10.00×	10.00×	25.00×
P25	31.80×	29.60×	29.60×	137.1×	65.00×
Median win	86.80×	85.50×	85.50×	230.1×	169.0×
Mean win	193.4×	154.9×	154.9×	296.0×	458.4×
P75	200.4×	161.4×	161.4×	357.3×	439.8×
P90	392.9×	301.4×	301.6×	512.4×	843.3×
P99	1,432.4×	966.7×	966.3×	1,354.7×	4,308.5×
P(win ≥ 100×)	44.7%	44.0%	44.0%	89.6%	63.7%
P(win ≥ 1,000×)	1.85%	0.930%	0.930%	1.85%	6.94%
P(win ≥ 10,000×)	0.0719%	0.0476%	0.0476%	0.0172%	0.670%
P(max win)	1 in 44,700	1 in 60,000	1 in 60,000	1 in 60,000	1 in 24,000
Standard deviation	703.8×	545.5×	545.6×	438.0×	1,733.1×

Wins in base bets. Each column is the distribution of the same bonus tier as published in a given mode.

The Grave Spell pays back its price 1 in 3.99 purchases (25.0%; specification: about 1 in 4). Bought: median 86.80×, minimum 10.00×; triggered in BASE: median 85.50×, minimum 10.00×. The Grave Spell of the Enhanced Spins 2 and of the Mystery are set for their mode (15× spin, 500× Mystery): means 296.0× and 458.4×.

CHAPTER 12

Mystery bonus (MYSTERY)

A 500× purchase: losing spin (65%), Grave Spell (25%) or Dark Enchantment (10%).

12.1 At a glance

Return to player **96.7000%** mean win 483.5×	Hit rate **35.00%** rounds paying more than 0	Win ≥ price **15.52%** 1 in 6.44	Median win **0×** 0.0% of the price
Standard deviation **4.38 × price** 2,188.0× in base bets	Win of 100× price or more **0.0324%** 1 in 3,083	Max win 50,000× **1 in 3,083** 0.0324%	Published books **599,345** 134,862 distinct payouts

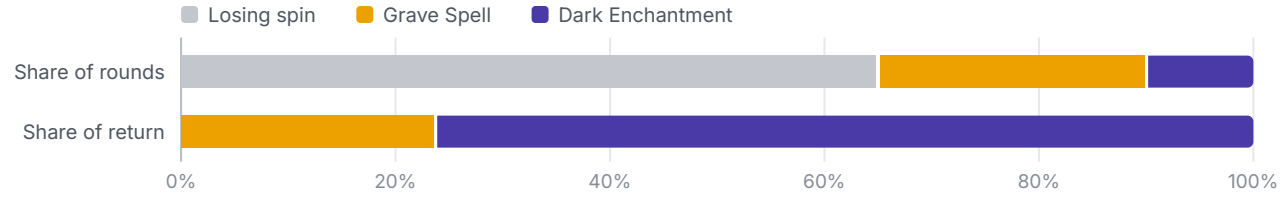
12.2 The three outcomes

Table 12.1 Outcome classes: frequency, value and conditional distribution EXACT

Class	Probability per round	Frequency	RTP points	Share of RTP	Mean win	Median	P10 – P90	P(no win)	Max win within class	Books
Losing spin	65.00%	1 in 1.54	0.00	0.0%	0×	0×	0× – 0×	100%	—	147,063
Grave Spell (4 scatters)	25.00%	1 in 4.00	22.92	23.7%	458.4×	169.0×	36.70× – 843.3×	0%	1 in 24,000	240,891
Dark Enchantment (5 scatters)	10.00%	1 in 10.00	73.78	76.3%	3,689.1×	1,789.8×	1,126.1× – 8,133.5×	0%	1 in 319	211,391
All rounds	**100%**		**96.70**	**100%**	**483.5×**	**0×**	**0× – 1,125.3×**	**65.0%**	**1 in 3,083**	**599,345**

Every round belongs to exactly one class. Mean, median, P10–P90, P(no win) and the max-win frequency are conditional on the class (i.e. per round of that class). RTP points add up to the mode's RTP.

Figure 12.1 MYSTERY: share of purchases and share of the return by outcome EXACT



The Dark Enchantment is 10% of purchases but returns 76.3% of the value; losing spins (two scatters, no win) are 65.0% of purchases.

The mixture is exact: $0.65 \times 0 + 0.25 \times 458.36 + 0.10 \times 3{,}689.09 = 483.50 = 96.7\% \times 500$. With the Grave Spell and Dark Enchantment triggered in BASE (means 154.9× and 1,965.3×) the same mixture would return only 235.3×, i.e. 47.1% of the price; this is why the Mystery bonuses are richer than the triggered ones (the price and split of the brief come first).

12.3 Result relative to the price

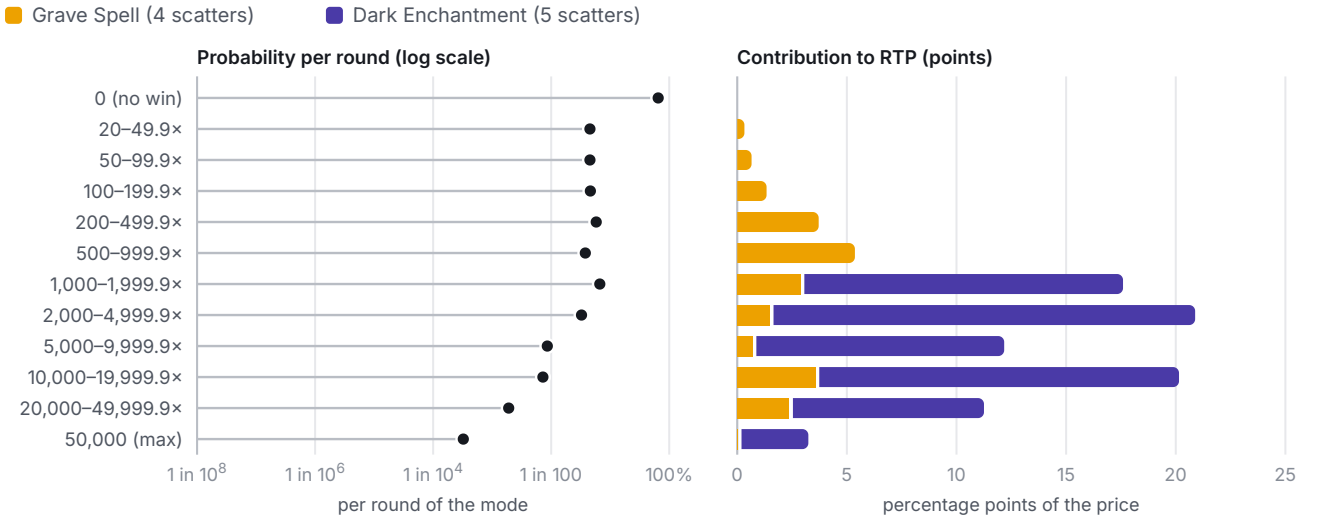
Table 12.2 MYSTERY: result relative to the price EXACT

Threshold	P(win below)	P(win at or above)	Frequency (at or above)
0.1× price (50×)	69.53%	30.47%	1 in 3.28
0.25× price (125×)	75.53%	24.47%	1 in 4.09
0.5× price (250×)	80.25%	19.75%	1 in 5.06
0.75× price (375×)	82.79%	17.21%	1 in 5.81
1× price (500×)	84.48%	15.52%	1 in 6.44
1.5× price (750×)	86.82%	13.18%	1 in 7.59
2× price (1,000×)	88.27%	11.73%	1 in 8.52
3× price (1,500×)	93.54%	6.461%	1 in 15.5
5× price (2,500×)	95.75%	4.249%	1 in 23.5
10× price (5,000×)	98.19%	1.807%	1 in 55.4
20× price (10,000×)	99.05%	0.9470%	1 in 106
50× price (25,000×)	99.86%	0.1386%	1 in 722
100× price (50,000×)	99.97%	0.03243%	1 in 3,083

Price: 500×. Mean win 483.5× = 96.70% of the price.

12.4 Payout distribution

Figure 12.2 MYSTERY: probability and RTP contribution of each payout band EXACT



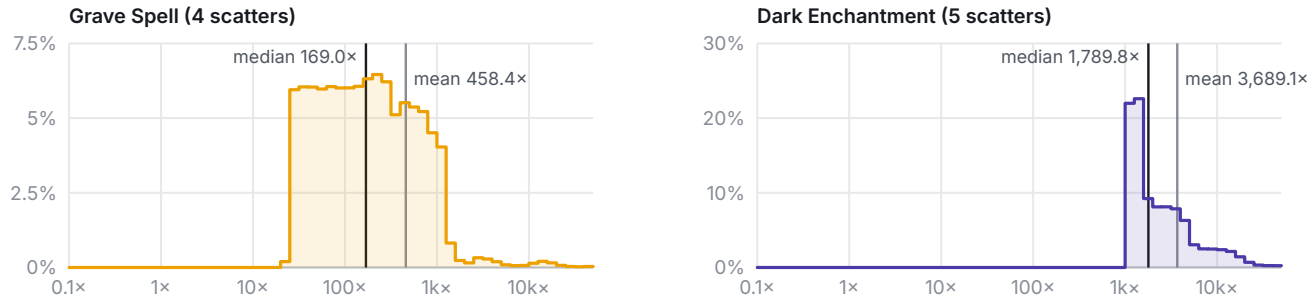
Left: probability that a round's total win falls in the band (dots on a log scale; empty bands are omitted). Right: the band's contribution to the RTP, coloured by outcome class. Frequent small wins carry little value; rare large wins carry a large part of it.

Table 12.3 MYSTERY: payout bands EXACT

Round win (× base bet)	Probability	Frequency	Per 10,000 rounds	RTP points	Mean in band	P(win ≥ band start)	Books
0 (no win)	65.00%	1 in 1.54	6,500	0.000	—	100%	147,063
20–49.9×	4.534%	1 in 22.1	453	0.327	36.08×	35.00%	38,469
50–99.9×	4.534%	1 in 22.1	453	0.654	72.10×	30.47%	42,792
100–199.9×	4.615%	1 in 21.7	461	1.338	144.9×	25.93%	35,603
200–499.9×	5.796%	1 in 17.3	580	3.715	320.5×	21.32%	30,725
500–999.9×	3.788%	1 in 26.4	379	5.367	708.5×	15.52%	13,304
1,000–1,999.9×	6.661%	1 in 15.0	666	17.595	1,320.6×	11.73%	59,472
2,000–4,999.9×	3.266%	1 in 30.6	327	20.885	3,196.9×	5.073%	92,915
5,000–9,999.9×	0.8596%	1 in 116	85.96	12.176	7,082.4×	1.807%	55,142
10,000–19,999.9×	0.7240%	1 in 138	72.40	20.148	13,914.0×	0.9470%	38,143
20,000–49,999.9×	0.1905%	1 in 525	19.05	11.253	29,530.9×	0.2230%	31,635
50,000 (max)	0.03243%	1 in 3,083	3.24	3.243	50,000×	0.03243%	14,082
Total	**100%**		**10,000**	**96.700**	**483.5×**		**599,345**

Bands are labelled by the smallest and largest win they can contain (wins are multiples of 0.1×). "P(win ≥ band start)" is the survival probability at the lower edge of the band. Per 10,000 rounds: expected number of rounds in the band.

Figure 12.3 MYSTERY: distribution of the bonus win (conditional on the outcome) EXACT



Probability per tenth of a decade of win; median (dark) and mean (grey). Max-win atoms: 1 in 24,000 and 1 in 319 for Grave Spell and Dark Enchantment.

12.5 Percentiles, thresholds and waiting times

Table 12.4 MYSTERY: percentiles of the round win EXACT

Percentile	P25	P50	P75	P90	P95	P99	P99.5	P99.9	P99.99	P99.999
Win (× base bet)	0×	0×	115.3×	1,125.3×	2,040.7×	9,553.1×	14,442.6×	30,854.6×	50,000×	50,000×
Win (× price)	0.000	0.000	0.231	2.251	4.081	19.106	28.885	61.709	100.000	100.000
Grave Spell (4 scatters) (×, conditional)	65.00×	169.0×	439.8×	843.3×	1,077×	4,308.5×	13,095.9×	24,692.4×	48,396.8×	50,000×
Dark Enchantment (5 scatters) (×, conditional)	1,294.4×	1,789.8×	3,638.1×	8,133.5×	13,099.4×	27,273.4×	42,012.8×	50,000×	50,000×	50,000×

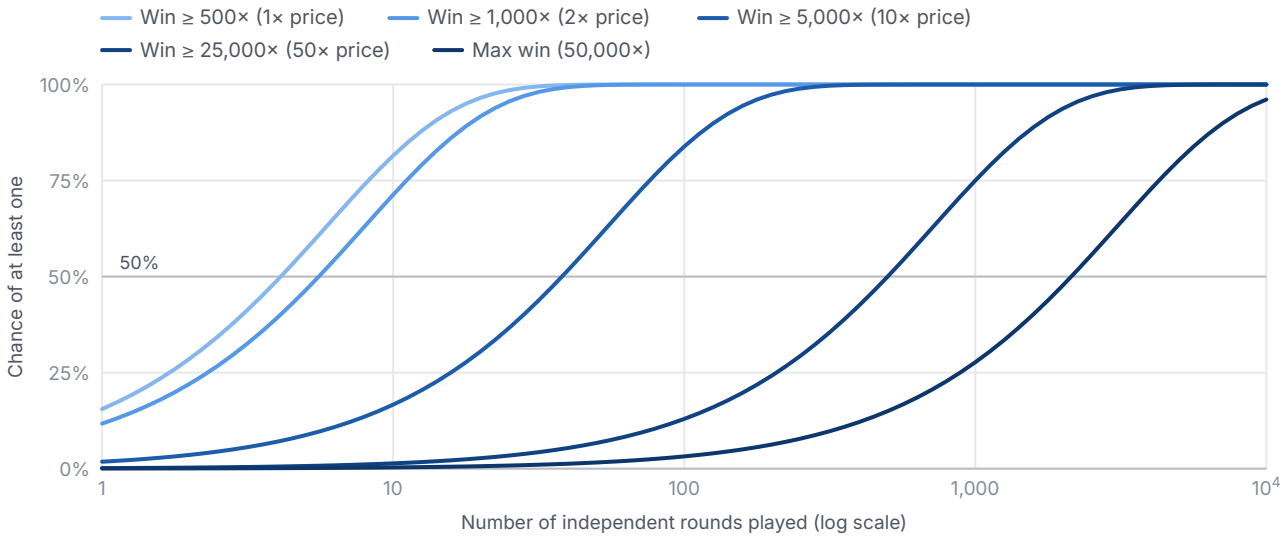
Pxx is the smallest win w with P(X ≤ w) ≥ xx%. Conditional rows give the percentiles within each outcome class.

Table 12.5 MYSTERY: thresholds, frequencies and waiting times EXACT FORMULA

Event per round	Probability	Frequency	Median wait (rounds)	At least once in 10 rounds	At least once in 100 rounds	At least once in 1,000 rounds
Any win (> 0)	35.00%	1 in 2.86	2	98.7%	100%	100%
Win ≥ 25× (0.05× price)	35.00%	1 in 2.86	2	98.7%	100%	100%
Win ≥ 50× (0.1× price)	30.47%	1 in 3.28	2	97.4%	100%	100%
Win ≥ 100× (0.2× price)	25.93%	1 in 3.86	3	95.0%	100%	100%
Win ≥ 250× (0.5× price)	19.75%	1 in 5.06	4	88.9%	100%	100%
Win ≥ 500× (1× price)	15.52%	1 in 6.44	5	81.5%	100%	100%
Win ≥ 1,000× (2× price)	11.73%	1 in 8.52	6	71.3%	100.0%	100%
Win ≥ 2,500× (5× price)	4.249%	1 in 23.5	16	35.2%	98.7%	100%
Win ≥ 5,000× (10× price)	1.807%	1 in 55.4	39	16.7%	83.8%	100%
Win ≥ 10,000× (20× price)	0.9470%	1 in 106	73	9.08%	61.4%	100.0%
Win ≥ 25,000× (50× price)	0.1386%	1 in 722	500	1.38%	12.9%	75.0%
Max win (50,000×)	0.03243%	1 in 3,083	2,137	0.324%	3.19%	27.7%

Probabilities are exact. Median wait: smallest n with P(event within n rounds) ≥ 50%, i.e. $\lceil \ln 0.5 / \ln(1-p) \rceil$. "At least once in n rounds" = $1-(1-p)^n$ (independent rounds).

Figure 12.4 MYSTERY: chance of at least one event within n rounds EXACT FORMULA



Curves $1-(1-p)^n$ for selected events of the table above (darker = rarer). Each curve crosses 50% at the median waiting time and 63% at $n = 1/p$ rounds.

12.6 Inside the Mystery bonuses

Table 12.6 MYSTERY: what happens inside the bonuses EXACT

Per bonus (conditional on its start)	Grave Spell	Dark Enchantment
Free spins played (mean)	10.75	8.60
P(at least one retrigger)	50.5%	43.6%
Extra spins from retriggers (mean)	0.75	0.61
Special symbols per bonus (mean)	14.33	26.50
Special symbols per free spin (mean)	1.334	3.082
P(the witch casts at least once)	78.4%	0.158%
P(4 identical special symbols at least once)	0.0881%	0.775%
P(no orb wild in the bonus)	4.47%	0.595%
P(an orb wild of ×10 or more)	39.5%	43.9%
P(an orb wild of ×100 or more)	4.58%	4.21%
Win of the triggering spin (mean)	0×	0×
Win of the free spins (mean)	458.4×	3,689.1×
P(bonus ends on the 50,000× cap)	0.00417% (1 in 24,000)	0.314% (1 in 319)

Weighted by the published lookup tables. A bought bonus has no triggering spin (0). Special symbols: those shown on the four squares during the free spins (four mirrors end the round at the cap).

Table 12.7 The Dark Enchantment in each mode where it occurs EXACT

Per bonus (conditional)	MYSTERY	BASE (triggered)	ENHANCE2 (triggered)
Frequency per round of its mode	1 in 10.00	1 in 20,000	1 in 300
Minimum win	1,000.1×	1,000.1×	1,000.1×
P25	1,294.4×	1,223.9×	1,224×
Median win	1,789.8×	1,460.1×	1,460.4×
Mean win	3,689.1×	1,965.3×	1,965.2×
P75	3,638.1×	1,826×	1,824.6×
P90	8,133.5×	2,906.4×	2,909.7×
P99	27,273.4×	9,694.7×	9,691.3×
P(win ≥ 100×)	100%	100%	100%
P(win ≥ 1,000×)	100%	100%	100%
P(win ≥ 10,000×)	7.79%	0.920%	0.920%
P(max win)	1 in 319	1 in 1,000	1 in 1,000
Standard deviation	5,344.9×	2,341.2×	2,340.8×

Wins in base bets. Each column is the distribution of the same bonus tier as published in a given mode.

CHAPTER 13

Inside the free spins

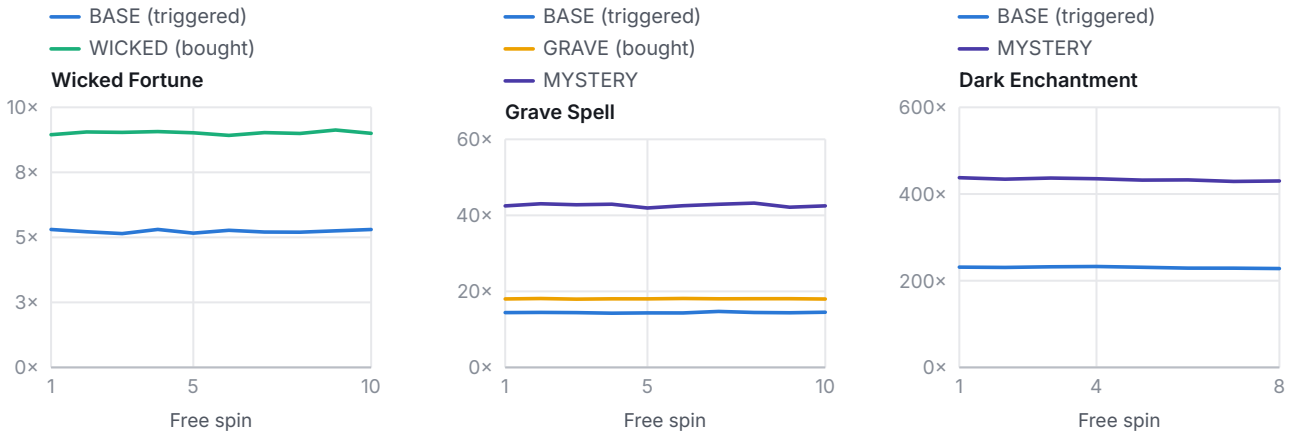
Special squares, orbs and retriggers, spin after spin.

13.1 Spin-by-spin win

The four special squares are unlocked for the whole bonus; the Grave Spell guarantees at least one special symbol on every free spin and the Dark Enchantment at least three. Nothing is carried from one free spin to the next: each spin is evaluated on its own board. The figure gives the mean win of the k-th free spin, among bonuses that reach spin k.

Figure 13.1 Mean win of the k-th free spin, by tier and mode

EXACT



Mean win of free spin k (base bets), among bonuses that reach that spin (10, 10, 8 spins for 3, 4, 5 scatters, plus retriggers). ENHANCE1 and ENHANCE2 are omitted for legibility (table below).

Table 13.1 Where the bonus win is made

EXACT

Tier · mode	Mean win spin 1	Last base spin	Share: spins 1–3	Spins 4–6	Spins 7+	Mean bonus win
Wicked Fortune · BASE (triggered)	5.30×	5.30×	27.9%	28.1%	44.0%	56.39×
Wicked Fortune · ENHANCE1 (triggered)	5.35×	5.19×	28.2%	27.9%	43.9%	56.43×
Wicked Fortune · WICKED (bought)	8.95×	9.00×	27.9%	27.9%	44.2%	96.70×
Grave Spell · BASE (triggered)	14.42×	14.52×	28.0%	27.7%	44.3%	154.9×
Grave Spell · ENHANCE2 (triggered)	28.09×	27.90×	28.0%	27.8%	44.2%	296.0×
Grave Spell · GRAVE (bought)	18.01×	17.99×	28.0%	28.0%	44.0%	193.4×
Grave Spell · MYSTERY	42.49×	42.49×	28.0%	27.8%	44.2%	458.4×
Dark Enchantment · BASE (triggered)	231.3×	228.1×	35.2%	35.1%	29.6%	1,965.3×
Dark Enchantment · ENHANCE2 (triggered)	232.8×	228.2×	35.4%	34.9%	29.6%	1,965.2×
Dark Enchantment · MYSTERY	437.6×	430.1×	35.3%	35.0%	29.7%	3,689.1×

Shares of the total free-spin win paid by each group of spins (they add up to 100%; spin wins are taken before the round cap). Last base spin: spin 10 (3 and 4 scatters) or 8 (5 scatters). Mean bonus win: mean round win of the tier in that mode.

13.2 Special symbols in the free spins

Table 13.2 Free spins: special symbols per spin EXACT

Spins	With a special symbol	Exactly 1	2	3	4	Broomsticks per 100 spins	Spellbooks per 100 spins	Orbs per 100 spins	Mirrors per 100 spins	4 identical
Wicked Fortune · BASE (triggered)	43.6%	26.6%	13.4%	3.17%	0.328%	15.29	19.90	13.11	16.00	1 in 17,252
Wicked Fortune · ENHANCE1 (triggered)	43.5%	26.6%	13.4%	3.19%	0.320%	15.33	19.90	13.12	15.90	1 in 19,176
Wicked Fortune · WICKED (bought)	44.1%	26.5%	13.8%	3.40%	0.364%	15.85	20.48	13.42	15.98	1 in 15,393
Grave Spell · BASE (triggered)	100%	75.1%	20.9%	3.68%	0.356%	31.95	38.83	25.97	32.57	1 in 16,418
Grave Spell · ENHANCE2 (triggered)	100%	71.4%	23.2%	4.83%	0.528%	33.68	42.70	27.04	31.07	1 in 17,415
Grave Spell · GRAVE (bought)	100%	74.8%	21.0%	3.83%	0.378%	32.27	39.01	26.09	32.43	1 in 17,236
Grave Spell · MYSTERY	100%	72.4%	22.4%	4.67%	0.539%	33.67	41.30	27.07	31.33	1 in 12,186
Dark Enchantment · BASE (triggered)	100%	0	0	92.2%	7.76%	74.89	88.60	55.46	88.81	1 in 1,290
Dark Enchantment · ENHANCE2 (triggered)	100%	0	0	92.3%	7.73%	75.18	88.41	55.56	88.58	1 in 1,248
Dark Enchantment · MYSTERY	100%	0	0	91.8%	8.20%	75.78	89.88	55.75	86.79	1 in 1,106

Per spin played in that context, weighted by the published lookup tables. 4 identical: the four squares show the same special symbol (its “all four” effect).

Table 13.3 Free spins: what the special symbols are worth EXACT

Spins	Mean win, no special symbol	Mean win, with one or more	Hit rate, no special symbol	Hit rate, with one or more	Share of the spin wins
Wicked Fortune · BASE (triggered)	0.43×	11.44×	34.2%	60.2%	95.3%
Wicked Fortune · ENHANCE1 (triggered)	0.43×	11.47×	34.1%	60.2%	95.4%
Wicked Fortune · WICKED (bought)	0.44×	19.91×	34.2%	61.0%	97.3%
Grave Spell · BASE (triggered)	—	14.45×	—	55.3%	100.0%
Grave Spell · ENHANCE2 (triggered)	—	27.47×	—	59.0%	100.0%
Grave Spell · GRAVE (bought)	—	18.06×	—	55.5%	100.0%
Grave Spell · MYSTERY	—	42.67×	—	57.9%	100.0%
Dark Enchantment · BASE (triggered)	—	229.3×	—	74.8%	100.0%
Dark Enchantment · ENHANCE2 (triggered)	—	229.3×	—	74.8%	100.0%
Dark Enchantment · MYSTERY	—	431.3×	—	75.3%	100.0%

Mean win of a spin in base bets (before the round cap). Share of the spin wins: part of the total paid by the spins of that context that comes from spins showing at least one special symbol.

Table 13.4 Free spins: effects of the special symbols EXACT

Spins	Board split (broomstick)	4 broomsticks: premium board	Wilds per spellbook	Wild nudges per 1,000 spins	Steps per nudge	Orb wilds per 100 spins	Mean orb multiplier
Wicked Fortune · BASE (triggered)	13.8%	1 in 77,295	2.51	0.09	1.96	13.11	6.4×
Wicked Fortune · ENHANCE1 (triggered)	13.8%	1 in 91,883	2.50	0.08	2.08	13.12	6.3×
Wicked Fortune · WICKED (bought)	14.3%	1 in 54,553	2.53	0.09	1.92	13.42	7.3×
Grave Spell · BASE (triggered)	29.8%	1 in 72,776	2.49	0.09	2.03	25.97	7.4×
Grave Spell · ENHANCE2 (triggered)	31.5%	1 in 68,729	2.64	0.08	1.90	27.04	8.3×
Grave Spell · GRAVE (bought)	30.1%	1 in 69,330	2.51	0.08	1.95	26.09	7.7×
Grave Spell · MYSTERY	31.4%	1 in 34,168	2.60	0.08	1.98	27.07	10.7×
Dark Enchantment · BASE (triggered)	58.2%	1 in 3,995	2.54	0.90	1.99	55.46	5.6×
Dark Enchantment · ENHANCE2 (triggered)	58.4%	1 in 3,825	2.54	0.89	2.00	55.56	5.7×
Dark Enchantment · MYSTERY	59.3%	1 in 3,388	2.59	0.87	2.05	55.75	6.5×

Board split: share of spins with at least one broomstick. Wild nudges come only from four spellbooks (2 per occurrence); the reel multiplier equals the number of steps. Weighted by the published lookup tables.

Table 13.5 Free spins: multiplier of the orb wilds EXACT

Spins	×2	×3	×5	×10	×20	×50	×100	×250	×500	×1,000
Wicked Fortune · BASE (triggered)	40.1%	28.3%	16.2%	9.17%	3.85%	1.55%	0.619%	0.163%	0.0490%	0.0208%
Wicked Fortune · ENHANCE1 (triggered)	40.4%	28.4%	16.1%	8.88%	3.84%	1.60%	0.619%	0.127%	0.0470%	0.0223%
Wicked Fortune · WICKED (bought)	39.9%	28.1%	16.2%	9.04%	4.02%	1.72%	0.730%	0.218%	0.0917%	0.0500%
Grave Spell · BASE (triggered)	40.2%	27.9%	16.0%	9.07%	3.92%	1.70%	0.788%	0.246%	0.106%	0.0503%
Grave Spell · ENHANCE2 (triggered)	38.5%	27.3%	16.2%	9.73%	4.51%	2.22%	1.05%	0.308%	0.126%	0.0480%
Grave Spell · GRAVE (bought)	39.9%	28.0%	16.0%	8.97%	4.02%	1.79%	0.820%	0.268%	0.116%	0.0589%
Grave Spell · MYSTERY	38.5%	27.4%	15.9%	9.41%	4.44%	2.22%	1.15%	0.489%	0.257%	0.168%
Dark Enchantment · BASE (triggered)	47.7%	28.0%	13.1%	6.09%	3.13%	1.26%	0.513%	0.139%	0.0562%	0.0184%
Dark Enchantment · ENHANCE2 (triggered)	47.7%	27.9%	13.1%	6.11%	3.18%	1.25%	0.526%	0.158%	0.0534%	0.0218%
Dark Enchantment · MYSTERY	47.0%	27.8%	13.1%	6.34%	3.31%	1.46%	0.615%	0.205%	0.0992%	0.0430%

Share of the orb wilds by multiplier (four orbs included: ×20 or more each).

13.3 Largest orb wild of the bonus

Table 13.6 Largest orb multiplier reached during the bonus EXACT

Tier · mode	No orb	×2	×3	×5	×10	×20	×50	×100	×250	×500	×1,000
Wicked Fortune · BASE (triggered)	24.5%	18.0%	21.2%	16.6%	11.2%	5.12%	2.13%	0.861%	0.227%	0.0680%	0.0290%
Wicked Fortune · ENHANCE1 (triggered)	24.7%	18.5%	21.1%	16.5%	10.7%	5.11%	2.21%	0.859%	0.177%	0.0652%	0.0311%
Wicked Fortune · WICKED (bought)	23.8%	18.2%	21.1%	16.6%	11.1%	5.36%	2.32%	0.969%	0.266%	0.109%	0.0655%
Grave Spell · BASE (triggered)	5.12%	12.8%	22.9%	23.6%	18.7%	9.52%	4.35%	2.01%	0.592%	0.251%	0.127%
Grave Spell · ENHANCE2 (triggered)	4.38%	10.8%	20.5%	22.9%	19.8%	11.2%	6.06%	2.95%	0.862%	0.352%	0.136%
Grave Spell · GRAVE (bought)	4.94%	12.5%	22.8%	23.8%	18.4%	9.77%	4.58%	2.10%	0.649%	0.275%	0.150%
Grave Spell · MYSTERY	4.47%	11.4%	21.6%	23.0%	19.0%	10.5%	5.41%	2.65%	0.983%	0.530%	0.422%
Dark Enchantment · BASE (triggered)	0.545%	6.89%	23.4%	27.7%	19.9%	12.6%	5.58%	2.36%	0.629%	0.253%	0.0825%
Dark Enchantment · ENHANCE2 (triggered)	0.573%	6.90%	23.1%	27.5%	20.0%	12.9%	5.54%	2.40%	0.722%	0.237%	0.0985%
Dark Enchantment · MYSTERY	0.595%	6.69%	22.1%	26.7%	20.3%	13.0%	6.33%	2.73%	0.875%	0.425%	0.189%

Share of bonuses by the largest multiplier of an orb wild at any free spin of the bonus.

13.4 Retriggers and bonus length

Table 13.7 Number of free spins played per bonus EXACT

Tier · mode	Base spins	Fewer (cap)	Base	+1	+2	+3	+4 or more	Mean	Longest
Wicked Fortune · BASE (triggered)	10	0.000815%	50.1%	32.9%	12.3%	3.54%	1.13%	10.73	19
Wicked Fortune · ENHANCE1 (triggered)	10	0.000840%	50.5%	32.3%	12.5%	3.56%	1.14%	10.73	18
Wicked Fortune · WICKED (bought)	10	0.00202%	50.1%	32.6%	12.4%	3.69%	1.17%	10.73	20
Grave Spell · BASE (triggered)	10	0.00138%	51.2%	32.3%	11.9%	3.44%	1.10%	10.71	20
Grave Spell · ENHANCE2 (triggered)	10	0.00137%	48.8%	32.7%	13.3%	3.99%	1.33%	10.77	19
Grave Spell · GRAVE (bought)	10	0.00183%	51.1%	32.3%	12.0%	3.44%	1.11%	10.71	19
Grave Spell · MYSTERY	10	0.00341%	49.4%	32.9%	12.7%	3.74%	1.19%	10.75	19
Dark Enchantment · BASE (triggered)	8	0.0806%	56.8%	30.2%	9.73%	2.45%	0.677%	8.60	16
Dark Enchantment · ENHANCE2 (triggered)	8	0.0819%	56.9%	30.0%	9.77%	2.53%	0.649%	8.60	16
Dark Enchantment · MYSTERY	8	0.258%	56.2%	30.4%	9.95%	2.47%	0.714%	8.60	15

Each scatter on the board of a free spin adds one spin (at most 40). “Fewer (cap)”: bonuses stopped early because the round reached 50,000×.

13.5 Triggered bonuses: summary of the experience

Table 13.8 What happens inside bonuses triggered in BASE EXACT

Per bonus (conditional on its start)	Wicked Fortune	Grave Spell	Dark Enchantment
Free spins played (mean)	10.73	10.71	8.60
P(at least one retrigger)	49.9%	48.8%	43.1%
Extra spins from retriggers (mean)	0.73	0.71	0.60
Special symbols per bonus (mean)	6.90	13.85	26.46
Special symbols per free spin (mean)	0.643	1.293	3.078
P(the witch casts at least once)	76.1%	73.4%	0.0463%
P(4 identical special symbols at least once)	0.0622%	0.0652%	0.655%
P(no orb wild in the bonus)	24.5%	5.12%	0.545%
P(an orb wild of ×10 or more)	19.7%	35.6%	41.5%
P(an orb wild of ×100 or more)	1.18%	2.98%	3.33%
Win of the triggering spin (mean)	0.33×	0.23×	0.23×
Win of the free spins (mean)	56.06×	154.7×	1,965.0×
P(bonus ends on the 50,000× cap)	0.00100% (1 in 100,000)	0.00167% (1 in 60,000)	0.1000% (1 in 1,000)

Weighted by the published lookup tables. A bought bonus has no triggering spin (0). Special symbols: those shown on the four squares during the free spins (four mirrors end the round at the cap).

The corresponding tables for the bought bonuses and the Mystery bonuses are in Chapters 10, 11 and 12.

CHAPTER 14

Sessions: waiting times and results over many rounds

Waiting times (exact), then session returns and bankroll life (Monte Carlo).

14.1 Waiting for a bonus

Rounds are independent and identically distributed, so the number of spins until the first occurrence of an event of probability p follows a geometric law: mean $1/p$, median $\lceil \ln 0.5 / \ln(1-p) \rceil$, and $P(\text{none in } n) = (1-p)^n$. Spend per event is the price of a spin divided by p.

Table 14.1 Waiting times for bonuses and for the witch

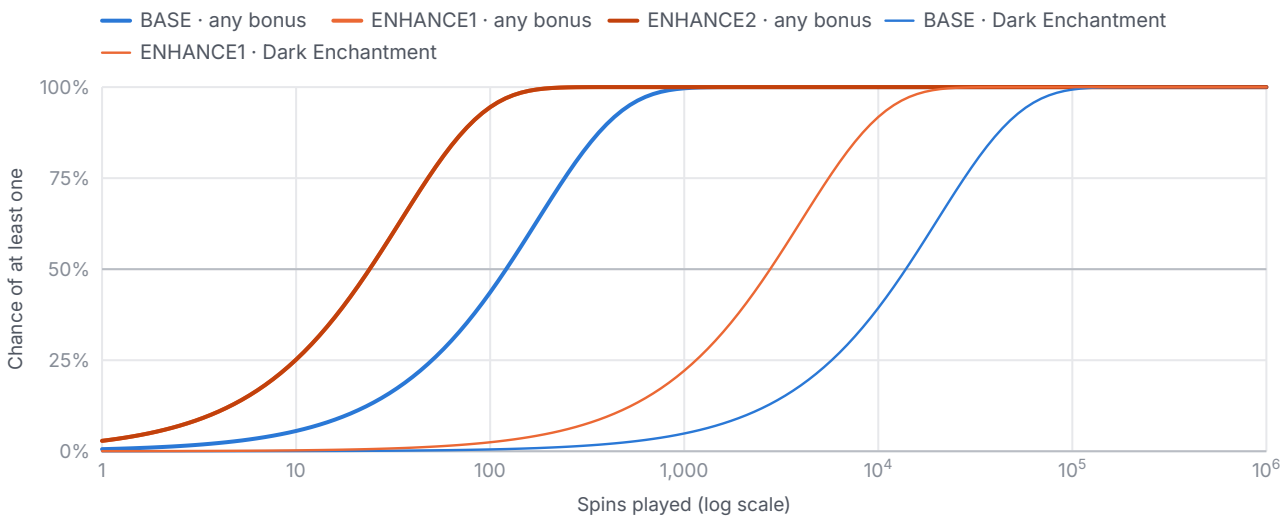
EXACT FORMULA

Mode	Event	Frequency	Median wait (spins)	90% of waits shorter than	P(none in 500 spins)	P(none in 2,000 spins)	Average spend per event
BASE	Any bonus	1 in 175	121	402 spins	5.69%	0.00105%	175×
BASE	Wicked Fortune (3 scatters)	1 in 200	139	460 spins	8.16%	0.00443%	200×
BASE	Grave Spell (4 scatters)	1 in 1,500	1,040	3,453 spins	71.6%	26.3%	1,500×
BASE	Dark Enchantment (5 scatters)	1 in 20,000	13,863	46,051 spins	97.5%	90.5%	20,000×
BASE	Witch unlock	1 in 50.0	35	114 spins	0.00410%	0%	50×
ENHANCE1	Any bonus	1 in 35.0	24	80 spins	5.04×10^{-5}%	0%	105×
ENHANCE1	Wicked Fortune (3 scatters)	1 in 40.0	28	91 spins	0.000318%	0%	120×
ENHANCE1	Grave Spell (4 scatters)	1 in 300	208	690 spins	18.8%	0.126%	900×
ENHANCE1	Dark Enchantment (5 scatters)	1 in 4,000	2,773	9,210 spins	88.2%	60.6%	12,000×
ENHANCE1	Witch unlock	1 in 50.0	35	114 spins	0.00410%	0%	150×
ENHANCE2	Any bonus	1 in 35.0	24	80 spins	5.08×10^{-5}%	0%	525×
ENHANCE2	Grave Spell (4 scatters)	1 in 39.6	28	91 spins	0.000281%	0%	594×
ENHANCE2	Dark Enchantment (5 scatters)	1 in 300	208	690 spins	18.8%	0.126%	4,500×
ENHANCE2	Witch unlock	1 in 50.0	35	114 spins	0.00410%	0%	750×

Average spend per event = price of the spin / probability per spin, in base bets. For comparison, the Wicked Fortune can be bought for 100× and the Grave Spell for 200× (Chapters 10 and 11).

Figure 14.1 Chance of at least one bonus within n spins

EXACT FORMULA



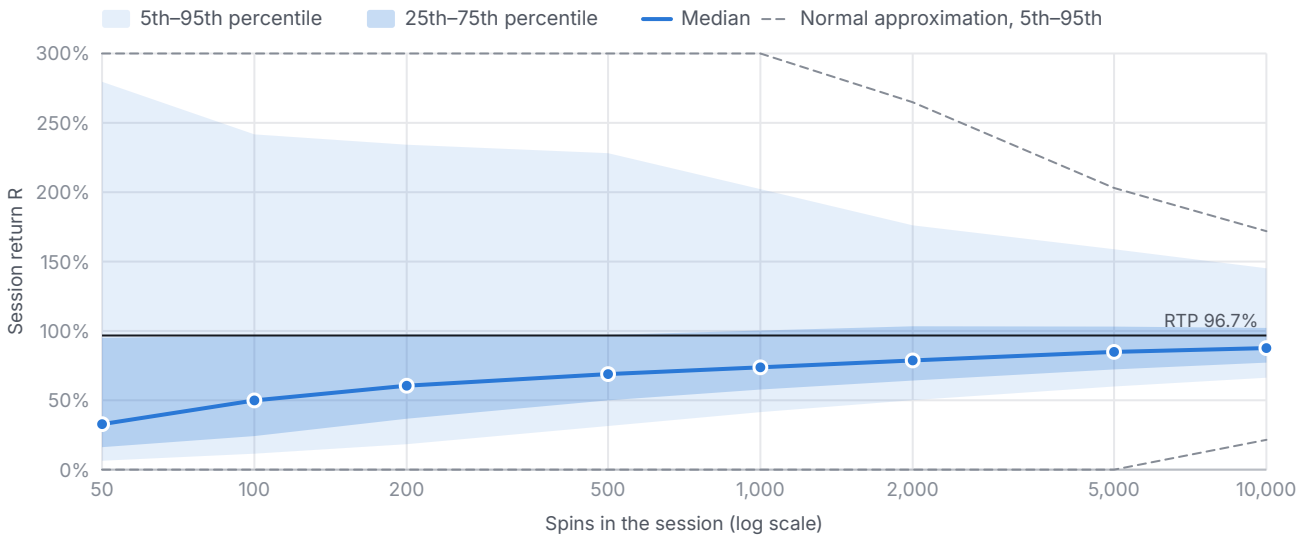
Thick lines: any bonus tier; thin lines: the Dark Enchantment (5 scatters). ENHANCE1 spins cost 3 times and ENHANCE2 spins 15 times as much as BASE spins, so equal numbers of spins are not equal spends.

14.2 Return over a session (Monte Carlo)

A session of N rounds in one mode at a constant stake has a *session return* $R_N = \frac{1}{Nc}\sum_{i=1}^{N} X_i$ (total won divided by total staked). Its expectation is the RTP exactly; its spread shrinks with N but very slowly, because of the heavy right tail of single-round wins. The distribution of R_N has no practical closed form, so it was estimated by Monte Carlo: rounds were drawn independently from the published lookup tables (alias method, fixed seeds), with the number of sessions shown in each table. Intervals are 95% confidence intervals: Wilson intervals for probabilities and order-statistic intervals for percentiles.

Figure 14.2 BASE: distribution of the session return by session length

MONTE CARLO



Shaded bands: simulated 5th–95th and 25th–75th percentiles of R_N; line: simulated median. The median stays well below the RTP because the mean is pulled up by rare large wins. Dashed: the normal (central-limit) approximation of the 5th–95th range, which is unreliable at these lengths — its lower bound is below 0 for short sessions while actual returns cannot be negative.

Table 14.2 BASE: simulated session returns

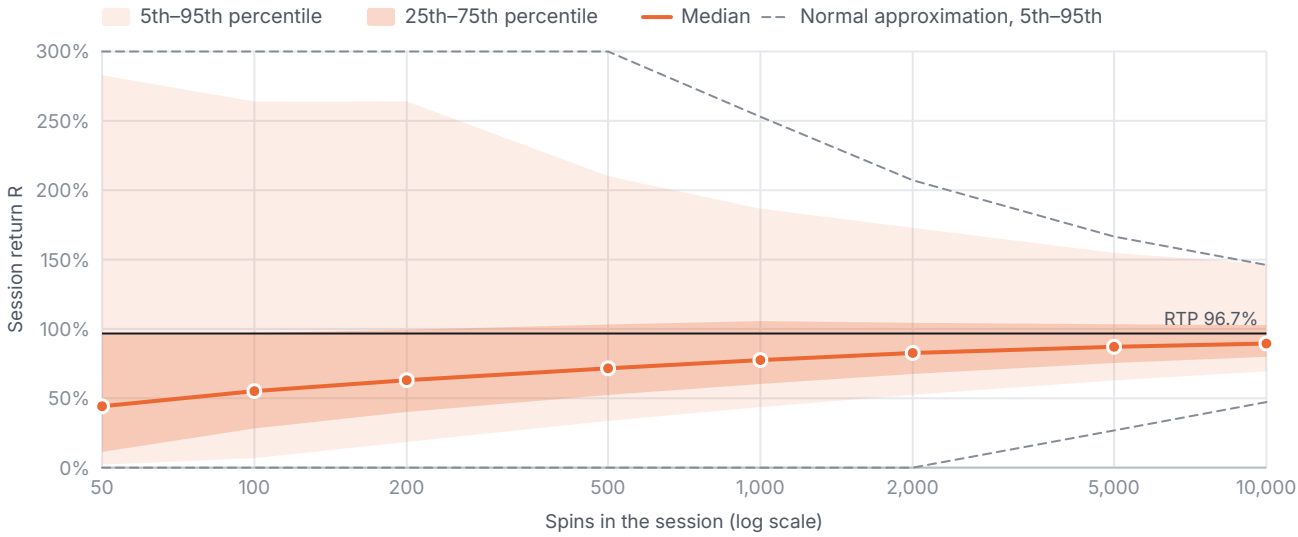
MONTE CARLO

Spins N	Sessions simulated	Mean R [95% CI]	P5	P25	Median [95% CI]	P75	P95	P(R ≥ 100%) [95% CI]	P(R < 50%) [95% CI]
50	1,000,000	97.18% [95.89%, 98.48%]	6.4%	16.2%	32.8% [32.6%, 32.8%]	95.0%	279.6%	23.87% [23.79%, 23.96%]	62.67% [62.57%, 62.76%]
100	1,000,000	96.85% [96.00%, 97.70%]	11.5%	24.2%	49.9% [49.7%, 50.1%]	96.5%	241.7%	23.83% [23.75%, 23.92%]	50.03% [49.94%, 50.13%]
200	1,000,000	96.63% [96.01%, 97.25%]	18.4%	36.7%	60.5% [60.4%, 60.7%]	97.4%	234.2%	23.85% [23.77%, 23.94%]	39.57% [39.47%, 39.66%]
500	400,000	96.37% [95.76%, 96.99%]	31.5%	50.0%	68.9% [68.7%, 69.0%]	97.2%	228.1%	23.45% [23.32%, 23.58%]	25.03% [24.90%, 25.17%]
1,000	400,000	96.53% [96.08%, 96.98%]	41.5%	57.8%	73.8% [73.7%, 73.9%]	100.3%	202.2%	25.15% [25.02%, 25.29%]	13.71% [13.61%, 13.82%]
2,000	200,000	96.65% [96.20%, 97.10%]	50.1%	64.2%	78.7% [78.6%, 78.9%]	103.4%	176.1%	27.33% [27.13%, 27.52%]	4.94% [4.85%, 5.04%]
5,000	100,000	96.80% [96.40%, 97.20%]	59.9%	72.2%	84.9% [84.7%, 85.1%]	103.1%	158.9%	28.26% [27.98%, 28.54%]	0.32% [0.28%, 0.35%]
10,000	100,000	96.49% [96.21%, 96.77%]	66.3%	77.1%	87.6% [87.5%, 87.8%]	102.2%	145.2%	27.73% [27.45%, 28.00%]	0.01% [0.00%, 0.01%]

R = total won / total staked over the session. P(R ≥ 100%): chance of ending the session at or above the amount staked. The exact expectation of R is 96.70% for every N; the simulated means agree with it within their confidence intervals (Chapter 16).

Figure 14.3 ENHANCE1: distribution of the session return by session length

MONTE CARLO



Shaded bands: simulated 5th–95th and 25th–75th percentiles of R_N; line: simulated median. The median stays well below the RTP because the mean is pulled up by rare large wins. Dashed: the normal (central-limit) approximation of the 5th–95th range, which is unreliable at these lengths — its lower bound is below 0 for short sessions while actual returns cannot be negative.

Table 14.3 ENHANCE1: simulated session returns

MONTE CARLO

Spins N	Sessions simulated	Mean R [95% CI]	P5	P25	Median [95% CI]	P75	P95	P(R ≥ 100%) [95% CI]	P(R < 50%) [95% CI]
50	1,000,000	96.38% [95.54%, 97.22%]	2.1%	11.3%	44.3% [44.1%, 44.4%]	96.3%	282.9%	23.83% [23.75%, 23.91%]	53.81% [53.71%, 53.90%]
100	1,000,000	96.56% [95.96%, 97.16%]	6.9%	28.3%	55.2% [55.0%, 55.3%]	97.0%	263.9%	23.88% [23.80%, 23.97%]	45.58% [45.48%, 45.68%]
200	1,000,000	96.75% [96.34%, 97.17%]	18.5%	40.2%	63.1% [62.9%, 63.2%]	99.4%	264.0%	24.76% [24.67%, 24.84%]	36.20% [36.10%, 36.29%]
500	400,000	96.54% [96.13%, 96.96%]	33.7%	52.4%	71.6% [71.5%, 71.7%]	103.2%	210.2%	26.58% [26.45%, 26.72%]	21.77% [21.64%, 21.90%]
1,000	400,000	96.59% [96.30%, 96.89%]	43.7%	60.3%	77.5% [77.4%, 77.7%]	105.7%	186.6%	28.69% [28.55%, 28.83%]	10.91% [10.81%, 11.00%]
2,000	200,000	96.59% [96.30%, 96.89%]	52.5%	67.5%	82.7% [82.5%, 82.8%]	104.4%	172.9%	28.89% [28.70%, 29.09%]	3.24% [3.17%, 3.32%]
5,000	100,000	96.51% [96.26%, 96.77%]	62.8%	75.5%	87.1% [86.9%, 87.3%]	103.4%	154.8%	28.91% [28.63%, 29.20%]	0.12% [0.10%, 0.14%]
10,000	100,000	96.56% [96.37%, 96.74%]	69.4%	79.9%	89.5% [89.3%, 89.6%]	102.8%	146.7%	28.93% [28.65%, 29.21%]	0.00% [0.00%, 0.01%]

R = total won / total staked over the session. P(R ≥ 100%): chance of ending the session at or above the amount staked. The exact expectation of R is 96.70% for every N; the simulated means agree with it within their confidence intervals (Chapter 16).

Table 14.4 ENHANCE2 and purchase modes: simulated returns over series of rounds

MONTE CARLO

Mode	Rounds N	Series simulated	Mean R	P5	P25	Median	P75	P95	P(R ≥ 100%) [95% CI]	P(R < 50%) [95% CI]
ENHANCE2	5	1,000,000	96.63%	0.0%	0.1%	0.4%	3.5%	476.5%	13.08% [13.02%, 13.15%]	85.57% [85.50%, 85.64%]
ENHANCE2	10	1,000,000	96.49%	0.0%	0.3%	1.4%	40.8%	451.5%	19.56% [19.48%, 19.64%]	75.50% [75.41%, 75.58%]
ENHANCE2	25	1,000,000	97.07%	0.3%	1.3%	23.6%	101.1%	422.1%	25.23% [25.14%, 25.31%]	60.94% [60.85%, 61.04%]
ENHANCE2	50	1,000,000	96.68%	0.8%	12.9%	48.4%	114.8%	329.6%	28.28% [28.20%, 28.37%]	50.98% [50.89%, 51.08%]
ENHANCE2	100	1,000,000	96.46%	4.8%	30.6%	63.5%	125.0%	270.6%	33.67% [33.57%, 33.76%]	41.09% [40.99%, 41.19%]
HEXED	5	1,000,000	96.96%	0.0%	0.0%	19.2%	95.7%	301.1%	24.07% [23.99%, 24.16%]	63.73% [63.64%, 63.83%]
HEXED	10	1,000,000	96.63%	0.0%	10.9%	43.7%	99.6%	249.9%	24.87% [24.78%, 24.95%]	53.45% [53.35%, 53.54%]
HEXED	25	1,000,000	96.49%	8.7%	32.3%	58.9%	97.1%	221.3%	23.74% [23.66%, 23.82%]	42.10% [42.00%, 42.20%]

Mode	Rounds N	Series simulated	Mean R	P5	P25	Median	P75	P95	P(R ≥ 100%) [95% CI]	P(R < 50%) [95% CI]
HEXED	50	1,000,000	96.51%	21.5%	43.5%	65.0%	95.5%	252.8%	22.57% [22.49%, 22.66%]	32.66% [32.57%, 32.75%]
HEXED	100	1,000,000	96.77%	33.3%	52.1%	69.7%	96.0%	298.1%	22.76% [22.68%, 22.84%]	22.12% [22.04%, 22.20%]
CURSED	5	1,000,000	96.49%	10.2%	27.6%	49.8%	90.4%	317.3%	21.89% [21.81%, 21.97%]	50.21% [50.12%, 50.31%]
CURSED	10	1,000,000	97.03%	19.6%	37.6%	58.7%	99.9%	301.1%	24.96% [24.88%, 25.05%]	40.57% [40.47%, 40.66%]
CURSED	25	1,000,000	96.64%	32.3%	49.8%	70.6%	109.5%	257.1%	29.34% [29.26%, 29.43%]	25.29% [25.20%, 25.38%]
CURSED	50	1,000,000	96.87%	41.7%	59.1%	79.1%	114.2%	224.0%	32.54% [32.45%, 32.63%]	13.23% [13.16%, 13.30%]
CURSED	100	1,000,000	96.77%	51.0%	67.7%	86.0%	115.0%	178.7%	35.73% [35.63%, 35.82%]	4.29% [4.25%, 4.33%]
WICKED	5	1,000,000	96.51%	25.8%	43.9%	64.1%	96.4%	211.4%	23.29% [23.21%, 23.37%]	32.92% [32.83%, 33.01%]
WICKED	10	1,000,000	96.79%	35.5%	52.5%	70.1%	98.5%	184.3%	24.16% [24.08%, 24.25%]	21.41% [21.33%, 21.49%]
WICKED	25	1,000,000	96.68%	47.3%	62.0%	76.7%	98.0%	160.7%	23.44% [23.36%, 23.53%]	7.53% [7.48%, 7.58%]
WICKED	50	1,000,000	96.74%	55.1%	67.9%	80.1%	96.9%	158.1%	21.95% [21.87%, 22.04%]	1.65% [1.63%, 1.68%]
WICKED	100	1,000,000	96.63%	61.8%	72.7%	82.5%	96.3%	177.7%	20.93% [20.85%, 21.01%]	0.10% [0.09%, 0.11%]
GRAVE	5	1,000,000	96.88%	24.7%	45.8%	69.3%	104.9%	212.1%	27.49% [27.40%, 27.58%]	29.69% [29.60%, 29.78%]
GRAVE	10	1,000,000	96.74%	36.1%	55.7%	75.4%	103.7%	195.8%	27.41% [27.32%, 27.50%]	17.93% [17.86%, 18.01%]
GRAVE	25	1,000,000	96.71%	49.5%	65.8%	80.9%	102.5%	198.6%	27.03% [26.95%, 27.12%]	5.33% [5.29%, 5.38%]
GRAVE	50	1,000,000	96.63%	58.1%	71.7%	84.2%	103.0%	175.4%	27.72% [27.63%, 27.81%]	0.90% [0.88%, 0.91%]
GRAVE	100	1,000,000	96.68%	65.3%	76.6%	87.4%	104.9%	156.3%	30.00% [29.91%, 30.09%]	0.03% [0.03%, 0.04%]
MYSTERY	5	1,000,000	96.71%	0.0%	6.3%	38.8%	98.0%	425.3%	24.56% [24.47%, 24.64%]	56.95% [56.85%, 57.05%]
MYSTERY	10	1,000,000	96.71%	2.9%	23.1%	51.8%	109.3%	353.4%	27.67% [27.59%, 27.76%]	48.82% [48.73%, 48.92%]

Mode	Rounds N	Series simulated	Mean R	P5	P25	Median	P75	P95	P(R ≥ 100%) [95% CI]	P(R < 50%) [95% CI]
MYSTERY	25	1,000,000	96.62%	17.1%	40.3%	68.9%	123.8%	265.5%	33.11% [33.02%, 33.20%]	34.28% [34.19%, 34.37%]
MYSTERY	50	1,000,000	96.59%	29.6%	53.0%	80.7%	122.4%	224.2%	36.68% [36.58%, 36.77%]	21.99% [21.91%, 22.07%]
MYSTERY	100	1,000,000	96.67%	42.2%	65.0%	88.0%	119.0%	181.4%	38.68% [38.59%, 38.78%]	10.39% [10.33%, 10.45%]

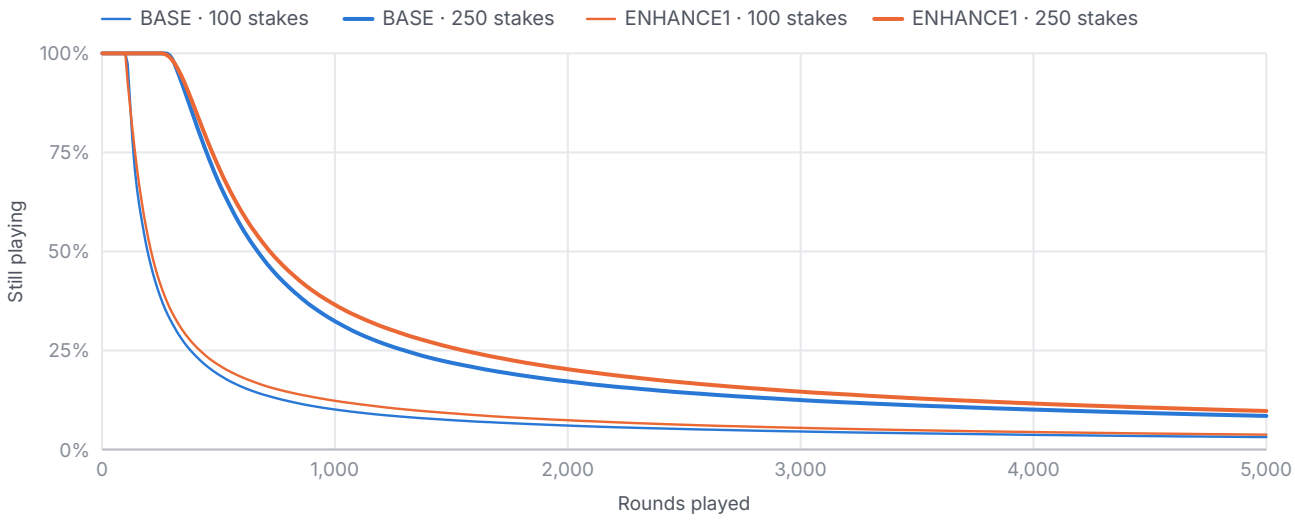
Same method as above, for series of N rounds at the price of the mode (ENHANCE2: 15× per spin). The 95% confidence interval of the simulated mean R is within ±1.12 percentage points in every row.

14.3 How long does a bankroll last? (Monte Carlo)

A player starts with a bankroll of B stakes of the mode, plays one round per stake, adds every win to the bankroll and stops when it falls below one stake, or after 5,000 rounds. The figure gives the share of players still playing after k rounds.

Figure 14.4 Share of players still playing after k rounds

MONTE CARLO



200,000 simulated players per curve. Thin lines: bankroll of 100 stakes; thick lines: 250 stakes. A bankroll lasts at least B rounds, since every round costs one stake and the bankroll can only grow from wins.

Table 14.5 Bankroll survival

MONTE CARLO

Mode	Bankroll	After 100	After 250	After 500	After 1,000	After 2,000	After 3,000	After 5,000	95% CI at 5,000
BASE	100 stakes (100×)	100.0%	38.6%	18.9%	10.1%	6.0%	4.6%	3.1%	[3.07%, 3.22%]
BASE	250 stakes (250×)	100.0%	100.0%	67.6%	32.4%	17.2%	12.5%	8.5%	[8.36%, 8.60%]
ENHANCE1	100 stakes (300×)	99.9%	41.8%	21.4%	12.3%	7.4%	5.5%	3.8%	[3.67%, 3.84%]
ENHANCE1	250 stakes (750×)	100.0%	100.0%	71.4%	36.5%	20.3%	14.6%	9.7%	[9.60%, 9.86%]

Share of simulated players with at least one stake left after k rounds (200,000 players per row, fixed seed).

CHAPTER 15

Maximum win and win cap

Frequency, origin and value of the 50,000× cap.

15.1 Probability and origin of the max win

Table 15.1 Max win (50,000×) per round, by mode and by origin EXACT

Mode	P(max win)	Frequency	From Base-game wins	From Witch	From Feature spin	From Wicked Fortune	From Grave Spell	From Dark Enchantment	Of which 4 mirrors
BASE	1.60×10^{-5}%	1 in 6,243,175	0.4%	30.2%	—	31.2%	6.9%	31.2%	8.0%
ENHANCE1	4.47×10^{-5}%	1 in 2,236,069	0.1%	10.8%	—	58.0%	12.4%	18.6%	4.9%
ENHANCE2	0.0003803%	1 in 262,948	0.0%	1.3%	—	—	11.1%	87.6%	19.2%
HEXED	0.006667%	1 in 15,000	—	—	100.0%	—	—	—	3.4%
CURSED	0.03226%	1 in 3,100	—	—	100.0%	—	—	—	19.8%
WICKED	0.002457%	1 in 40,700	—	—	—	100.0%	—	—	0.6%
GRAVE	0.002237%	1 in 44,700	—	—	—	—	100.0%	—	0.7%
MYSTERY	0.03243%	1 in 3,083	—	—	—	—	3.2%	96.8%	17.5%

Origin: share of the mode's max wins produced by each outcome class. Of which 4 mirrors: share of the mode's max wins given directly by four mirrors (all classes). Base-game wins without the witch reach the cap only in BASE, ENHANCE1, ENHANCE2 (0.449% of the BASE max wins, 0.146% of the ENHANCE1 max wins, 0.0165% of the ENHANCE2 max wins).

15.2 Value carried by the cap and effect of the truncation

When a round reaches 50,000× it stops and pays exactly the cap; the remaining free spins are not played. The uncapped value of such a round is therefore not defined by the published data. What can be measured is the total of the way wins already accumulated in each capped round when it stopped, each recomputed from the book before the cap (pay per way × multiplied ways): it exceeds 50,000× by the amount the cap removed. For a one-spin round (base game, HEXED, CURSED) this amount is exact; for a bonus it is a *lower bound*, since the spins not played are not counted. A max win from four mirrors removes nothing.

Table 15.2 RTP carried by max wins and measured truncation EXACT

Mode	RTP points paid at the cap	Share of the RTP	Mean amount removed by the cap	RTP points removed (lower bound)
BASE	0.801	0.83%	277,909×	4.451
ENHANCE1	0.745	0.77%	255,879×	3.814
ENHANCE2	1.268	1.31%	201,050×	5.097
HEXED	3.333	3.45%	460,985×	30.732
CURSED	3.226	3.34%	378,176×	24.398
WICKED	1.229	1.27%	268,039×	6.586
GRAVE	0.559	0.58%	266,109×	2.977
MYSTERY	3.243	3.35%	204,462×	13.263

Amount removed: uncapped way wins accumulated by the round minus 50,000×, averaged over capped rounds (weighted). Without a cap, the same books would pay at least this much more; for bonuses the true uncapped value would also include the spins not played. The cap is part of the design: the lookup tables were built with it.

15.3 Waiting for a max win

Table 15.3 Rounds needed to see a max win

EXACT FORMULA

Mode	Frequency	Median wait (rounds)	At least one in 10,000 rounds	in 100,000	in 1,000,000	Average spend per max win
BASE	1 in 6,243,175	4,327,439	0.160%	1.59%	14.8%	6,243,175×
ENHANCE1	1 in 2,236,069	1,549,925	0.446%	4.37%	36.1%	6,708,206×
ENHANCE2	1 in 262,948	182,262	3.73%	31.6%	97.8%	3,944,221×
HEXED	1 in 15,000	10,397	48.7%	99.9%	100%	1,500,000×
CURSED	1 in 3,100	2,149	96.0%	100%	100%	1,550,000×
WICKED	1 in 40,700	28,211	21.8%	91.4%	100%	4,070,000×
GRAVE	1 in 44,700	30,984	20.0%	89.3%	100%	8,940,000×
MYSTERY	1 in 3,083	2,137	96.1%	100%	100%	1,541,552×

Average spend per max win = price / P(max win), in base bets. Independent rounds.

CHAPTER 16

Verification and consistency

Checks on the published files, on the computations and against the specification.

16.1 Integrity of the published files

Table 16.1 File-level checks

EXACT

Mode	Books read	Books declared	Lookup rows	ID mismatches	Payout mismatches	Zero weights	Event inconsistencies	SHA-256 vs declared
BASE	1,266,915	1,266,915	1,266,915	0	0	0	0	match
ENHANCE1	1,266,729	1,266,729	1,266,729	0	0	0	0	match
ENHANCE2	984,947	984,947	984,947	0	0	0	0	match
HEXED	493,605	493,605	493,605	0	0	0	0	match
CURSED	495,382	495,382	495,382	0	0	0	0	match
WICKED	499,551	499,551	499,551	0	0	0	0	match
GRAVE	499,553	499,553	499,553	0	0	0	0	match
MYSTERY	599,345	599,345	599,345	0	0	0	0	match

Books declared: count in the verification file used by the RGS. ID and payout mismatches: book i of the book file compared with row i of the lookup table. Event inconsistencies: final-win event different from the payout, or max-win event inconsistent with the payout. SHA-256 of each book file compared with the hash declared in its verification file.

16.2 Re-evaluation of every round

A second program, separate from the game engine (it shares no code with it, enumerates every combination of cells of every way explicitly in integer arithmetic — the engine uses a closed formula — and takes the rules from the brief, not from the engine configuration), replays every published book and checks it against the rules:

- boards: 4-2-4-2-4; at most one scatter per reel; no wild on the reels; never three scatters on a paid spin without a bonus; no scatter in HEXED and CURSED; Mystery losing spins: two scatters on reels 1–2 and no win;
- special squares: the four squares in order, locked or unlocked as the context requires; at least 1 special symbol on every Grave Spell free spin, 3 on every Dark Enchantment free spin, 2 with the witch and in HEXED, 4 in CURSED; no special symbol on the spin that starts a bought or Mystery bonus;
- effects, in order: four mirrors give exactly the max win and end the round; spellbook wilds (×1, distinct cells, never a scatter nor a wild) or, with four spellbooks, two wild nudges (2 of reels 1, 3, 5 without a scatter, 1 to 3 steps, multiplier = steps, whole reel wild); orb wilds (never on a scatter or a multiplied wild, multiplier in the brief's list, ×20 or more with four orbs); board split with any broomstick, a premium-only paying board with four;
- ways: every win recomputed (symbol, reels, ways, multiplied ways with the wild multipliers added, pay per way, amount, positions) and every spin total; a way made only of wilds never pays;
- bonuses: tier = number of scatters (the witch's included), 10 / 10 / 8 free spins, every scatter of a free spin adds exactly one spin, never more than 40 spins;
- amounts: running total, bonus total and final win consistent; final win equal to the payout, never above the cap; floors and ceilings of the simulation pools; Dark Enchantment never below 1,000×.

Table 16.2 Result of the re-evaluation (run on 12 October 2026) EXACT

Mode	Books checked	Defects
BASE	1,266,915	0
ENHANCE1	1,266,729	0
ENHANCE2	984,947	0
HEXED	493,605	0
CURSED	495,382	0
WICKED	499,551	0
GRAVE	499,553	0
MYSTERY	599,345	0

Output of `verify --lib library --threads 10`: “OK : 0 defaut”. The verifier was itself validated on 18 deliberate corruptions of different types (way amount, multiplied ways, spellbook wild on a scatter, orb multiplier outside the list, four orbs below ×20, premium board without four broomsticks, Grave Spell free spin without a special symbol, retrigger count, mirror max win, final win, wild on the reels, wild nudge of 4 steps, special symbol on the spin starting a bought bonus, unlocked squares in the base game, paying Mystery losing spin, wrong number of free spins, CURSED with fewer than 4 special symbols, spellbook without its wilds): all 18 were detected, no sound book was flagged.

16.3 Closure of the decompositions

The same RTP is obtained by summing the published payouts, the outcome classes, the payout bands, or the base-game and free-spin parts. The table gives each sum in RTP points; differences are rounding of the double-precision conversions only.

Table 16.3 RTP points obtained by four separate summations EXACT

Mode	Payouts × weights	Σ classes	Σ bands	Base + free spins	Largest difference
BASE	96.7000000	96.7000000	96.7000000	96.7000000	$< 10^{-6}$
ENHANCE1	96.7000000	96.7000000	96.7000000	96.7000000	$< 10^{-6}$
ENHANCE2	96.7000000	96.7000000	96.7000000	96.7000000	$< 10^{-6}$
HEXED	96.7000000	96.7000000	96.7000000	96.7000000	$< 10^{-6}$
CURSED	96.7000000	96.7000000	96.7000000	96.7000000	$< 10^{-6}$
WICKED	96.7000000	96.7000000	96.7000000	96.7000000	$< 10^{-6}$
GRAVE	96.7000000	96.7000000	96.7000000	96.7000000	$< 10^{-6}$
MYSTERY	96.7000000	96.7000000	96.7000000	96.7000000	$< 10^{-6}$

Payouts × weights is the reference: $\Sigma\, w_i\, x_i\, /\, (W\, c)$ computed in integer arithmetic.

16.4 Cross-check with the engine’s summary statistics

The publishing pipeline writes its own summary of each lookup table (`library/stats_summary.json`). Its standard deviation, max-win frequency and hit rate agree with this document to every digit it reports.

Table 16.4 Engine statistics file against this document: spread, max win, hit frequency EXACT

Mode	SD (engine)	SD (this document)	Max win 1 in (engine)	Max win 1 in (this document)	Any win 1 in (engine)	Any win 1 in (this document)
BASE	45.722	45.722	6,243,174.883	6,243,174.883	4.545	4.545
ENHANCE1	90.134	90.134	2,236,068.760	2,236,068.760	6.457	6.457
ENHANCE2	197.992	197.992	262,948.086	262,948.086	4.118	4.118
HEXED	954.776	954.776	15,000.000	15,000.000	5.000	5.000
CURSED	2,035.095	2,035.095	3,100.000	3,100.000	1.429	1.429
WICKED	608.919	608.919	40,700.000	40,700.000	1.000	1.000
GRAVE	703.794	703.794	44,700.000	44,700.000	1.000	1.000
MYSTERY	2,188.015	2,188.015	3,083.105	3,083.105	2.857	2.857

Both sources at the engine's precision (three decimals). SD in base bets.

For skewness and kurtosis the engine file agrees with this document for BASE (price 1) but not for the other modes, where its values are larger by exactly the price cubed and the price to the fourth power. Skewness and kurtosis are dimensionless and cannot depend on the unit of the wins; the engine file standardises the third and fourth moments with a standard deviation expressed per unit of price instead of in base bets. Rescaled, its values coincide with this document's.

Table 16.5 Skewness and excess kurtosis: engine file, engine file rescaled, this document EXACT

Mode	Price c	Skewness (engine)	Engine ÷ c^3	Skewness (this document)	Kurtosis (engine)	(Engine + 3) ÷ c^4 − 3	Kurtosis (this document)
BASE	1×	702.4	702.40	702.40	664,865	664,865	664,865
ENHANCE1	3×	8,449.8	312.96	312.96	11,251,770	138,908	138,908
ENHANCE2	15×	342,623.5	101.52	101.52	1,037,083,165	20,483	20,483
HEXED	100×	35,771,010.8	35.77	35.77	152,475,174,334	1,522	1,522
CURSED	500×	1,800,935,216.5	14.41	14.41	17,145,273,399,470	271	271
WICKED	100×	52,487,007.8	52.49	52.49	341,246,453,219	3,409	3,409
GRAVE	200×	263,463,155.3	32.93	32.93	2,705,027,750,454	1,688	1,688
MYSTERY	500×	1,465,812,330.8	11.73	11.73	12,244,780,841,163	193	193

Excess kurtosis = fourth standardised moment − 3. Rescaled engine values are given to the precision of the engine file (3 decimals on the raw value).

16.5 Monte Carlo consistency

The session simulations can be checked against exact values. (1) The expected session return is exactly the RTP: the 95% confidence interval of the simulated mean contains it in 44 of the 46 simulated configurations (about 95% expected by construction of the intervals). (2) The probability of at least one large win in a session is exactly $1 - (1 - p)^N$. The table compares it with the simulated proportion through the standard score z = (simulated − exact) / √(exact (1 − exact) / sessions): $|z| \le 1.96$ in 44 of 46 configurations (about 95% expected under pure sampling noise) and the largest $|z|$ is 2.30, so the simulations show no bias.

Table 16.6 Exact and simulated chance of at least one large win in a session MONTE CARLO

Mode	Rounds N	Sessions	Large win ≥	Exact	Simulated	z
BASE	50	1,000,000	100×	5.6705%	5.6897%	0.83
BASE	100	1,000,000	100×	11.019%	11.047%	0.88
BASE	200	1,000,000	100×	20.825%	20.854%	0.73
BASE	500	400,000	100×	44.220%	44.148%	-0.92
BASE	1,000	400,000	100×	68.886%	68.950%	0.86

Mode	Rounds N	Sessions	Large win ≥	Exact	Simulated	z
BASE	2,000	200,000	100×	90.319%	90.380%	0.91
BASE	5,000	100,000	100×	99.708%	99.721%	0.74
BASE	10,000	100,000	100×	99.999%	100%	0.92
ENHANCE1	50	1,000,000	300×	5.3394%	5.3311%	-0.37
ENHANCE1	100	1,000,000	300×	10.394%	10.380%	-0.44
ENHANCE1	200	1,000,000	300×	19.707%	19.709%	0.06
ENHANCE1	500	400,000	300×	42.231%	42.219%	-0.16
ENHANCE1	1,000	400,000	300×	66.628%	66.456%	-2.30
ENHANCE1	2,000	200,000	300×	88.863%	88.777%	-1.23
ENHANCE1	5,000	100,000	300×	99.586%	99.617%	1.52
ENHANCE1	10,000	100,000	300×	99.998%	99.998%	-0.22
ENHANCE2	5	1,000,000	150×	10.050%	10.040%	-0.34
ENHANCE2	10	1,000,000	150×	19.090%	19.066%	-0.61
ENHANCE2	25	1,000,000	150×	41.115%	41.165%	1.01
ENHANCE2	50	1,000,000	150×	65.326%	65.365%	0.83
ENHANCE2	100	1,000,000	150×	87.977%	87.924%	-1.61
HEXED	5	1,000,000	1,000×	7.2672%	7.2628%	-0.17
HEXED	10	1,000,000	1,000×	14.006%	14.023%	0.49
HEXED	25	1,000,000	1,000×	31.425%	31.386%	-0.84
HEXED	50	1,000,000	1,000×	52.975%	52.993%	0.36
HEXED	100	1,000,000	1,000×	77.886%	77.908%	0.53
CURSED	5	1,000,000	5,000×	7.1477%	7.1406%	-0.28
CURSED	10	1,000,000	5,000×	13.785%	13.847%	1.80
CURSED	25	1,000,000	5,000×	30.982%	30.962%	-0.43
CURSED	50	1,000,000	5,000×	52.365%	52.469%	2.08
CURSED	100	1,000,000	5,000×	77.309%	77.364%	1.32
WICKED	5	1,000,000	1,000×	2.8294%	2.8022%	-1.64
WICKED	10	1,000,000	1,000×	5.5787%	5.5874%	0.38
WICKED	25	1,000,000	1,000×	13.369%	13.369%	0.00
WICKED	50	1,000,000	1,000×	24.950%	24.898%	-1.21
WICKED	100	1,000,000	1,000×	43.675%	43.679%	0.07
GRAVE	5	1,000,000	2,000×	2.8655%	2.8692%	0.22
GRAVE	10	1,000,000	2,000×	5.6489%	5.6653%	0.71
GRAVE	25	1,000,000	2,000×	13.530%	13.549%	0.57
GRAVE	50	1,000,000	2,000×	25.229%	25.156%	-1.66
GRAVE	100	1,000,000	2,000×	44.093%	44.038%	-1.10
MYSTERY	5	1,000,000	5,000×	8.7123%	8.7191%	0.24
MYSTERY	10	1,000,000	5,000×	16.666%	16.668%	0.08
MYSTERY	25	1,000,000	5,000×	36.604%	36.600%	-0.09
MYSTERY	50	1,000,000	5,000×	59.810%	59.724%	-1.74
MYSTERY	100	1,000,000	5,000×	83.847%	83.864%	0.44

Large win: 100× the price for BASE and ENHANCE1, 10× the price for ENHANCE2 and the purchase modes.

16.6 Compliance with the design specification

Table 16.7 Specification against the published model

SPECIFICATION

Item	Specification	Published (measured)	Status
RTP of every mode	96.70%, never above	96.7000% / 96.7000% / 96.7000% / 96.7000% / 96.7000% / 96.7000% / 96.7000% / 96.7000%	met
Maximum win	50,000× in every mode	50,000× / 50,000× / 50,000× / 50,000× / 50,000× / 50,000× / 50,000× / 50,000×	met
BASE bonus frequencies	3S 1/200 · 4S 1/1,500 · 5S 1/20,000	1 in 200 · 1 in 1,500 · 1 in 20,000	met
ENHANCE1 bonus frequencies	5 × BASE for each tier	×5.0000 · ×5.0000 · ×5.0000	met
ENHANCE2	only 4+ scatters; a bonus about 1 in 35; Dark Enchantment 1 in 300	3 scatters never; any bonus 1 in 35.0 (1 in 39.6 · 1 in 300)	met
BASE split base game / bonuses	≈ 50 / 50	50.0% / 50.0%	met
Tiers ordered	more scatters: rarer and richer on average	56× < 155× < 1,965×	met
BASE hit rate	≈ 22%	22.00%	met
Witch (BASE)	unlocks the squares 1 in 50 spins; drops the scatters of about 30% of the triggered bonuses	1 in 50.0; 30.1% of the bonuses	met
Triggered Wicked Fortune / Grave Spell medians	same as the buys	43.10× vs 43.70× · 85.50× vs 86.80×	met
Dark Enchantment (5 scatters)	never below 1,000× (all modes), re-simulated, never topped up; median 1,400–1,600× in the triggered modes	minimum 1,000× (BASE), 1,000× (ENHANCE1), 1,000× (ENHANCE2), 1,000× (MYSTERY); median 1,460.1×	met
Dark Enchantment spread	no pile-up at the floor: 5th percentile ≥ 1,040×; middle half wide (P75 / P25 ≥ 1.4)	BASE 1,046× · ENHANCE1 1,047× · ENHANCE2 1,047× · MYSTERY 1,071×; P75 / P25 1.49	met
GRAVE buy pays its price back	about 1 in 4 (25%)	1 in 3.99 (25.0%)	met
Lowest possible win (not shown to players)	WICKED 5×, GRAVE 10×; Mystery Grave Spell ≥ 25×; triggered tiers as their buys	5×, 10×; 25×; 5× / 10×	met
HEXED / CURSED shape	HEXED 80%, CURSED 30% of spins pay exactly 0; CURSED median at least a quarter of the price; at least 10.5% pay the price back	HEXED: 80.00% pay 0, median 0×, 12.3% ≥ price; CURSED: 30.00% pay 0, median 125.6×, 17.5% ≥ price	met
Volatility (brief)	ENHANCE1 3.5/5 · ENHANCE2 4.5/5 · HEXED 5/5 · CURSED 4/5 · WICKED 5/5 · GRAVE 4/5 · MYSTERY 5/5 (targets on SD / price, Chapter 4)	ENHANCE1 30.04 · ENHANCE2 13.20 · HEXED 9.55 · CURSED 4.07 · WICKED 6.09 · GRAVE 3.52 · MYSTERY 4.38	met
Free-spin retrigger	1 scatter = +1 free spin, at most 40 spins	longest bonus 20 spins	met
Mystery mix	65% / 25% / 10%	65.00% / 25.00% / 10.00%	met
Max-win frequencies of the purchases	WICKED 1 in 40,700 · GRAVE 1 in 44,700 · HEXED 1 in 15,000 · CURSED 1 in 3,100	1 in 40,700 · 1 in 44,700 · 1 in 15,000 · 1 in 3,100	met
Large wins without a bonus (BASE)	≥ 100× at most 1 in 8,000 spins	1 in 8,000	met
Round totals	no multiple of 100× between 100× and 49,900×	0 lookup rows (all modes)	met

Item	Specification	Published (measured)	Status
Pools	≥ 100,000 books per bonus / feature; ≥ 10,000 max-win books	smallest pool 126,729 books; max-win books 19,012 / 19,031 / 13,704 / 10,290 / 12,667 / 12,705 / 13,062 / 14,082	met
Rule compliance	every book follows the rules	0 defects (re-evaluation of every round)	met

Specification: game math contract and recorded design decisions of 12 October 2026. Measured values are exact (published lookup tables).

APPENDIX A

Conditional distributions by outcome class

Payout bands within each outcome class, and their RTP points.

The four single-outcome modes (HEXED, CURSED, WICKED, GRAVE) are not repeated here: their band tables in Chapters 8 to 11 are already conditional on their only outcome.

A.1 BASE: probability of each band within each class

Table A.1 BASE: P(band | class) EXACT

Round win (×)	Base-game wins	Witch	Wicked Fortune	Grave Spell	Dark Enchantment
0 (no win)		7.143%			
0.1–0.4×	68.39%				
0.5–0.9×	5.035%				
1–1.9×	20.09%	17.72%			
2–2.9×	2.305%	4.778%			
3–4.9×	0.3618%	25.53%			
5–9.9×	1.156%	38.69%	15.04%		
10–19.9×	1.156%	5.173%	15.04%	15.95%	
20–49.9×	1.156%	0.8617%	27.04%	21.08%	
50–99.9×	0.2909%	0.08819%	28.90%	19.00%	
100–199.9×	0.009965%	0.01749%	11.77%	25.73%	
200–499.9×	0.004204%	0.002897%	1.957%	14.91%	
500–999.9×	0.04254%	0.0002960%	0.2002%	2.403%	
1,000–1,999.9×	0.002749%	5.80×10^{-5}%	0.03959%	0.4764%	79.13%
2,000–4,999.9×	0.001552%	9.65×10^{-6}%	0.006545%	0.2411%	16.43%
5,000–9,999.9×	0.0002938%	9.93×10^{-7}%	0.0006654%	0.1650%	3.524%
10,000–19,999.9×	0.0001493%	1.98×10^{-7}%	0.0001317%	0.03942%	0.7033%
20,000–49,999.9×	0.0002730%	4.13×10^{-8}%	2.21×10^{-5}%	0.006557%	0.1171%
50,000 (max)	3.68×10^{-7}%	0.0002417%	0.001000%	0.001667%	0.10000%

Each column sums to 100% (probabilities conditional on the class).

Table A.2 BASE: RTP points by band and class

EXACT

Round win (×)	Base-game wins	Witch	Wicked Fortune	Grave Spell	Dark Enchantment	All classes
0.1–0.4×	2.332					2.332
0.5–0.9×	0.691					0.691
1–1.9×	5.284	0.472				5.756
2–2.9×	1.052	0.235				1.286
3–4.9×	0.277	2.141				2.419
5–9.9×	1.434	4.881	0.541			6.856
10–19.9×	3.506	1.356	1.083	0.153		6.099
20–49.9×	8.987	0.481	4.764	0.460		14.692
50–99.9×	3.606	0.119	10.049	0.951		14.724
100–199.9×	0.239	0.046	7.711	2.425		10.420
200–499.9×	0.338	0.016	2.738	3.020		6.112
500–999.9×	5.781	0.004	0.674	1.079		7.538
1,000–1,999.9×	0.649	0.002	0.260	0.416	5.441	6.768
2,000–4,999.9×	1.162	0.001	0.092	0.526	2.323	4.103
5,000–9,999.9×	0.396	0.000	0.023	0.772	1.188	2.379
10,000–19,999.9×	0.443	0.000	0.009	0.347	0.461	1.260
20,000–49,999.9×	2.175	0.000	0.003	0.123	0.164	2.465
50,000 (max)	0.004	0.242	0.250	0.056	0.250	0.801
Total	**38.355**	**9.995**	**28.196**	**10.327**	**9.826**	**96.700**

A.2 ENHANCE1: probability of each band within each class

Table A.3 ENHANCE1: P(band | class)

EXACT

Round win (×)	Base-game wins	Witch	Wicked Fortune	Grave Spell	Dark Enchantment
0 (no win)		7.143%			
0.1–0.4×	71.14%				
0.5–0.9×	4.532%				
1–1.9×	16.62%	17.72%			
2–2.9×	1.768%	4.818%			
3–4.9×	5.306%	25.49%			
5–9.9×	0.3336%	38.69%	15.04%		
10–19.9×	0.007287%	5.173%	15.04%	15.95%	
20–49.9×	0.0001320%	0.8617%	27.04%	21.08%	
50–99.9×	1.97×10^{-5}%	0.08819%	28.90%	19.00%	
100–199.9×	0.0009306%	0.01749%	11.77%	25.73%	
200–499.9×	0.1206%	0.002897%	1.957%	14.91%	
500–999.9×	0.1520%	0.0002960%	0.2002%	2.403%	
1,000–1,999.9×	0.006026%	5.80×10^{-5}%	0.03959%	0.4764%	79.18%
2,000–4,999.9×	0.009117%	9.65×10^{-6}%	0.006545%	0.2411%	16.44%
5,000–9,999.9×	0.001358%	9.93×10^{-7}%	0.0006654%	0.1650%	3.526%
10,000–19,999.9×	0.001594%	1.98×10^{-7}%	0.0001317%	0.03942%	0.7038%
20,000–49,999.9×	0.002634%	4.14×10^{-8}%	2.21×10^{-5}%	0.006557%	0.1172%
50,000 (max)	6.06×10^{-7}%	0.0002421%	0.001037%	0.001667%	0.03333%

Each column sums to 100% (probabilities conditional on the class).

Table A.4 ENHANCE1: RTP points by band and class EXACT

Round win (×)	Base-game wins	Witch	Wicked Fortune	Grave Spell	Dark Enchantment	All classes
0.1–0.4×	0.443					0.443
0.5–0.9×	0.111					0.111
1–1.9×	0.802	0.157				0.959
2–2.9×	0.159	0.079				0.238
3–4.9×	0.689	0.713				1.402
5–9.9×	0.070	1.625	0.902			2.597
10–19.9×	0.003	0.452	1.805	0.256		2.516
20–49.9×	0.000	0.160	7.940	0.767		8.867
50–99.9×	0.000	0.040	16.756	1.584		18.380
100–199.9×	0.006	0.015	12.858	4.042		16.920
200–499.9×	1.727	0.005	4.566	5.035		11.333
500–999.9×	3.434	0.001	1.123	1.796		6.354
1,000–1,999.9×	0.274	0.001	0.433	0.694	9.075	10.476
2,000–4,999.9×	0.881	0.000	0.153	0.876	3.874	5.785
5,000–9,999.9×	0.401	0.000	0.038	1.287	1.981	3.707
10,000–19,999.9×	0.821	0.000	0.014	0.578	0.768	2.182
20,000–49,999.9×	3.201	0.000	0.005	0.205	0.273	3.684
50,000 (max)	0.001	0.081	0.432	0.093	0.139	0.745
Total	**13.022**	**3.329**	**47.025**	**17.213**	**16.111**	**96.700**

A.3 ENHANCE2: probability of each band within each class

Table A.5 ENHANCE2: P(band | class) EXACT

Round win (×)	Base-game wins	Witch	Grave Spell	Dark Enchantment
0 (no win)		7.143%		
0.1–0.4×	68.39%			
0.5–0.9×	5.036%			
1–1.9×	20.09%	17.72%		
2–2.9×	2.305%	4.778%		
3–4.9×	0.3619%	25.53%		
5–9.9×	1.156%	38.69%		
10–19.9×	1.156%	5.173%	0.1331%	
20–49.9×	1.156%	0.8617%	1.573%	
50–99.9×	0.2909%	0.08819%	8.685%	
100–199.9×	0.009965%	0.01749%	35.01%	
200–499.9×	0.004204%	0.002897%	44.08%	
500–999.9×	0.04254%	0.0002960%	8.673%	
1,000–1,999.9×	0.002749%	5.80×10^{-5}%	1.425%	79.13%
2,000–4,999.9×	0.001552%	9.65×10^{-6}%	0.3534%	16.43%
5,000–9,999.9×	0.0002938%	9.93×10^{-7}%	0.04960%	3.524%
10,000–19,999.9×	0.0001493%	1.98×10^{-7}%	0.01152%	0.7033%
20,000–49,999.9×	0.0002730%	4.14×10^{-8}%	0.003959%	0.1171%
50,000 (max)	3.21×10^{-7}%	0.0002422%	0.001667%	0.1000%

Each column sums to 100% (probabilities conditional on the class).

Table A.6 ENHANCE2: RTP points by band and class

EXACT

Round win (×)	Base-game wins	Witch	Grave Spell	Dark Enchantment	All classes
0.1–0.4×	0.155				0.155
0.5–0.9×	0.046				0.046
1–1.9×	0.352	0.031			0.384
2–2.9×	0.070	0.016			0.086
3–4.9×	0.018	0.143			0.161
5–9.9×	0.096	0.325			0.421
10–19.9×	0.234	0.090	0.004		0.328
20–49.9×	0.599	0.032	0.099		0.730
50–99.9×	0.241	0.008	1.157		1.406
100–199.9×	0.016	0.003	8.595		8.614
200–499.9×	0.022	0.001	24.274		24.297
500–999.9×	0.383	0.000	9.555		9.938
1,000–1,999.9×	0.044	0.000	3.227	24.181	27.451
2,000–4,999.9×	0.077	0.000	1.734	10.324	12.136
5,000–9,999.9×	0.027	0.000	0.565	5.281	5.872
10,000–19,999.9×	0.029	0.000	0.263	2.048	2.340
20,000–49,999.9×	0.144	0.000	0.195	0.728	1.067
50,000 (max)	0.000	0.016	0.140	1.111	1.268
Total	**2.554**	**0.666**	**49.808**	**43.672**	**96.700**

A.4 MYSTERY: probability of each band within each class

Table A.7 MYSTERY: P(band | class)

EXACT

Round win (×)	Grave Spell	Dark Enchantment
20–49.9×	18.13%	
50–99.9×	18.13%	
100–199.9×	18.46%	
200–499.9×	23.18%	
500–999.9×	15.15%	
1,000–1,999.9×	5.089%	53.89%
2,000–4,999.9×	0.9559%	30.27%
5,000–9,999.9×	0.2222%	8.040%
10,000–19,999.9×	0.5041%	5.980%
20,000–49,999.9×	0.1620%	1.500%
50,000 (max)	0.004167%	0.3139%

Each column sums to 100% (probabilities conditional on the class).

Table A.8 MYSTERY: RTP points by band and class

EXACT

Round win (×)	Grave Spell	Dark Enchantment	All classes
20–49.9×	0.327		0.327
50–99.9×	0.654		0.654
100–199.9×	1.338		1.338
200–499.9×	3.715		3.715
500–999.9×	5.367		5.367
1,000–1,999.9×	2.970	14.624	17.595
2,000–4,999.9×	1.563	19.321	20.885
5,000–9,999.9×	0.781	11.395	12.176
10,000–19,999.9×	3.650	16.498	20.148
20,000–49,999.9×	2.449	8.804	11.253
50,000 (max)	0.104	3.139	3.243
Total	**22.918**	**73.782**	**96.700**

APPENDIX B

Way wins by symbol in the bonuses

RTP points by symbol and length in the free spins.

Table B.1 ENHANCE1: free-spin way wins by symbol — EXACT

Symbol	3 reels	4 reels	5 reels	Total (pts)	Symbol wins per 1,000 rounds	Mean multiplier
H1 Premium 1	2.351	1.957	11.230	15.538	20.09	1.76
H2 Premium 2	2.518	1.835	11.342	15.695	25.69	1.69
H3 Premium 3	2.417	1.918	12.239	16.574	34.47	1.72
H4 Premium 4	2.554	2.230	14.007	18.790	44.37	1.67
L1 A	0.545	0.362	2.181	3.088	13.86	1.93
L2 K	0.640	0.355	2.226	3.222	16.83	1.94
L3 Q	0.511	0.356	2.481	3.348	21.19	1.88
L4 J	0.578	0.323	2.478	3.378	24.50	1.86
L5 10	0.362	0.411	2.752	3.525	29.54	1.86
All symbols				**83.158**	**230.5**	

RTP points paid by the way wins of each symbol and length, wild multipliers and board splits included, before the round cap. Mean multiplier: multiplied ways divided by ways, i.e. the average wild multiplier of a winning way (1 without a multiplied wild; the board split doubles both and is not included).

Table B.2 ENHANCE2: free-spin way wins by symbol — EXACT

Symbol	3 reels	4 reels	5 reels	Total (pts)	Symbol wins per 1,000 rounds	Mean multiplier
H1 Premium 1	1.232	1.696	16.797	19.725	35.09	2.43
H2 Premium 2	1.254	1.513	16.820	19.588	41.85	2.40
H3 Premium 3	1.178	1.413	17.557	20.148	52.93	2.52
H4 Premium 4	1.187	1.462	19.727	22.377	63.85	2.49
L1 A	0.287	0.260	2.801	3.348	25.00	2.63
L2 K	0.312	0.232	2.673	3.217	28.71	2.51
L3 Q	0.245	0.229	2.856	3.330	34.62	2.47
L4 J	0.274	0.201	2.706	3.181	38.98	2.53
L5 10	0.169	0.245	2.815	3.228	45.41	2.59
All symbols				**98.141**	**366.4**	

RTP points paid by the way wins of each symbol and length, wild multipliers and board splits included, before the round cap. Mean multiplier: multiplied ways divided by ways, i.e. the average wild multiplier of a winning way (1 without a multiplied wild; the board split doubles both and is not included).

Table B.3 WICKED: free-spin way wins by symbol EXACT

Symbol	3 reels	4 reels	5 reels	Total (pts)	Symbol wins per 1,000 rounds	Mean multiplier
H1 Premium 1	2.780	2.412	15.142	20.333	680.50	1.89
H2 Premium 2	2.937	2.232	14.362	19.530	877.14	1.84
H3 Premium 3	2.732	2.347	15.326	20.406	1,187.82	1.80
H4 Premium 4	2.818	2.620	17.147	22.585	1,527.96	1.75
L1 A	0.637	0.469	2.944	4.049	459.92	2.12
L2 K	0.723	0.450	2.877	4.049	555.57	2.06
L3 Q	0.573	0.450	3.104	4.126	706.81	2.01
L4 J	0.640	0.378	3.008	4.026	820.64	1.97
L5 10	0.392	0.488	3.292	4.172	999.19	1.92
All symbols				**103.279**	**7,815.6**	

RTP points paid by the way wins of each symbol and length, wild multipliers and board splits included, before the round cap. Mean multiplier: multiplied ways divided by ways, i.e. the average wild multiplier of a winning way (1 without a multiplied wild; the board split doubles both and is not included).

Table B.4 GRAVE: free-spin way wins by symbol EXACT

Symbol	3 reels	4 reels	5 reels	Total (pts)	Symbol wins per 1,000 rounds	Mean multiplier
H1 Premium 1	2.231	2.164	15.083	19.478	973.19	2.23
H2 Premium 2	2.356	1.967	15.286	19.608	1,203.39	2.20
H3 Premium 3	2.186	1.926	16.188	20.300	1,559.64	2.22
H4 Premium 4	2.235	2.127	18.413	22.774	1,924.42	2.16
L1 A	0.529	0.373	2.542	3.444	712.85	2.34
L2 K	0.605	0.357	2.551	3.512	840.38	2.32
L3 Q	0.471	0.353	2.760	3.583	1,032.19	2.28
L4 J	0.526	0.319	2.617	3.462	1,170.55	2.30
L5 10	0.318	0.394	2.800	3.512	1,384.77	2.25
All symbols				**99.673**	**10,801.4**	

RTP points paid by the way wins of each symbol and length, wild multipliers and board splits included, before the round cap. Mean multiplier: multiplied ways divided by ways, i.e. the average wild multiplier of a winning way (1 without a multiplied wild; the board split doubles both and is not included).

Table B.5 MYSTERY: free-spin way wins by symbol

EXACT

Symbol	3 reels	4 reels	5 reels	Total (pts)	Symbol wins per 1,000 rounds	Mean multiplier
H1 Premium 1	0.833	1.254	26.262	28.350	440.28	3.28
H2 Premium 2	0.786	1.018	22.080	23.884	531.61	3.14
H3 Premium 3	0.688	0.915	19.175	20.778	657.82	3.07
H4 Premium 4	0.658	0.903	19.045	20.606	787.40	2.97
L1 A	0.183	0.205	3.178	3.566	322.55	3.29
L2 K	0.200	0.191	2.927	3.318	372.14	3.20
L3 Q	0.155	0.171	2.923	3.248	444.46	3.13
L4 J	0.165	0.143	2.577	2.884	497.20	3.07
L5 10	0.093	0.169	2.513	2.776	575.06	2.96
All symbols				**109.410**	**4,628.5**	

RTP points paid by the way wins of each symbol and length, wild multipliers and board splits included, before the round cap. Mean multiplier: multiplied ways divided by ways, i.e. the average wild multiplier of a winning way (1 without a multiplied wild; the board split doubles both and is not included).